

Enhanced Pulse Width Modulator (ePWM) Module

The enhanced pulse width modulator (ePWM) peripheral is a key element in controlling many of the power electronic systems found in both commercial and industrial equipments. The features supported by the ePWM make it especially suitable for digital motor control.

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19.1 Introduction

An effective PWM peripheral must be able to generate complex pulse width waveforms with minimal CPU overhead or intervention. It needs to be highly programmable and very flexible while being easy to understand and use. The ePWM unit described here addresses these requirements by allocating all needed timing and control resources on a per PWM channel basis. Cross coupling or sharing of resources has been avoided; instead, the ePWM is built up from smaller single channel modules with separate resources that can operate together as required to form a system. This modular approach results in an orthogonal architecture and provides a more transparent view of the peripheral structure, helping users to understand its operation quickly.

In this document the letter x within a signal or module name is used to indicate a generic ePWM instance on a device. For example, output signals EPWMxA and EPWMxB refer to the output signals from the ePWMx instance. Thus, EPWM1A and EPWM1B belong to ePWM1 and likewise EPWM4A and EPWM4B belong to ePWM4.

19.1.1 Submodule Overview

The ePWM module represents one complete PWM channel composed of two PWM outputs: EPWMxA and EPWMxB. Multiple ePWM modules are instanced within a device as shown in [Figure 19-1](#). Each ePWM instance is identical and is indicated by a numerical value starting with 1. For example, ePWM1 is the first instance and ePWM3 is the third instance in the system and ePWMx indicates any instance.

The ePWM modules are chained together via a clock synchronization scheme that allows them to operate as a single system when required. Additionally, this synchronization scheme can be extended to the capture peripheral modules (eCAP). Modules can also operate stand-alone.

Each ePWM module supports the following features:

- Dedicated 16-bit time-base counter with period and frequency control
- Two PWM outputs (EPWMxA and EPWMxB) that can be used in the following configurations:
 - Two independent PWM outputs with single-edge operation
 - Two independent PWM outputs with dual-edge symmetric operation
 - One independent PWM output with dual-edge asymmetric operation
- Asynchronous override control of PWM signals through software.
- Programmable phase-control support for lag or lead operation relative to other ePWM modules.
- Hardware-locked (synchronized) phase relationship on a cycle-by-cycle basis.
- Dead-band generation with independent rising and falling edge delay control.
- Programmable trip zone allocation of both cycle-by-cycle trip and one-shot trip on fault conditions.
- A trip condition can force either high, low, or high-impedance state logic levels at PWM outputs.
- All events can trigger both CPU interrupts and ADC start of conversion (SOC)
- Programmable event prescaling minimizes CPU overhead on interrupts.
- PWM chopping by high-frequency carrier signal, useful for pulse transformer gate drives.

Each ePWM module is connected to the input/output signals shown in [Figure 19-1](#). The signals are described in detail in subsequent sections.

Each ePWM module consists of eight submodules and is connected within a system via the signals shown in [Figure 19-2](#).

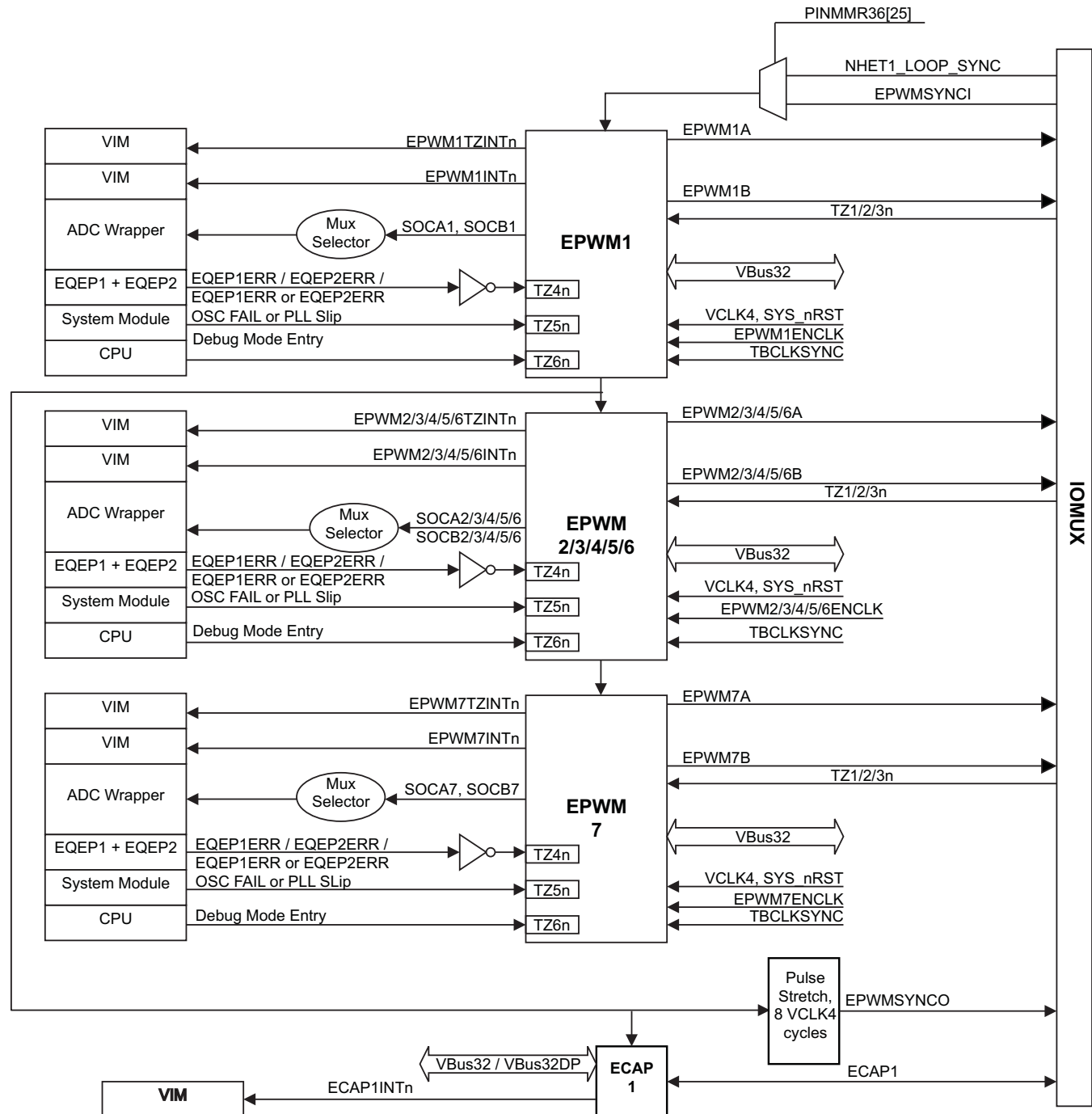
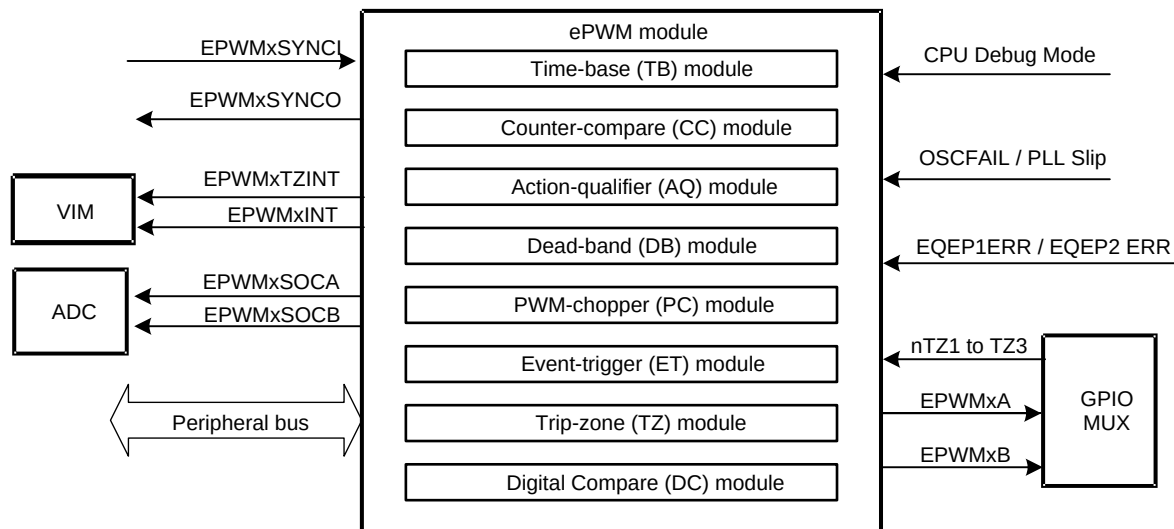
Figure 19-1. Multiple ePWM Modules


Figure 19-2. Submodules and Signal Connections for an ePWM Module


The main signals used by the ePWM module are:

- **PWM output signals (EPWMxA and EPWMxB).**

The PWM output signals are made available external to the device through the I/O Multiplexing Module (IOMM) as described in the IOMM chapter of the device's technical reference manual.

- **Trip-zone signals ($\overline{TZ1}$ to $\overline{TZ6}$).**

These input signals alert the ePWM module of fault conditions external to the ePWM module. Each ePWM module can be configured to either use or ignore any of the trip-zone signals. The $\overline{TZ1}$ to $\overline{TZ3}$ trip-zone signals can be configured as asynchronous inputs, or double-synchronized using VCLK4, or double-synchronized and filtered through a 6-VCLK4-cycle counter before connecting to the ePWM modules. This selection is done by configuring registers in the IOMM. $\overline{TZ4}$ is connected to an inverted eQEP1 error signal (EQEP1ERR), or to an inverted eQEP2 error signal (EQEP2ERR), or an OR-combination of EQEP1ERR and EQEP2ERR. This selection is also done via the IOMM registers. $\overline{TZ5}$ is connected to the system clock fail status. This is asserted whenever an oscillator failure is detected, or a PLL slip is detected. $\overline{TZ6}$ is connected to the debug mode entry indicator output from the CPU. This allows you to configure a trip action when the CPU halts.

- **Time-base synchronization input (EPWMxSYNCl) and output (EPWMxSYNCO) signals.**

The synchronization signals daisy chain the ePWM modules together. Each module can be configured to either use or ignore its synchronization input. The clock synchronization input and output signal are brought out to pins only for ePWM1 (ePWM module #1). The synchronization output for ePWM1 (EPWM1SYNCO) is also connected to the SYNCl of the first enhanced capture module (eCAP1).

- **ADC start-of-conversion signals (EPWMxSOCA and EPWMxSOCB).**

Each ePWM module has two ADC start of conversion signals. Any ePWM module can trigger a start of conversion. Which event triggers the start of conversion is configured in the Event-Trigger submodule of the ePWM.

- **Peripheral Bus**

The peripheral bus is 32-bits wide and allows both 16-bit and 32-bit writes to the ePWM register file.

19.1.2 Register Mapping

The complete ePWM module control and status register set is grouped by submodule as shown in [Table 19-1](#). Each register set is duplicated for each instance of the ePWM module. The start address for each ePWM register file instance on a device is specified in the specific part's datasheet.

Table 19-1. ePWM Module Control and Status Register Set Grouped by Submodule

Name	Address Offset ⁽¹⁾	Size (×16)	Shadow	Privileged Mode Write Only?	Description
Time-Base Submodule Registers					
TBCTL	0x0000	1	No	No	Time-Base Control Register
TBSTS	0x0002	1	No	No	Time-Base Status Register
Reserved	0x0004	1	–	–	Reserved
TBPHS	0x0006	1	No	No	Time-Base Phase Register
TBCTR	0x0008	1	No	No	Time-Base Counter Register
TBPRD	0x000A	1	Yes	No	Time-Base Period Register
Reserved	0x000C	1	–	–	Reserved
Counter-Compare Submodule Registers					
CMPCTL	0x000E	1	No	No	Counter-Compare Control Register
Reserved	0x0010	1	–	–	Reserved
CMPA	0x0012	1	Yes	No	Counter-Compare A Register
CMPB	0x0014	1	Yes	No	Counter-Compare B Register
Action-Qualifier Submodule Registers					
AQCTLA	0x0016	1	No	No	Action-Qualifier Control Register for Output A (EPWMxA)
AQCTLB	0x0018	1	No	No	Action-Qualifier Control Register for Output B (EPWMxB)
AQSFRC	0x001A	1	No	No	Action-Qualifier Software Force Register
AQCSFRC	0x001C	1	Yes	No	Action-Qualifier Continuous S/W Force Register Set
Dead-Band Generator Submodule Registers					
DBCTL	0x001E	1	No	No	Dead-Band Generator Control Register
DBRED	0x0020	1	No	No	Dead-Band Generator Rising Edge Delay Count Register
DBFED	0x0022	1	No	No	Dead-Band Generator Falling Edge Delay Count Register
Trip-Zone Submodule Registers					
TZSEL	0x0024	1	No	Yes	Trip-Zone Select Register
TZDCSEL	0x0026	1	No	Yes	Trip Zone Digital Compare Select Register
TZCTL	0x0028	1	No	Yes	Trip-Zone Control Register
TZEINT	0x002A	1	No	Yes	Trip-Zone Enable Interrupt Register
TZFLG	0x002C	1	No	No	Trip-Zone Flag Register
TZCLR	0x002E	1	No	Yes	Trip-Zone Clear Register
TZFRC	0x0030	1	No	Yes	Trip-Zone Force Register
Event-Trigger Submodule Registers					
ETSEL	0x0032	1	No	No	Event-Trigger Selection Register
ETPS	0x0034	1	No	No	Event-Trigger Pre-Scale Register
ETFLG	0x0036	1	No	No	Event-Trigger Flag Register
ETCLR	0x0038	1	No	No	Event-Trigger Clear Register
ETFRC	0x003A	1	No	No	Event-Trigger Force Register
PWM-Chopper Submodule Registers					
PCCTL	0x003C	1	No	No	PWM-Chopper Control Register

⁽¹⁾ Locations not shown are reserved.

Table 19-1. ePWM Module Control and Status Register Set Grouped by Submodule (continued)

Name	Address Offset ⁽¹⁾	Size (×16)	Shadow	Privileged Mode Write Only?	Description
Digital Compare Event Registers					
DCTRIPSEL	0x0060	1	No	Yes	Digital Compare Trip Select Register
DCACTL	0x0062	1	No	Yes	Digital Compare A Control Register
DCBCTL	0x0064	1	No	Yes	Digital Compare B Control Register
DCFCTL	0x0066	1	No	Yes	Digital Compare Filter Control Register
DCCAPCTL	0x0068	1	No	Yes	Digital Compare Capture Control Register
DCFOFFSET	0x006A	1	Writes	No	Digital Compare Filter Offset Register
DCFOFFSETCNT	0x006C	1	No	No	Digital Compare Filter Offset Counter Register
DCFWINDOW	0x006E	1	No	No	Digital Compare Filter Window Register
DCFWINDOWCNT	0x0070	1	No	No	Digital Compare Filter Window Counter Register
DCCAP	0x0072	1	Yes	No	Digital Compare Counter Capture Register

19.2 ePWM Submodules

Eight submodules are included in every ePWM peripheral. Each of these submodules performs specific tasks that can be configured by software.

19.2.1 Overview

Table 19-2 lists the eight key submodules together with a list of their main configuration parameters. For example, if you need to adjust or control the duty cycle of a PWM waveform, then you should see the counter-compare submodule in Section 19.2.3 for relevant details.

Table 19-2. Submodule Configuration Parameters

Submodule	Configuration Parameter or Option
Time-base (TB)	- Scale the time-base clock (TBCLK) relative to the system clock (VCLK4). - Configure the PWM time-base counter (TBCTR) frequency or period. - Set the mode for the time-base counter: - count-up mode: used for asymmetric PWM - count-down mode: used for asymmetric PWM - count-up-and-down mode: used for symmetric PWM - Configure the time-base phase relative to another ePWM module. - Synchronize the time-base counter between modules through hardware or software. - Configure the direction (up or down) of the time-base counter after a synchronization event. - Configure how the time-base counter will behave when the device is halted by an emulator. - Specify the source for the synchronization output of the ePWM module: - Synchronization input signal - Time-base counter equal to zero - Time-base counter equal to counter-compare B (CMPB) - No output synchronization signal generated.
Counter-compare (CC)	- Specify the PWM duty cycle for output EPWMxA and/or output EPWMxB - Specify the time at which switching events occur on the EPWMxA or EPWMxB output

Table 19-2. Submodule Configuration Parameters (continued)

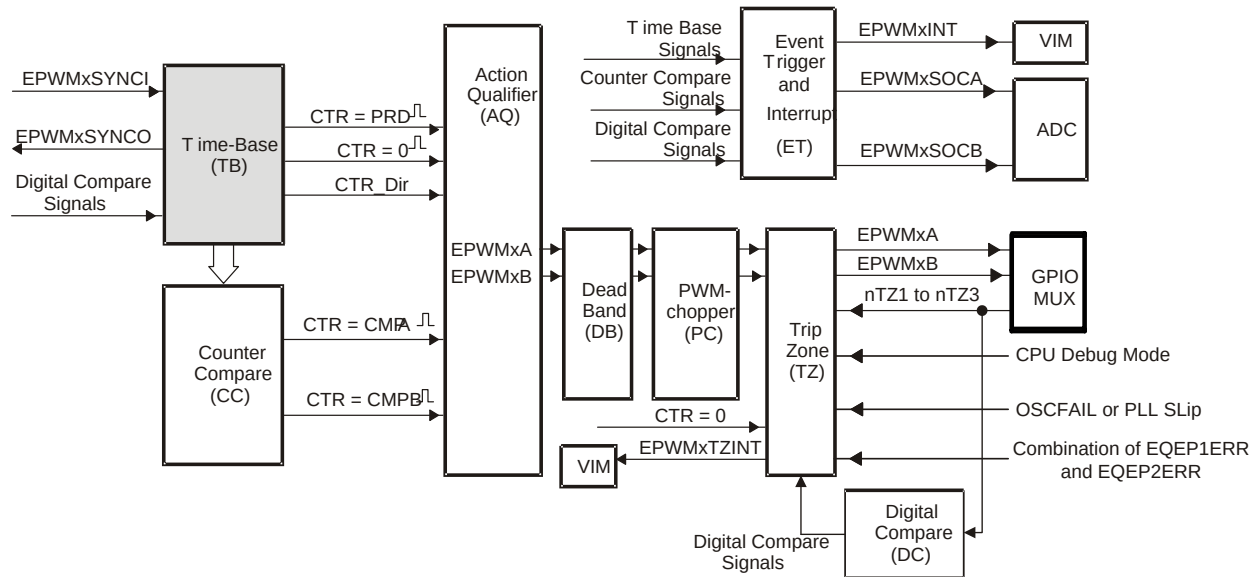
Submodule	Configuration Parameter or Option
Action-qualifier (AQ)	- Specify the type of action taken when a time-base or counter-compare submodule event occurs: - No action taken - Output EPWMxA and/or EPWMxB switched high - Output EPWMxA and/or EPWMxB switched low - Output EPWMxA and/or EPWMxB toggled - Force the PWM output state through software control - Configure and control the PWM dead-band through software
Dead-band (DB)	- Control of traditional complementary dead-band relationship between upper and lower switches - Specify the output rising-edge-delay value - Specify the output falling-edge delay value - Bypass the dead-band module entirely. In this case the PWM waveform is passed through without modification. - Option to enable half-cycle clocking for double resolution.
PWM-chopper (PC)	- Create a chopping (carrier) frequency. - Pulse width of the first pulse in the chopped pulse train. - Duty cycle of the second and subsequent pulses. - Bypass the PWM-chopper module entirely. In this case the PWM waveform is passed through without modification.
Trip-zone (TZ)	- Configure the ePWM module to react to one, all, or none of the trip-zone signals or digital compare events. - Specify the tripping action taken when a fault occurs: - Force EPWMxA and/or EPWMxB high - Force EPWMxA and/or EPWMxB low - Force EPWMxA and/or EPWMxB to a high-impedance state - Configure EPWMxA and/or EPWMxB to ignore any trip condition. - Configure how often the ePWM will react to each trip-zone signal: - One-shot - Cycle-by-cycle - Enable the trip-zone to initiate an interrupt. - Bypass the trip-zone module entirely.
Event-trigger (ET)	- Enable the ePWM events that will trigger an interrupt. - Enable ePWM events that will trigger an ADC start-of-conversion event. - Specify the rate at which events cause triggers (every occurrence or every second or third occurrence) - Poll, set, or clear event flags
Digital-compare (DC)	- Enables trip zone signals to create events and filtered events - Specify event-filtering options to capture TBCTR counter or generate blanking window

Code examples are provided in the remainder of this document that show how to implement various ePWM module configurations. These examples use the constant definitions in the device *EPwm_defines.h* file in the device-specific header file and peripheral examples software package.

19.2.2 Time-Base (TB) Submodule

Each ePWM module has its own time-base submodule that determines all of the event timing for the ePWM module. Built-in synchronization logic allows the time-base of multiple ePWM modules to work together as a single system. Figure 19-3 illustrates the time-base module's place within the ePWM.

Figure 19-3. Time-Base Submodule Block Diagram



19.2.2.1 Purpose of the Time-Base Submodule

You can configure the time-base submodule for the following:

- Specify the ePWM time-base counter (TBCTR) frequency or period to control how often events occur.
- Manage time-base synchronization with other ePWM modules.
- Maintain a phase relationship with other ePWM modules.
- Set the time-base counter to count-up, count-down, or count-up-and-down mode.
- Generate the following events:
 - CTR = PRD: Time-base counter equal to the specified period (TBCTR = TBPRD) .
 - CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000).
- Configure the rate of the time-base clock; a prescaled version of the device peripheral clock domain (VCLK4). This allows the time-base counter to increment/decrement at a slower rate.

19.2.2.2 Controlling and Monitoring the Time-base Submodule

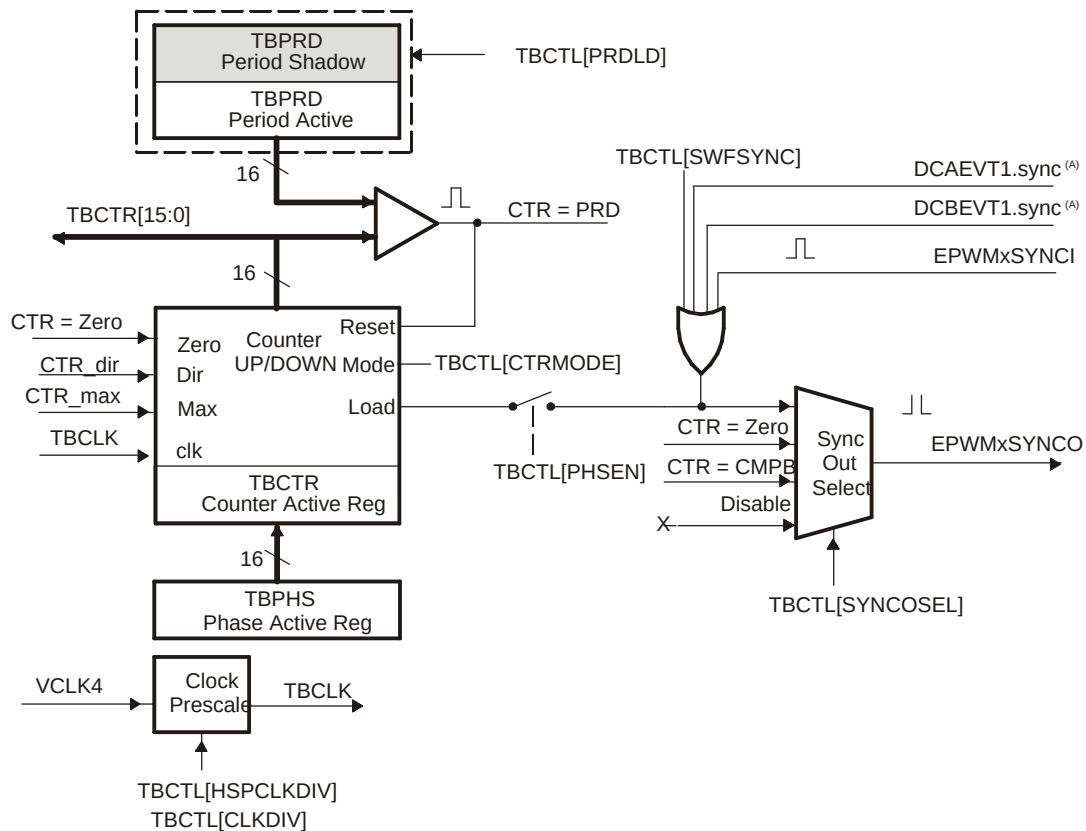
Table 19-3 shows the registers used to control and monitor the time-base submodule.

Table 19-3. Time-Base Submodule Registers

Register Name	Address Offset	Shadowed	Description
TBCTL	0x0000	No	Time-Base Control Register
TBSTS	0x0002	No	Time-Base Status Register
TBPHS	0x0006	No	Time-Base Phase Register
TBCTR	0x0008	No	Time-Base Counter Register
TBPRD	0x000A	Yes	Time-Base Period Register

The block diagram in Figure 19-4 shows the critical signals and registers of the time-base submodule. Table 19-4 provides descriptions of the key signals associated with the time-base submodule.

Figure 19-4. Time-Base Submodule Signals and Registers



A. These signals are generated by the digital compare (DC) submodule.

Table 19-4. Key Time-Base Signals

Signal	Description
EPWMxSYNCl	Time-base synchronization input. Input pulse used to synchronize the time-base counter with the counter of ePWM module earlier in the synchronization chain. An ePWM peripheral can be configured to use or ignore this signal. For the first ePWM module (EPWM1), this signal comes from a device pin or from the N2HET1 module. For subsequent ePWM modules, this signal is passed from another ePWM peripheral. For example, EPWM2SYNCl is generated by the ePWM1 peripheral, EPWM3SYNCl is generated by ePWM2 and so forth. See Section 19.2.2.3.3 for information on the synchronization order of a particular device.
EPWMxSYNCO	Time-base synchronization output. This output pulse is used to synchronize the counter of an ePWM module later in the synchronization chain. The ePWM module generates this signal from one of three event sources: 1. EPWMxSYNCl (Synchronization input pulse) 2. CTR = Zero: The time-base counter equal to zero (TBCTR = 0x0000). 3. CTR = CMPB: The time-base counter equal to the counter-compare B (TBCTR = CMPB) register.
CTR = PRD	Time-base counter equal to the specified period. This signal is generated whenever the counter value is equal to the active period register value. That is when TBCTR = TBPRD.
CTR = Zero	Time-base counter equal to zero This signal is generated whenever the counter value is zero. That is when TBCTR equals 0x0000.
CTR = CMPB	Time-base counter equal to active counter-compare B register (TBCTR = CMPB). This event is generated by the counter-compare submodule and used by the synchronization out logic
CTR_dir	Time-base counter direction. Indicates the current direction of the ePWM's time-base counter. This signal is high when the counter is increasing and low when it is decreasing.
CTR_max	Time-base counter equal max value. (TBCTR = 0xFFFF) Generated event when the TBCTR value reaches its maximum value. This signal is only used only as a status bit
TBCLK	Time-base clock. This is a prescaled version of the system clock (VCLK4) and is used by all submodules within the ePWM. This clock determines the rate at which time-base counter increments or decrements.

19.2.2.3 Calculating PWM Period and Frequency

The frequency of PWM events is controlled by the time-base period (TBPRD) register and the mode of the time-base counter. [Figure 19-5](#) shows the period (T_{pwm}) and frequency (F_{pwm}) relationships for the up-count, down-count, and up-down-count time-base counter modes when the period is set to 4 (TBPRD = 4). The time increment for each step is defined by the time-base clock (TBCLK) which is a prescaled version of the system clock (VCLK4).

The time-base counter has three modes of operation selected by the time-base control register (TBCTL):

- **Up-Down-Count Mode:**

In up-down-count mode, the time-base counter starts from zero and increments until the period (TBPRD) value is reached. When the period value is reached, the time-base counter then decrements until it reaches zero. At this point the counter repeats the pattern and begins to increment.

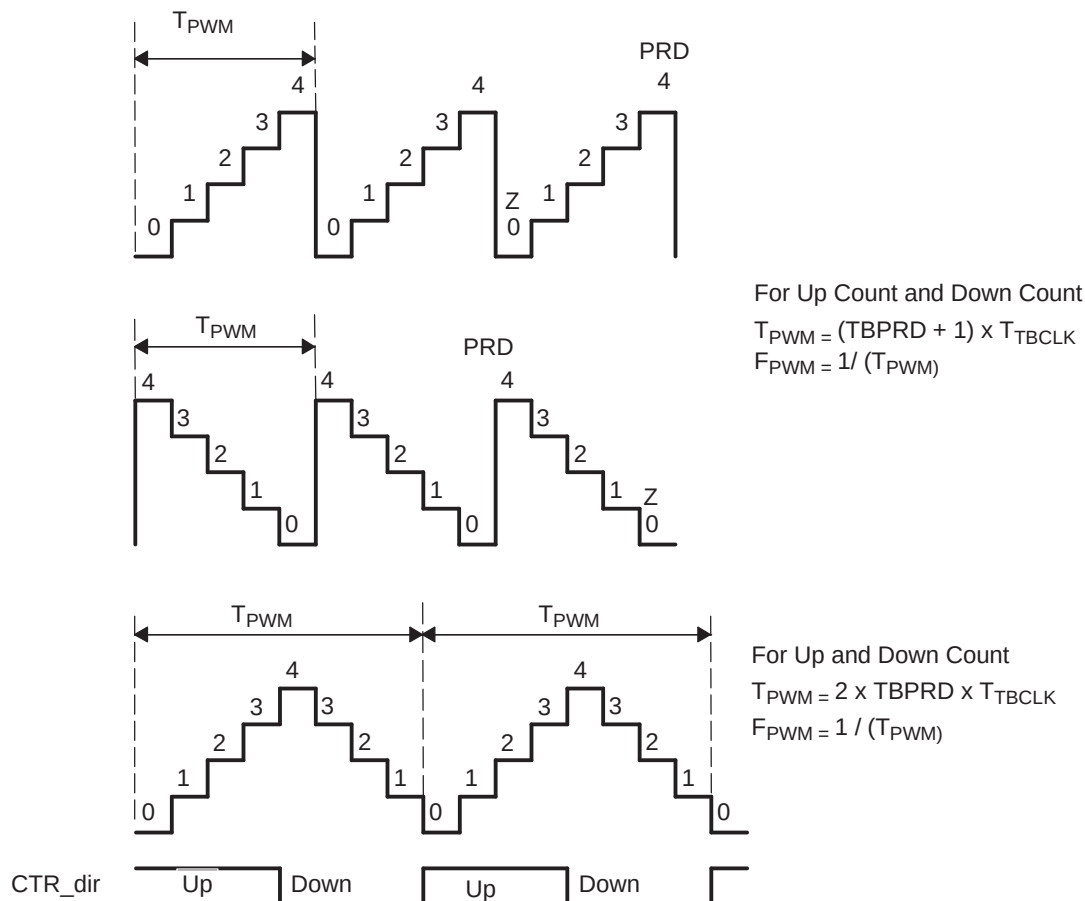
- **Up-Count Mode:**

In this mode, the time-base counter starts from zero and increments until it reaches the value in the period register (TBPRD). When the period value is reached, the time-base counter resets to zero and begins to increment once again.

- **Down-Count Mode:**

In down-count mode, the time-base counter starts from the period (TBPRD) value and decrements until it reaches zero. When it reaches zero, the time-base counter is reset to the period value and it begins to decrement once again.

Figure 19-5. Time-Base Frequency and Period



19.2.2.3.1 Time-Base Period Shadow Register

The time-base period register (TBPRD) has a shadow register. Shadowing allows the register update to be synchronized with the hardware. The following definitions are used to describe all shadow registers in the ePWM module:

- **Active Register**

The active register controls the hardware and is responsible for actions that the hardware causes or invokes.

- **Shadow Register**

The shadow register buffers or provides a temporary holding location for the active register. It has no direct effect on any control hardware. At a strategic point in time the shadow register's content is transferred to the active register. This prevents corruption or spurious operation due to the register being asynchronously modified by software.

The memory address of the shadow period register is the same as the active register. Which register is written to or read from is determined by the TBCTL[PRDL] bit. This bit enables and disables the TBPRD shadow register as follows:

- **Time-Base Period Shadow Mode:**

The TBPRD shadow register is enabled when TBCTL[PRDL] = 0. Reads from and writes to the TBPRD memory address go to the shadow register. The shadow register contents are transferred to the active register (TBPRD (Active) ← TBPRD (shadow)) when the time-base counter equals zero (TBCTR = 0x0000). By default the TBPRD shadow register is enabled.

- **Time-Base Period Immediate Load Mode:**

If immediate load mode is selected (TBCTL[PRDL] = 1), then a read from or a write to the TBPRD memory address goes directly to the active register.

19.2.2.3.2 Time-Base Clock Synchronization

Bit 1 of the device-level multiplexing control module (IOMM) register PINMMR37 is defined as the TBCLKSYNC bit. The TBCLKSYNC bit allows users to globally synchronize all enabled ePWM modules to the time-base clock (TBCLK). When set, all enabled ePWM module clocks are started with the first rising edge of TBCLK aligned. For perfectly synchronized TBCLKs, the prescalers for each ePWM module must be set identically.

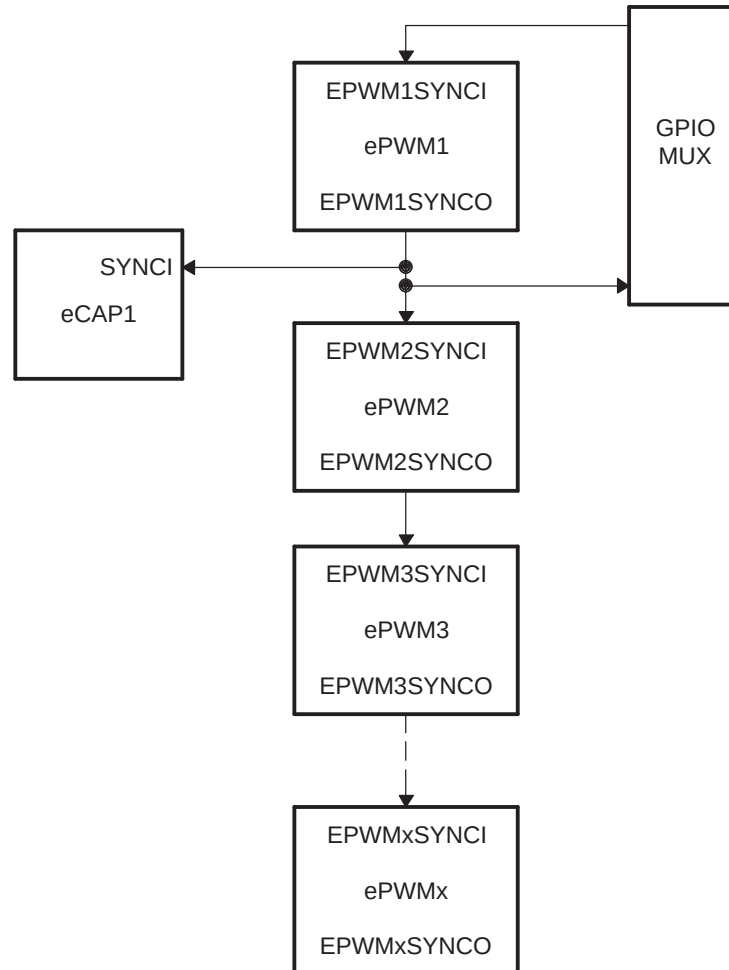
The proper procedure for enabling ePWM clocks is as follows:

1. Enable ePWM module clocks using the IOMM control registers for each ePWM module instance
2. Set TBCLKSYNC = 0. This will stop the time-base clock within any enabled ePWM module.
3. Configure ePWM modules: prescaler values and ePWM modes.
4. Set TBCLKSYNC = 1.

19.2.2.3.3 Time-Base Counter Synchronization

A time-base synchronization scheme connects all of the ePWM modules on a device. Each ePWM module has a synchronization input (EPWMxSYNCl) and a synchronization output (EPWMxSYNCO). The input synchronization for the first instance (ePWM1) comes from an external pin. The synchronization connections for the remaining ePWM modules are shown in Figure 19-6.

Figure 19-6. Time-Base Counter Synchronization Scheme



Each ePWM module can be configured to use or ignore the synchronization input. If the TBCTL[PHSEN] bit is set, then the time-base counter (TBCTR) of the ePWM module will be automatically loaded with the phase register (TBPHS) contents when one of the following conditions occur:

- **EPWMxSYNCl: Synchronization Input Pulse:**

The value of the phase register is loaded into the counter register when an input synchronization pulse is detected (TBPHS → TBCTR). This operation occurs on the next valid time-base clock (TBCLK) edge.

The delay from internal master module to slave modules is given by:

- if (TBCLK = VCLK4): $2 \times VCLK4$
- if (TBCLK != VCLK4): 1 TBCLK

- **Software Forced Synchronization Pulse:**

Writing a 1 to the TBCTL[SWFSYNC] control bit invokes a software forced synchronization. This pulse is ORed with the synchronization input signal, and therefore has the same effect as a pulse on EPWMxSYNCl.

- **Digital Compare Event Synchronization Pulse:**

DCAEVT1 and DCBEVT1 digital compare events can be configured to generate synchronization pulses which have the same affect as EPWMxSYNCl.

This feature enables the ePWM module to be automatically synchronized to the time base of another ePWM module. Lead or lag phase control can be added to the waveforms generated by different ePWM modules to synchronize them. In up-down-count mode, the TBCTL[PSHDIR] bit configures the direction of the time-base counter immediately after a synchronization event. The new direction is independent of the direction prior to the synchronization event. The PSHDIR bit is ignored in count-up or count-down modes. See [Figure 19-7](#) through [Figure 19-10](#) for examples.

Clearing the TBCTL[PHSEN] bit configures the ePWM to ignore the synchronization input pulse. The synchronization pulse can still be allowed to flow-through to the EPWMxSYNCO and be used to synchronize other ePWM modules. In this way, you can set up a master time-base (for example, ePWM1) and downstream modules (ePWM2 - ePWMx) may elect to run in synchronization with the master.

19.2.2.4 Phase Locking the Time-Base Clocks of Multiple ePWM Modules

The TBCLKSYNC bit can be used to globally synchronize the time-base clocks of all enabled ePWM modules on a device. When TBCLKSYNC = 0, the time-base clock of all ePWM modules is stopped (default). When TBCLKSYNC = 1, all ePWM time-base clocks are started with the rising edge of TBCLK aligned. For perfectly synchronized TBCLKs, the prescaler bits in the TBCTL register of each ePWM module must be set identically. The proper procedure for enabling the ePWM clocks is as follows:

1. Enable ePWM module clocks using the IOMM control registers for each ePWM module instance
2. Set TBCLKSYNC = 0. This will stop the time-base clock within any enabled ePWM module.
3. Configure ePWM modules: prescaler values and ePWM modes.
4. Set TBCLKSYNC = 1.

19.2.2.5 Time-Base Counter Modes and Timing Waveforms

The time-base counter operates in one of four modes:

- Up-count mode which is asymmetrical.
- Down-count mode which is asymmetrical.
- Up-down-count which is symmetrical
- Frozen where the time-base counter is held constant at the current value

To illustrate the operation of the first three modes, the following timing diagrams show when events are generated and how the time-base responds to an EPWMxSYNCl signal.

Figure 19-7. Time-Base Up-Count Mode Waveforms

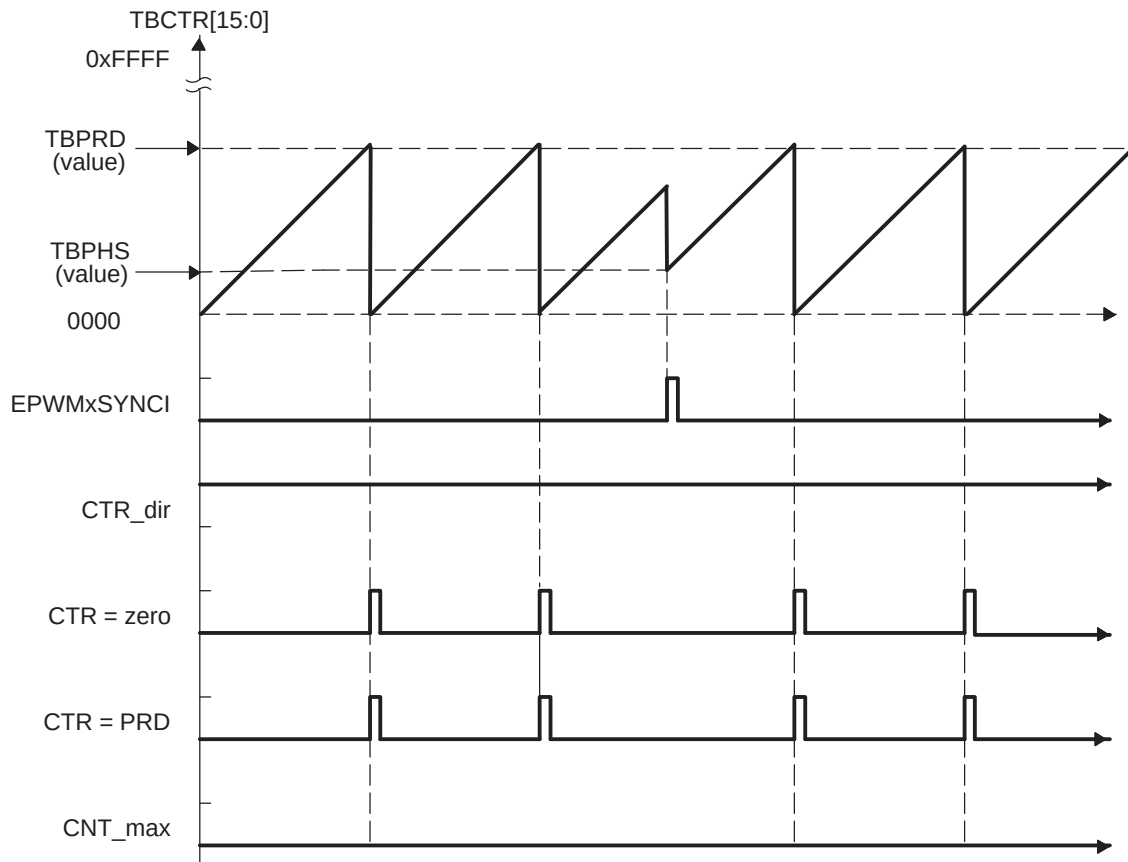


Figure 19-8. Time-Base Down-Count Mode Waveforms

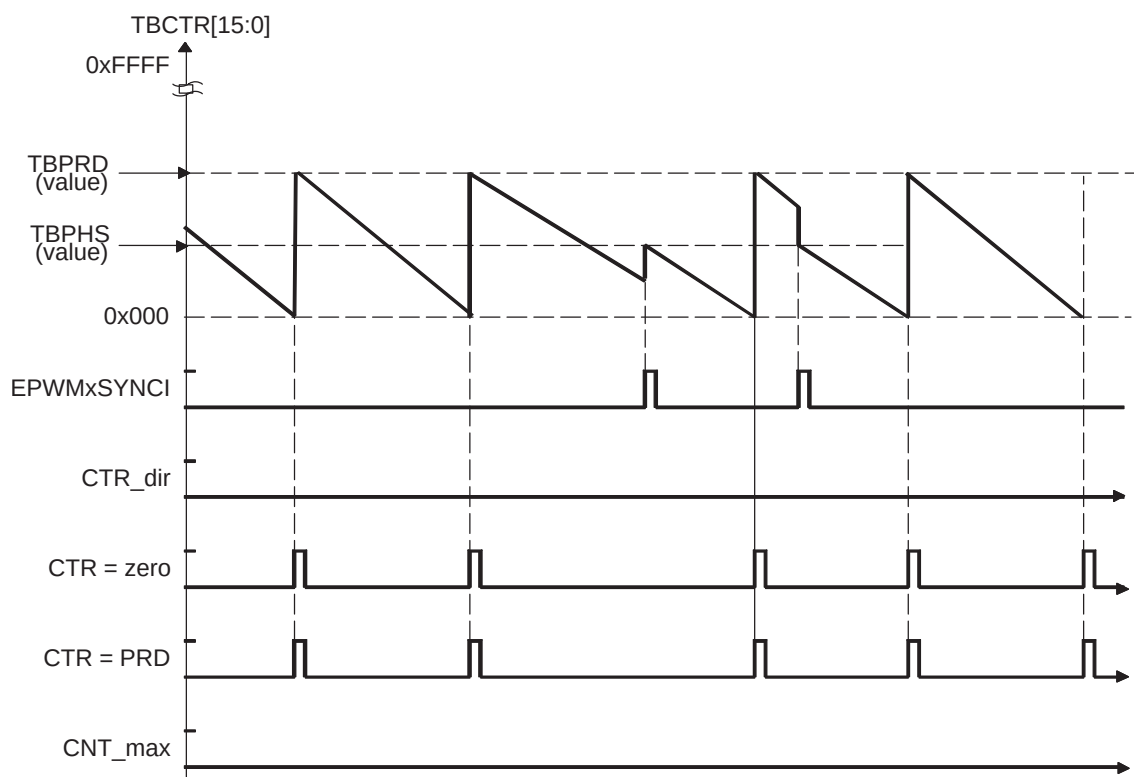


Figure 19-9. Time-Base Up-Down-Count Waveforms, TBCTL[PHSDIR = 0] Count Down On Synchronization Event

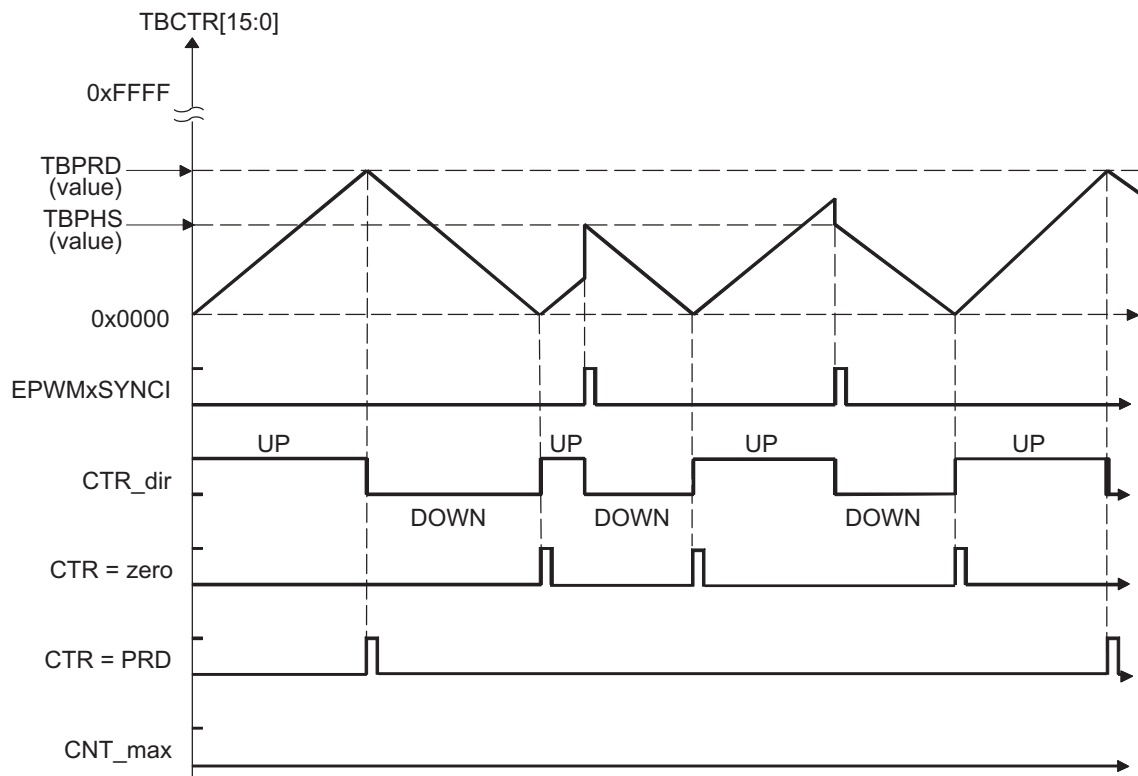
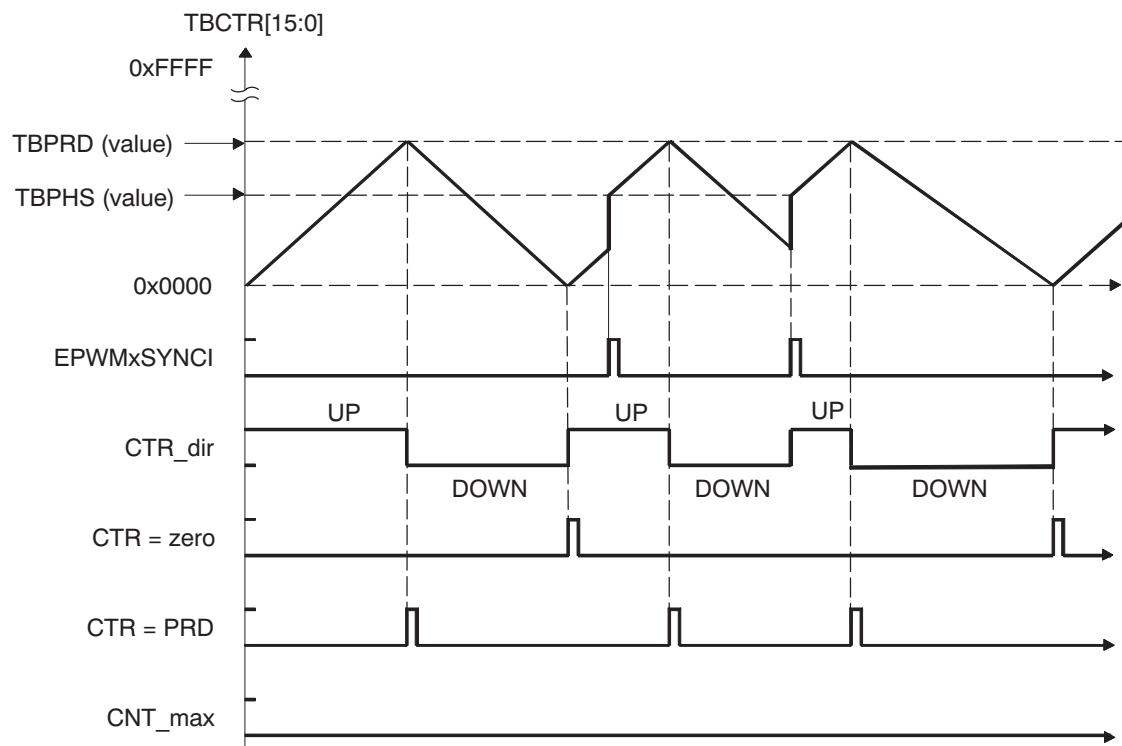


Figure 19-10. Time-Base Up-Down Count Waveforms, TBCTL[PHSDIR = 1] Count Up On Synchronization Event



19.2.3 Counter-Compare (CC) Submodule

Figure 19-11 illustrates the counter-compare submodule within the ePWM.

Figure 19-12 shows the basic structure of the counter-compare submodule.

Figure 19-11. Counter-Compare Submodule

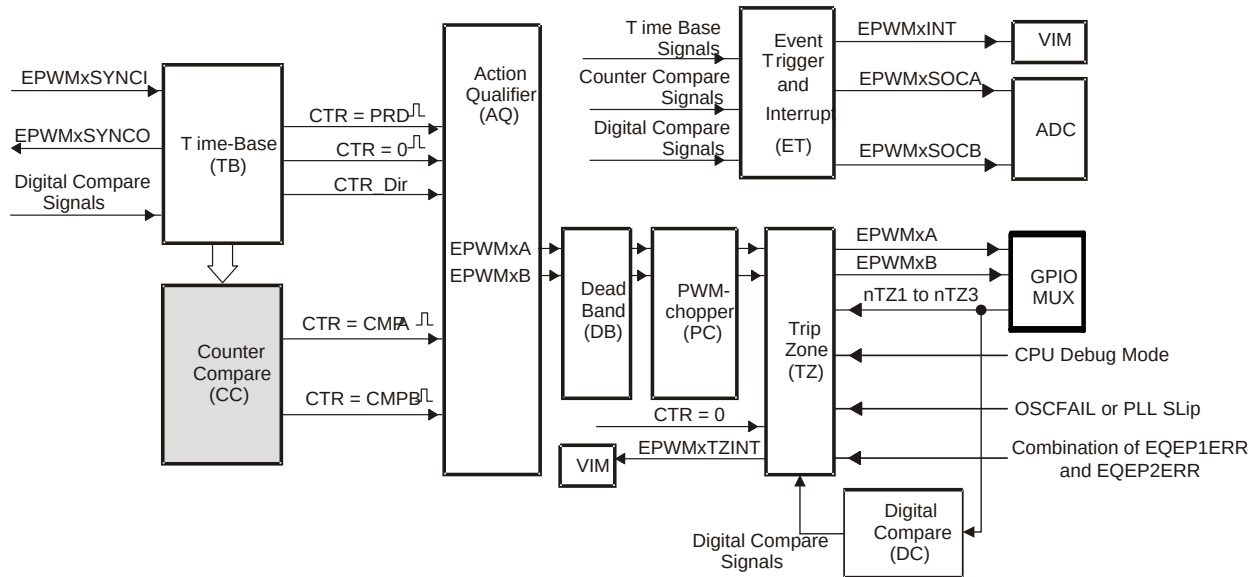
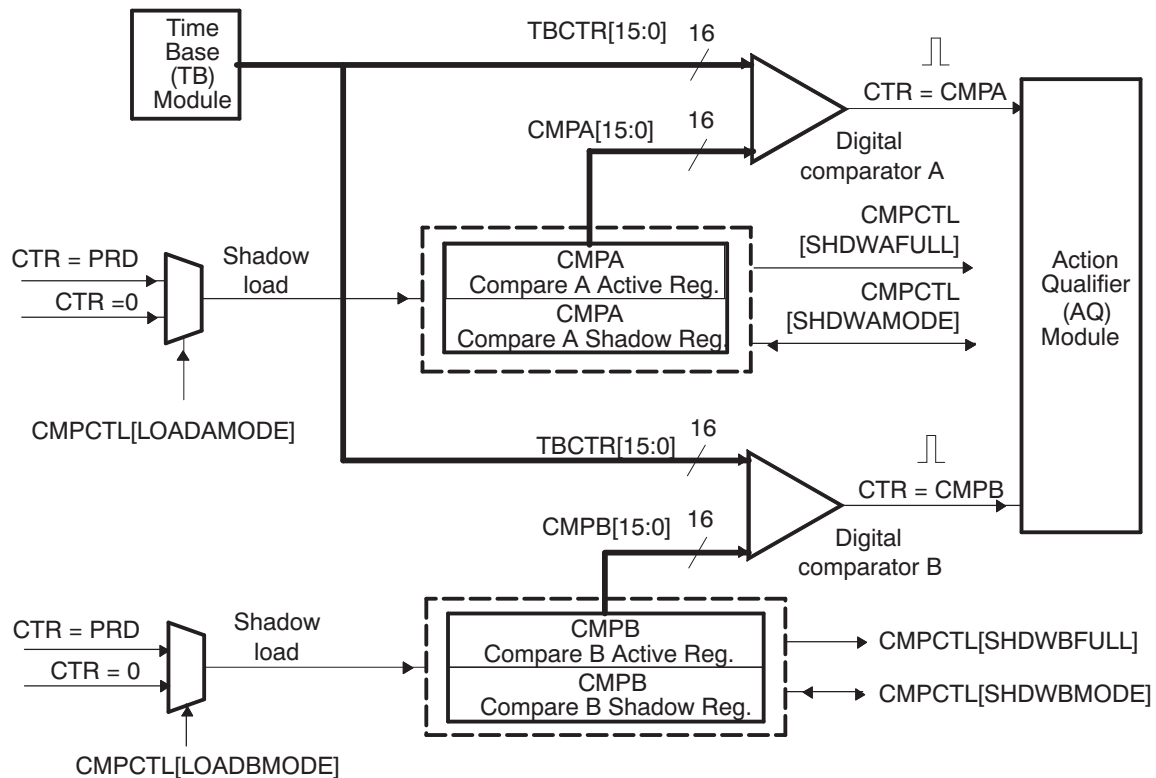


Figure 19-12. Detailed View of the Counter-Compare Submodule



19.2.3.1 Purpose of the Counter-Compare Submodule

The counter-compare submodule takes as input the time-base counter value. This value is continuously compared to the counter-compare A (CMPA) and counter-compare B (CMPB) registers. When the time-base counter is equal to one of the compare registers, the counter-compare unit generates an appropriate event.

The counter-compare:

- Generates events based on programmable time stamps using the CMPA and CMPB registers
 - CTR = CMPA: Time-base counter equals counter-compare A register (TBCTR = CMPA).
 - CTR = CMPB: Time-base counter equals counter-compare B register (TBCTR = CMPB)
- Controls the PWM duty cycle if the action-qualifier submodule is configured appropriately
- Shadows new compare values to prevent corruption or glitches during the active PWM cycle

19.2.3.2 Controlling and Monitoring the Counter-Compare Submodule

The counter-compare submodule operation is controlled and monitored by the registers shown in [Table 19-5](#):

Table 19-5. Counter-Compare Submodule Registers

Register Name	Address Offset	Shadowed	Description
CMPCTL	0x000E	No	Counter-Compare Control Register.
CMPA	0x0012	Yes	Counter-Compare A Register
CMPB	0x0014	Yes	Counter-Compare B Register

The key signals associated with the counter-compare submodule are described in [Table 19-6](#).

Table 19-6. Counter-Compare Submodule Key Signals

Signal	Description of Event	Registers Compared
CTR = CMPA	Time-base counter equal to the active counter-compare A value	TBCTR = CMPA
CTR = CMPB	Time-base counter equal to the active counter-compare B value	TBCTR = CMPB
CTR = PRD	Time-base counter equal to the active period. Used to load active counter-compare A and B registers from the shadow register	TBCTR = TBPRD
CTR = ZERO	Time-base counter equal to zero. Used to load active counter-compare A and B registers from the shadow register	TBCTR = 0x0000

19.2.3.3 Operational Highlights for the Counter-Compare Submodule

The counter-compare submodule is responsible for generating two independent compare events based on two compare registers:

1. CTR = CMPA: Time-base counter equal to counter-compare A register (TBCTR = CMPA).
2. CTR = CMPB: Time-base counter equal to counter-compare B register (TBCTR = CMPB).

For up-count or down-count mode, each event occurs only once per cycle. For up-down-count mode each event occurs twice per cycle if the compare value is between 0x0000-TBPRD and once per cycle if the compare value is equal to 0x0000 or equal to TBPRD. These events are fed into the action-qualifier submodule where they are qualified by the counter direction and converted into actions if enabled. Refer to [Section 19.2.4.1](#) for more details.

The counter-compare registers CMPA and CMPB each have an associated shadow register. Shadowing provides a way to keep updates to the registers synchronized with the hardware. When shadowing is used, updates to the active registers only occur at strategic points. This prevents corruption or spurious operation due to the register being asynchronously modified by software. The memory address of the active register and the shadow register is identical. Which register is written to or read from is determined by the CMPCTL[SHDWAMODE] and CMPCTL[SHDWBMODE] bits. These bits enable and disable the CMPA shadow register and CMPB shadow register respectively. The behavior of the two load modes is as described:

Shadow Mode:

The shadow mode for the CMPA is enabled by clearing the CMPCTL[SHDWAMODE] bit and the shadow register for CMPB is enabled by clearing the CMPCTL[SHDWBMODE] bit. Shadow mode is enabled by default for both CMPA and CMPB.

If the shadow register is enabled then the content of the shadow register is transferred to the active register on one of the following events as specified by the CMPCTL[LOADAMODE] and CMPCTL[LOADBMODE] register bits:

- CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD).
- CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)
- Both CTR = PRD and CTR = Zero

Only the active register contents are used by the counter-compare submodule to generate events to be sent to the action-qualifier.

Immediate Load Mode:

If immediate load mode is selected (that is, TBCTL[SHADWAMODE] = 1 or TBCTL[SHADWBMODE] = 1), then a read from or a write to the register will go directly to the active register.

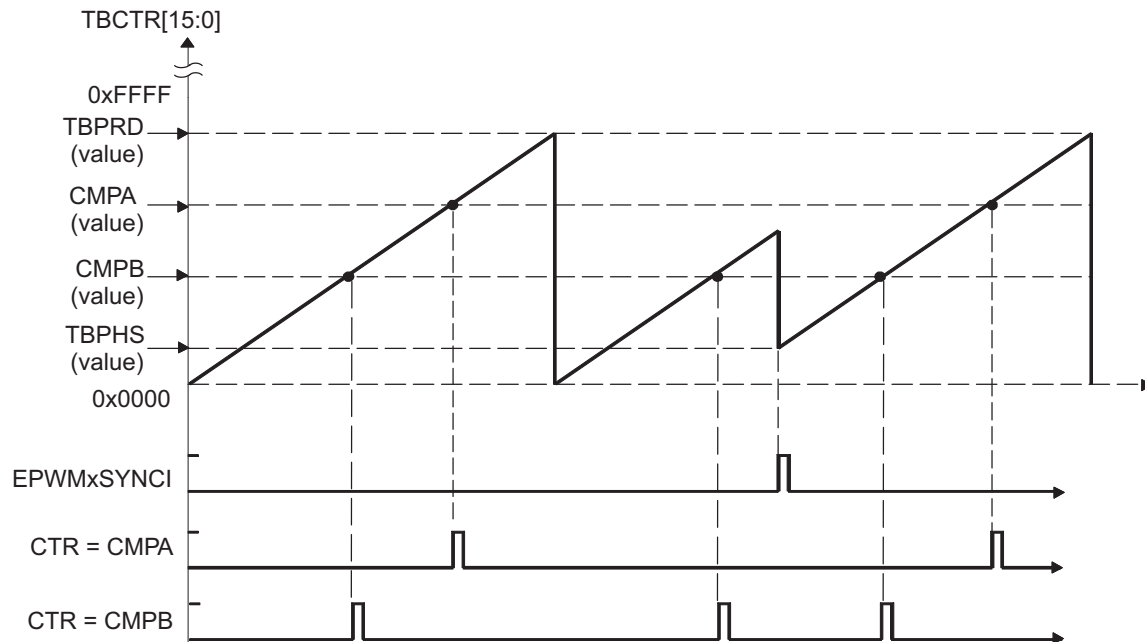
19.2.3.4 Count Mode Timing Waveforms

The counter-compare module can generate compare events in all three count modes:

- Up-count mode: used to generate an asymmetrical PWM waveform.
- Down-count mode: used to generate an asymmetrical PWM waveform.
- Up-down-count mode: used to generate a symmetrical PWM waveform.

To best illustrate the operation of the first three modes, the timing diagrams in [Figure 19-13](#) through [Figure 19-16](#) show when events are generated and how the EPWMxSYNCl signal interacts.

Figure 19-13. Counter-Compare Event Waveforms in Up-Count Mode



NOTE: An EPWMxSYNCl external synchronization event can cause a discontinuity in the TBCTR count sequence. This can lead to a compare event being skipped. This skipping is considered normal operation and must be taken into account.

Figure 19-14. Counter-Compare Events in Down-Count Mode

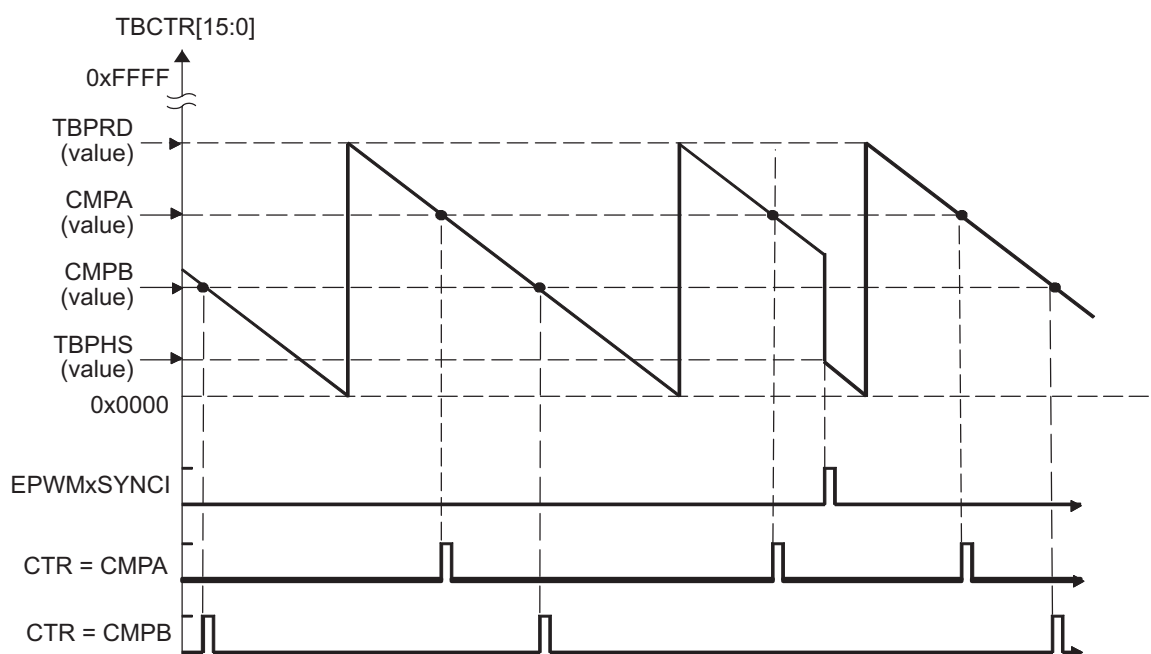


Figure 19-15. Counter-Compare Events In Up-Down-Count Mode, TBCTL[PHSDIR = 0] Count Down On Synchronization Event

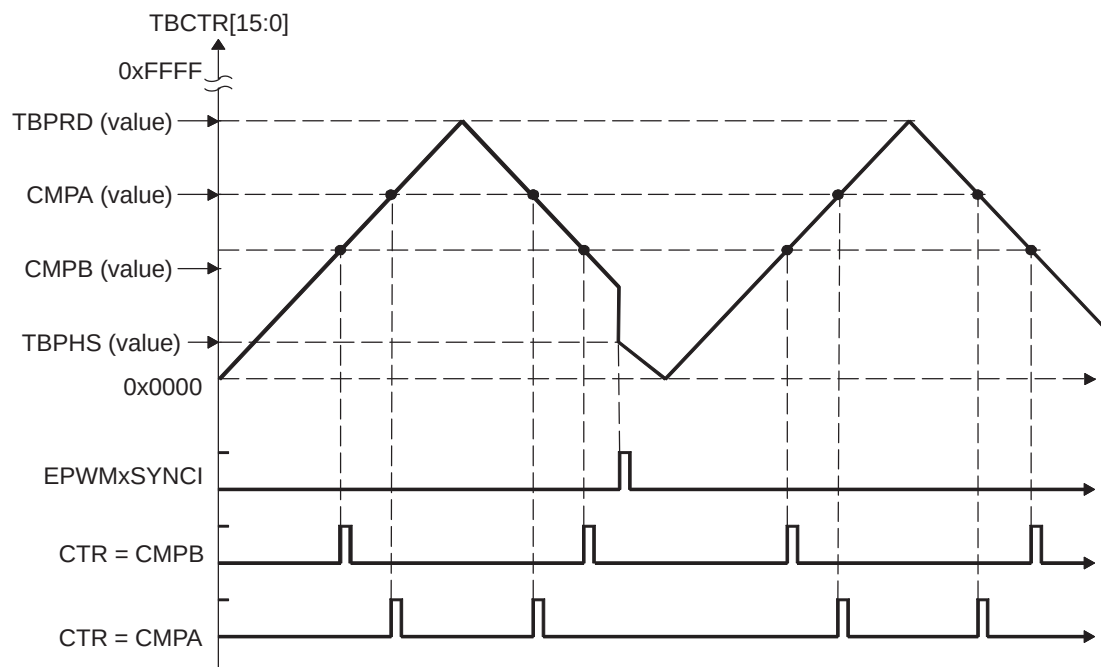
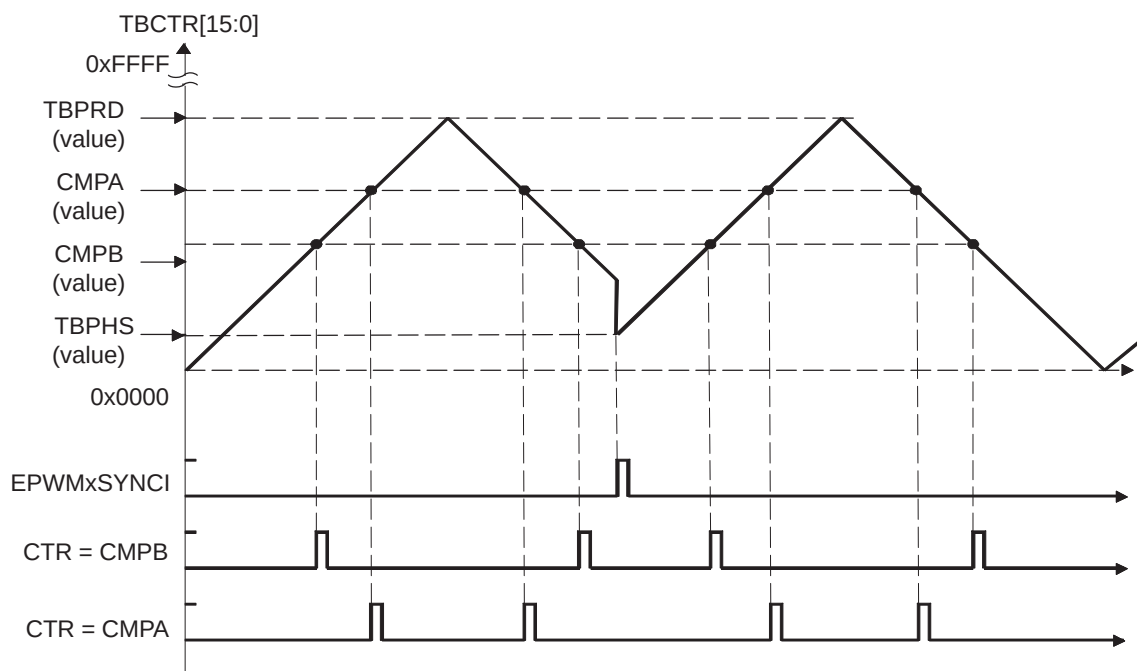


Figure 19-16. Counter-Compare Events In Up-Down-Count Mode, TBCTL[PHSDIR = 1] Count Up On Synchronization Event

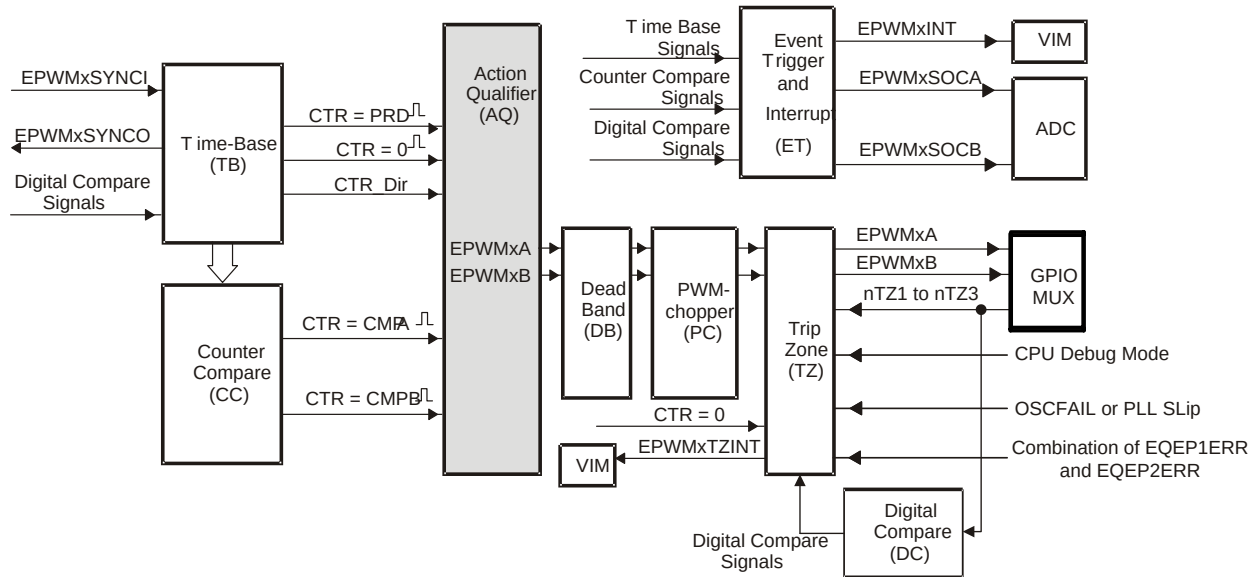


19.2.4 Action-Qualifier (AQ) Submodule

Figure 19-17 shows the action-qualifier (AQ) submodule (see shaded block) in the ePWM system.

The action-qualifier submodule has the most important role in waveform construction and PWM generation. It decides which events are converted into various action types, thereby producing the required switched waveforms at the EPWMxA and EPWMxB outputs.

Figure 19-17. Action-Qualifier Submodule



19.2.4.1 Purpose of the Action-Qualifier Submodule

The action-qualifier submodule is responsible for the following:

- Qualifying and generating actions (set, clear, toggle) based on the following events:
 - CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD)
 - CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)
 - CTR = CMPA: Time-base counter equal to the counter-compare A register (TBCTR = CMPA)
 - CTR = CMPB: Time-base counter equal to the counter-compare B register (TBCTR = CMPB)
- Managing priority when these events occur concurrently
- Providing independent control of events when the time-base counter is increasing and when it is decreasing

19.2.4.2 Action-Qualifier Submodule Control and Status Register Definitions

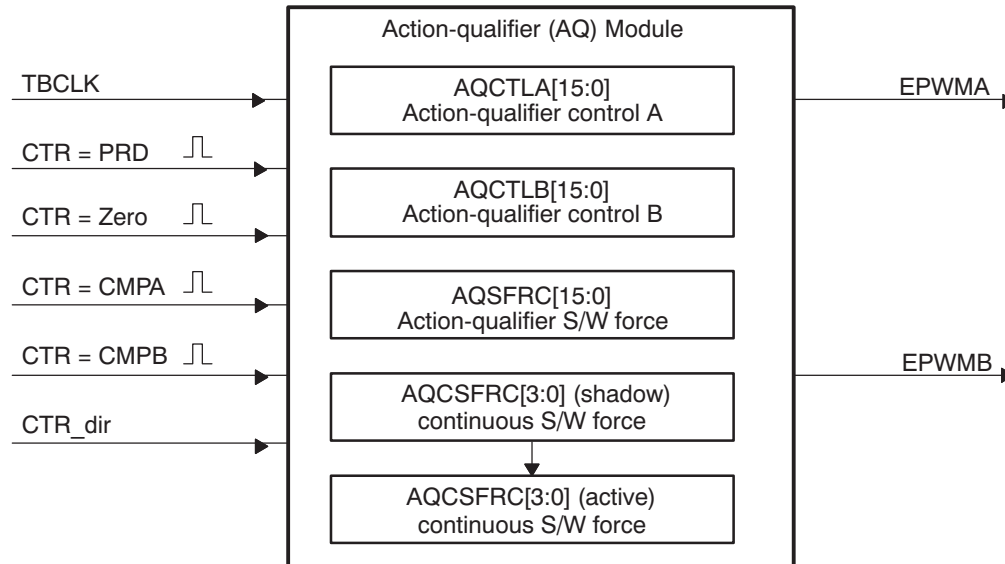
The action-qualifier submodule operation is controlled and monitored via the registers in Table 19-7.

Table 19-7. Action-Qualifier Submodule Registers

Register Name	Address Offset	Shadowed	Description
AQCTLA	0x0016	No	Action-Qualifier Control Register For Output A (EPWMxA)
AQCTLB	0x0018	No	Action-Qualifier Control Register For Output B (EPWMxB)
AQSFRC	0x001A	No	Action-Qualifier Software Force Register
AQCSFRC	0x001C	Yes	Action-Qualifier Continuous Software Force

The action-qualifier submodule is based on event-driven logic. It can be thought of as a programmable cross switch with events at the input and actions at the output, all of which are software controlled via the set of registers shown in [Table 19-7](#).

Figure 19-18. Action-Qualifier Submodule Inputs and Outputs



For convenience, the possible input events are summarized again in [Table 19-8](#).

Table 19-8. Action-Qualifier Submodule Possible Input Events

Signal	Description	Registers Compared
CTR = PRD	Time-base counter equal to the period value	TBCTR = TBPRD
CTR = Zero	Time-base counter equal to zero	TBCTR = 0x0000
CTR = CMPA	Time-base counter equal to the counter-compare A	TBCTR = CMPA
CTR = CMPB	Time-base counter equal to the counter-compare B	TBCTR = CMPB
Software forced event	Asynchronous event initiated by software	

The software forced action is a useful asynchronous event. This control is handled by registers AQSFR and AQCSFRC.

The action-qualifier submodule controls how the two outputs EPWMxA and EPWMxB behave when a particular event occurs. The event inputs to the action-qualifier submodule are further qualified by the counter direction (up or down). This allows for independent action on outputs on both the count-up and count-down phases.

The possible actions imposed on outputs EPWMxA and EPWMxB are:

- **Set High:**
Set output EPWMxA or EPWMxB to a high level.
- **Clear Low:**
Set output EPWMxA or EPWMxB to a low level.
- **Toggle:**
If EPWMxA or EPWMxB is currently pulled high, then pull the output low. If EPWMxA or EPWMxB is currently pulled low, then pull the output high.
- **Do Nothing:**
Keep outputs EPWMxA and EPWMxB at same level as currently set. Although the "Do Nothing" option prevents an event from causing an action on the EPWMxA and EPWMxB outputs, this event can still trigger interrupts and ADC start of conversion. See the Event-trigger Submodule description in [Section 19.2.8](#) for details.

Actions are specified independently for either output (EPWMxA or EPWMxB). Any or all events can be configured to generate actions on a given output. For example, both CTR = CMPA and CTR = CMPB can operate on output EPWMxA. All qualifier actions are configured via the control registers found at the end of this section.

For clarity, the drawings in this document use a set of symbolic actions. These symbols are summarized in [Figure 19-19](#). Each symbol represents an action as a marker in time. Some actions are fixed in time (zero and period) while the CMPA and CMPB actions are moveable and their time positions are programmed via the counter-compare A and B registers, respectively. To turn off or disable an action, use the "Do Nothing option"; it is the default at reset.

Figure 19-19. Possible Action-Qualifier Actions for EPWMxA and EPWMxB Outputs

S/W force	TB Counter equals:				Actions
	Zero	Comp A	Comp B	Period	
					Do Nothing
					Clear Low
					Set High
					Toggle

19.2.4.3 Action-Qualifier Event Priority

It is possible for the ePWM action qualifier to receive more than one event at the same time. In this case events are assigned a priority by the hardware. The general rule is events occurring later in time have a higher priority and software forced events always have the highest priority. The event priority levels for up-down-count mode are shown in [Table 19-9](#). A priority level of 1 is the highest priority and level 7 is the lowest. The priority changes slightly depending on the direction of TBCTR.

Table 19-9. Action-Qualifier Event Priority for Up-Down-Count Mode

Priority Level	Event If TBCTR is Incrementing TBCTR = Zero up to TBCTR = TBPRD	Event If TBCTR is Decrementing TBCTR = TBPRD down to TBCTR = 1
1 (Highest)	Software forced event	Software forced event
2	Counter equals CMPB on up-count (CBU)	Counter equals CMPB on down-count (CBD)
3	Counter equals CMPA on up-count (CAU)	Counter equals CMPA on down-count (CAD)
4	Counter equals zero	Counter equals period (TBPRD)
5	Counter equals CMPB on down-count (CBD)	Counter equals CMPB on up-count (CBU)
6 (Lowest)	Counter equals CMPA on down-count (CAD)	Counter equals CMPA on up-count (CBU)

[Table 19-10](#) shows the action-qualifier priority for up-count mode. In this case, the counter direction is always defined as up and thus down-count events will never be taken.

Table 19-10. Action-Qualifier Event Priority for Up-Count Mode

Priority Level	Event
1 (Highest)	Software forced event
2	Counter equal to period (TBPRD)
3	Counter equal to CMPB on up-count (CBU)
4	Counter equal to CMPA on up-count (CAU)
5 (Lowest)	Counter equal to Zero

[Table 19-11](#) shows the action-qualifier priority for down-count mode. In this case, the counter direction is always defined as down and thus up-count events will never be taken.

Table 19-11. Action-Qualifier Event Priority for Down-Count Mode

Priority Level	Event
1 (Highest)	Software forced event
2	Counter equal to Zero
3	Counter equal to CMPB on down-count (CBD)
4	Counter equal to CMPA on down-count (CAD)
5 (Lowest)	Counter equal to period (TBPRD)

It is possible to set the compare value greater than the period. In this case the action will take place as shown in [Table 19-12](#).

Table 19-12. Behavior if CMPA/CMPB is Greater than the Period

Counter Mode	Compare on Up-Count Event CAD/CBD	Compare on Down-Count Event CAD/CBD
Up-Count Mode	If $CMPA/CMPB \leq TBPRD$ period, then the event occurs on a compare match (TBCTR=CMPA or CMPB). If $CMPA/CMPB > TBPRD$, then the event will not occur.	Never occurs.
Down-Count Mode	Never occurs.	If $CMPA/CMPB < TBPRD$, the event will occur on a compare match (TBCTR=CMPA or CMPB). If $CMPA/CMPB \geq TBPRD$, the event will occur on a period match (TBCTR=TBPRD).
Up-Down-Count Mode	If $CMPA/CMPB < TBPRD$ and the counter is incrementing, the event occurs on a compare match (TBCTR=CMPA or CMPB). If $CMPA/CMPB \geq TBPRD$, the event will occur on a period match (TBCTR = TBPRD).	If $CMPA/CMPB < TBPRD$ and the counter is decrementing, the event occurs on a compare match (TBCTR=CMPA or CMPB). If $CMPA/CMPB \geq TBPRD$, the event occurs on a period match (TBCTR=TBPRD).

19.2.4.4 Waveforms for Common Configurations

NOTE: The waveforms in this document show the ePWMs behavior for a static compare register value. In a running system, the active compare registers (CMPA and CMPB) are typically updated from their respective shadow registers once every period. The user specifies when the update will take place; either when the time-base counter reaches zero or when the time-base counter reaches period. There are some cases when the action based on the new value can be delayed by one period or the action based on the old value can take effect for an extra period. Some PWM configurations avoid this situation. These include, but are not limited to, the following:

Use up-down-count mode to generate a symmetric PWM:

- If you load CMPA/CMPB on zero, then use CMPA/CMPB values greater than or equal to 1.
- If you load CMPA/CMPB on period, then use CMPA/CMPB values less than or equal to TBPRD - 1.

This means there will always be a pulse of at least one TBCLK cycle in a PWM period which, when very short, tend to be ignored by the system.

Use up-down-count mode to generate an asymmetric PWM:

- To achieve 50%-0% asymmetric PWM use the following configuration: Load CMPA/CMPB on period and use the period action to clear the PWM and a compare-up action to set the PWM. Modulate the compare value from 0 to TBPRD to achieve 50%-0% PWM duty.

When using up-count mode to generate an asymmetric PWM:

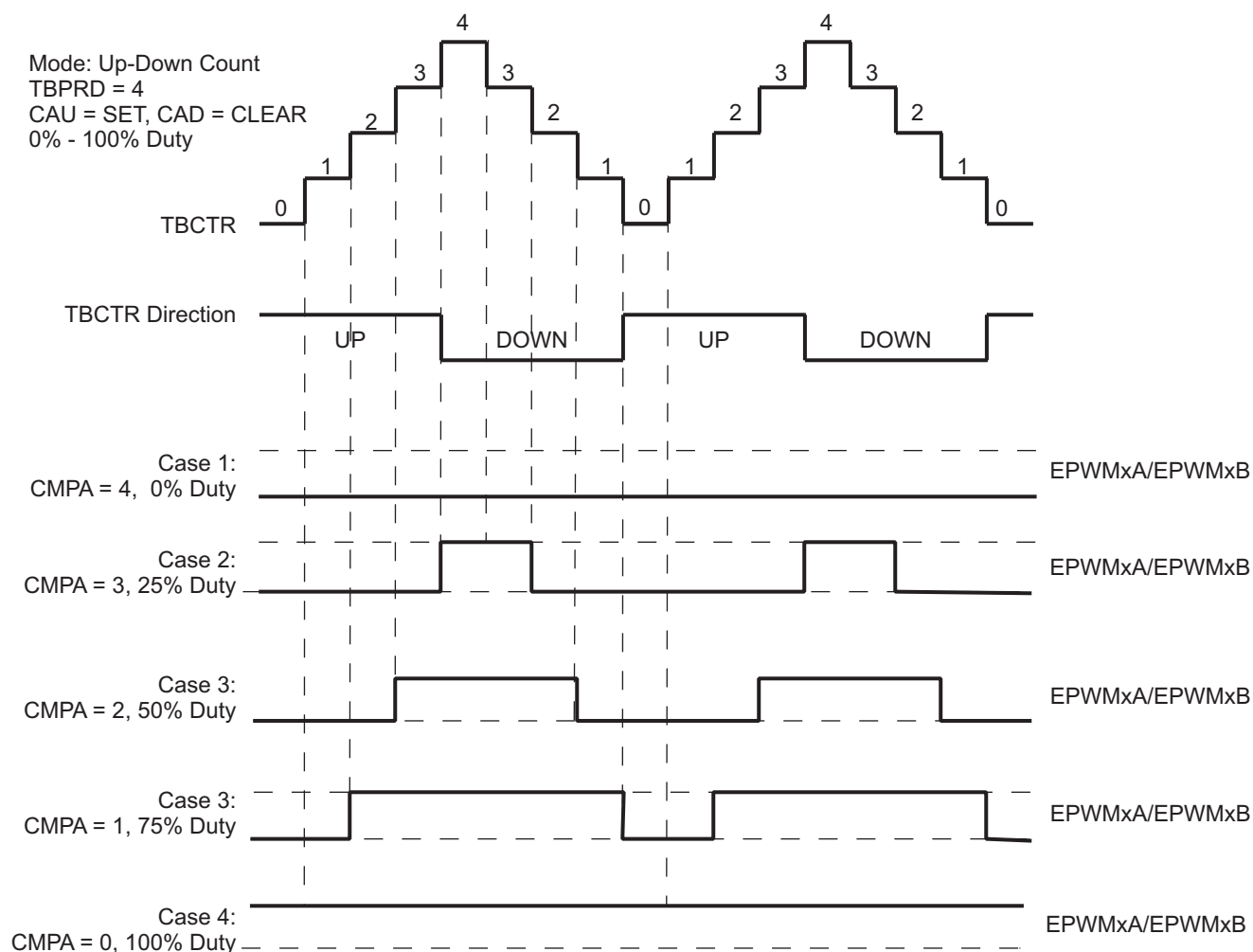
- To achieve 0-100% asymmetric PWM use the following configuration: Load CMPA/CMPB on TBPRD. Use the Zero action to set the PWM and a compare-up action to clear the PWM. Modulate the compare value from 0 to TBPRD + 1 to achieve 0-100% PWM duty.

See the *Using Enhanced Pulse Width Modulator (ePWM) Module for 0-100% Duty Cycle Control Application Report* ([SPRAA11](#))

Figure 19-20 shows how a symmetric PWM waveform can be generated using the up-down-count mode of the TBCTR. In this mode 0%-100% DC modulation is achieved by using equal compare matches on the up count and down count portions of the waveform. In the example shown, CMPA is used to make the comparison. When the counter is incrementing the CMPA match will pull the PWM output high. Likewise, when the counter is decrementing the compare match will pull the PWM signal low. When CMPA = 0, the PWM signal is low for the entire period giving the 0% duty waveform. When CMPA = TBPRD, the PWM signal is high achieving 100% duty.

When using this configuration in practice, if you load CMPA/CMPB on zero, then use CMPA/CMPB values greater than or equal to 1. If you load CMPA/CMPB on period, then use CMPA/CMPB values less than or equal to TBPRD - 1. This means there will always be a pulse of at least one TBCLK cycle in a PWM period which, when very short, tend to be ignored by the system.

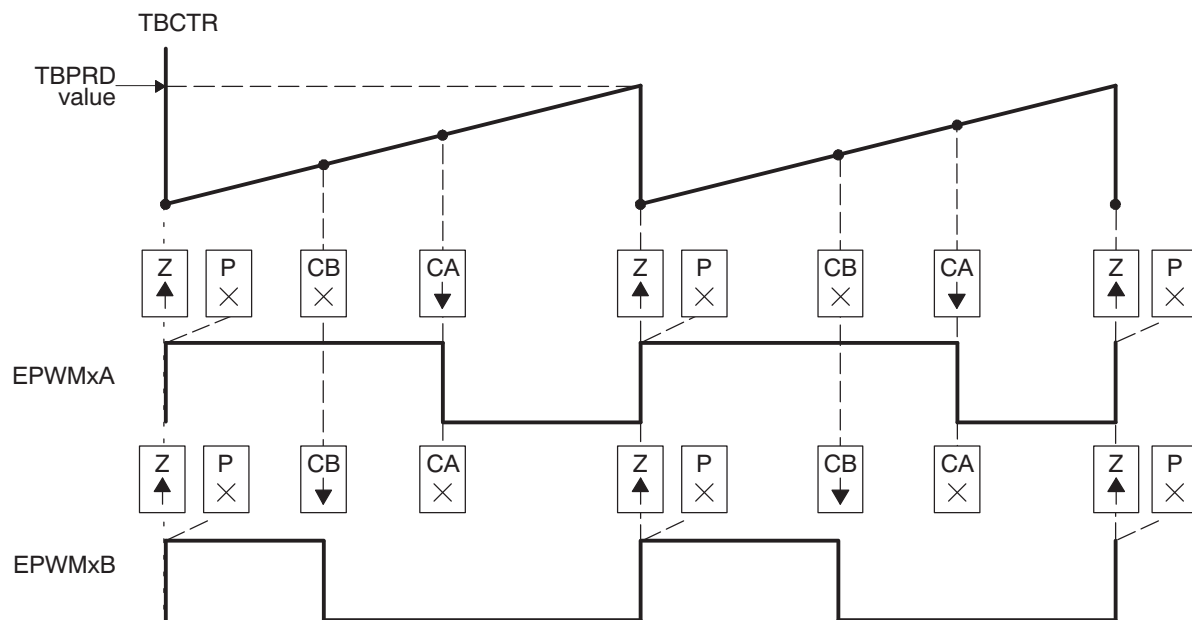
Figure 19-20. Up-Down-Count Mode Symmetrical Waveform



The PWM waveforms in [Figure 19-21](#) through [Figure 19-26](#) show some common action-qualifier configurations. The C-code samples in [Example 19-1](#) through [Example 19-6](#) shows how to configure an ePWM module for each case. Some conventions used in the figures and examples are as follows:

- TBPRD, CMPA, and CMPB refer to the value written in their respective registers. The active register, not the shadow register, is used by the hardware.
- CMPx, refers to either CMPA or CMPB
- EPWMxA and EPWMxB refer to the output signals from ePWMx
- Up-Down means Count-up-and-down mode, Up means up-count mode and Dwn means down-count mode
- Sym = Symmetric, Asym = Asymmetric

Figure 19-21. Up, Single Edge Asymmetric Waveform, With Independent Modulation on EPWMxA and EPWMxB—Active High



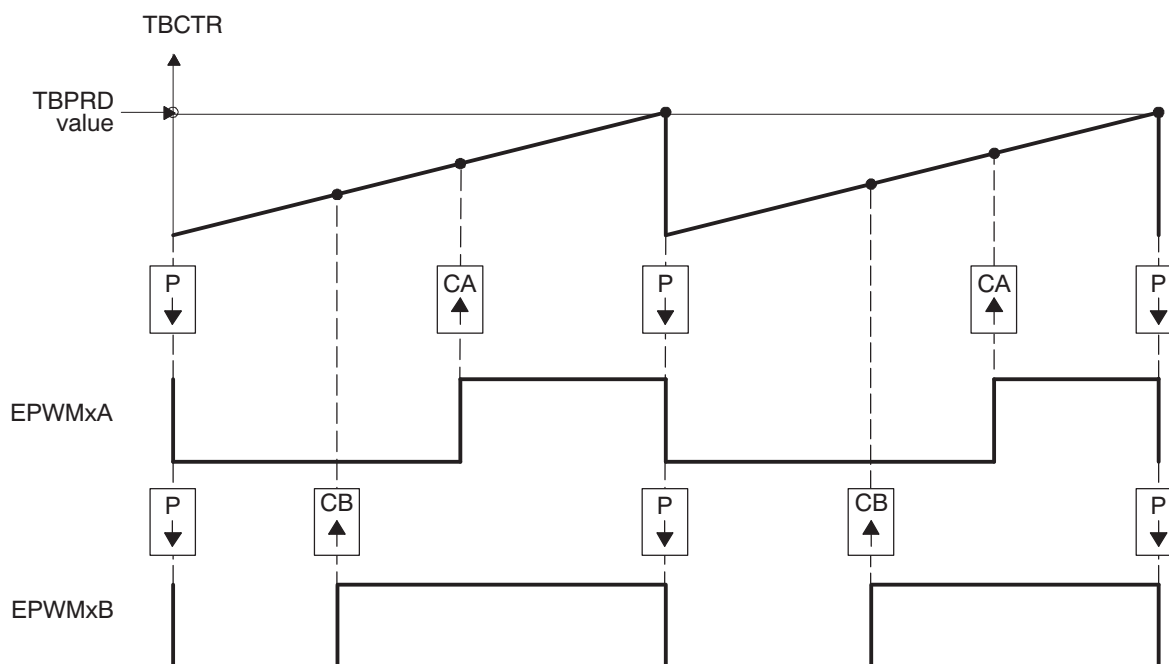
- PWM period = $(TBPRD + 1) \times T_{TBCLK}$
- Duty modulation for EPWMxA is set by CMPA, and is active high (that is, high time duty proportional to CMPA).
- Duty modulation for EPWMxB is set by CMPB and is active high (that is, high time duty proportional to CMPB).
- The "Do Nothing" actions (X) are shown for completeness, but will not be shown on subsequent diagrams.
- Actions at zero and period, although appearing to occur concurrently, are actually separated by one TBCLK period. TBCTR wraps from period to 0000.

[Example 19-1](#) contains a code sample showing initialization and run time for the waveforms in [Figure 19-21](#).

Example 19-1. Code Sample for [Figure 19-21](#)

```
// Initialization Time
// = = = = =
EPwm1Regs.TBPRD = 600;                // Period = 601 TBCLK counts
EPwm1Regs.CMPA.half.CMPA = 350;      // Compare A = 350 TBCLK counts
EPwm1Regs.CMPB = 200;                // Compare B = 200 TBCLK counts
EPwm1Regs.TBPHS = 0;                 // Set Phase register to zero
EPwm1Regs.TBCTR = 0;                 // clear TB counter
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; // TBCLK = SYSCLK
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.AQCTLA.bit.ZRO = AQ_SET;
EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.ZRO = AQ_SET;
EPwm1Regs.AQCTLB.bit.CBU = AQ_CLEAR;
//
// Run Time
// = = = = =
EPwm1Regs.CMPA.half.CMPA = Duty1A;    // adjust duty for output EPWM1A
EPwm1Regs.CMPB = Duty1B;              // adjust duty for output EPWM1B
```

Figure 19-22. Up, Single Edge Asymmetric Waveform With Independent Modulation on EPWMxA and EPWMxB—Active Low



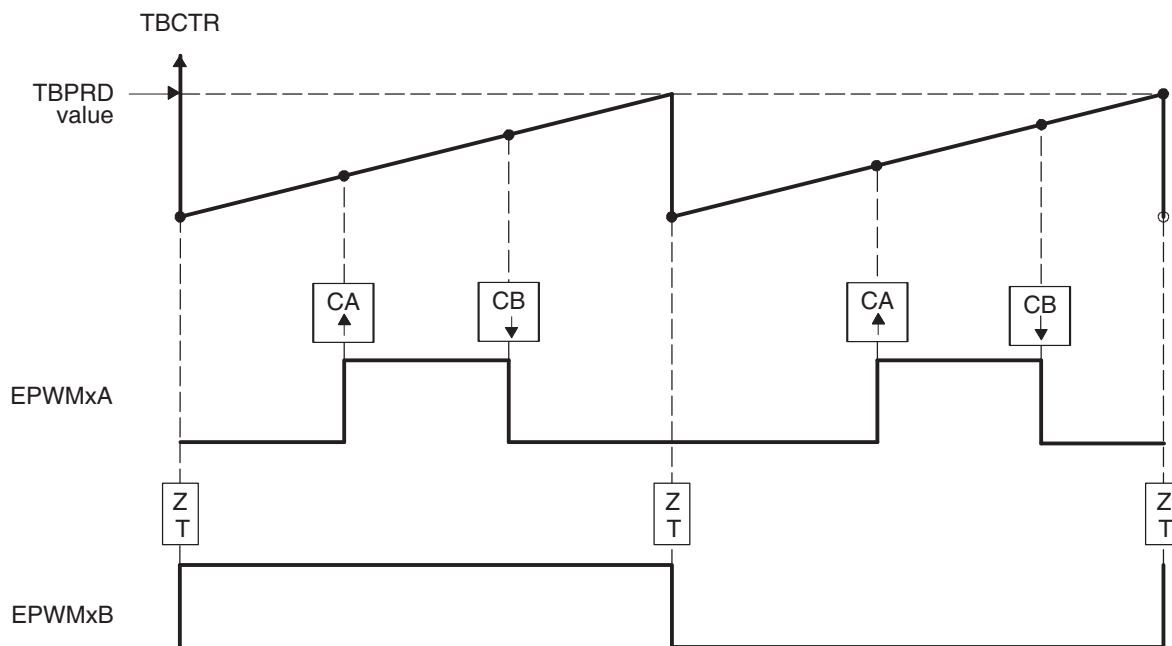
- A $\text{PWM period} = (\text{TBPRD} + 1) \times T_{\text{TBCLK}}$
- B Duty modulation for EPWMxA is set by CMPA, and is active low (that is, the low time duty is proportional to CMPA).
- C Duty modulation for EPWMxB is set by CMPB and is active low (that is, the low time duty is proportional to CMPB).
- D Actions at zero and period, although appearing to occur concurrently, are actually separated by one TBCLK period. TBCTR wraps from period to 0000.

[Example 19-2](#) contains a code sample showing initialization and run time for the waveforms in [Figure 19-22](#).

Example 19-2. Code Sample for Figure 19-22

```
// Initialization Time
// =====
EPwm1Regs.TBPRD = 600;                // Period = 601 TBCLK counts
EPwm1Regs.CMPA.half.CMPA = 350;       // Compare A = 350 TBCLK counts
EPwm1Regs.CMPB = 200;                 // Compare B = 200 TBCLK counts
EPwm1Regs.TBPHS = 0;                  // Set Phase register to zero
EPwm1Regs.TBCTR = 0;                  // clear TB counter
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; // TBCLK = VCLK4
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on TBCTR = Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on TBCTR = Zero
EPwm1Regs.AQCTLA.bit.PR = AQ_CLEAR;
EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;
EPwm1Regs.AQCTLB.bit.PR = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.CBU = AQ_SET;
//
// Run Time
// =====
EPwm1Regs.CMPA.half.CMPA = Duty1A;    // adjust duty for output EPWM1A
EPwm1Regs.CMPB = Duty1B;              // adjust duty for output EPWM1B
```

Figure 19-23. Up-Count, Pulse Placement Asymmetric Waveform With Independent Modulation on EPWMxA



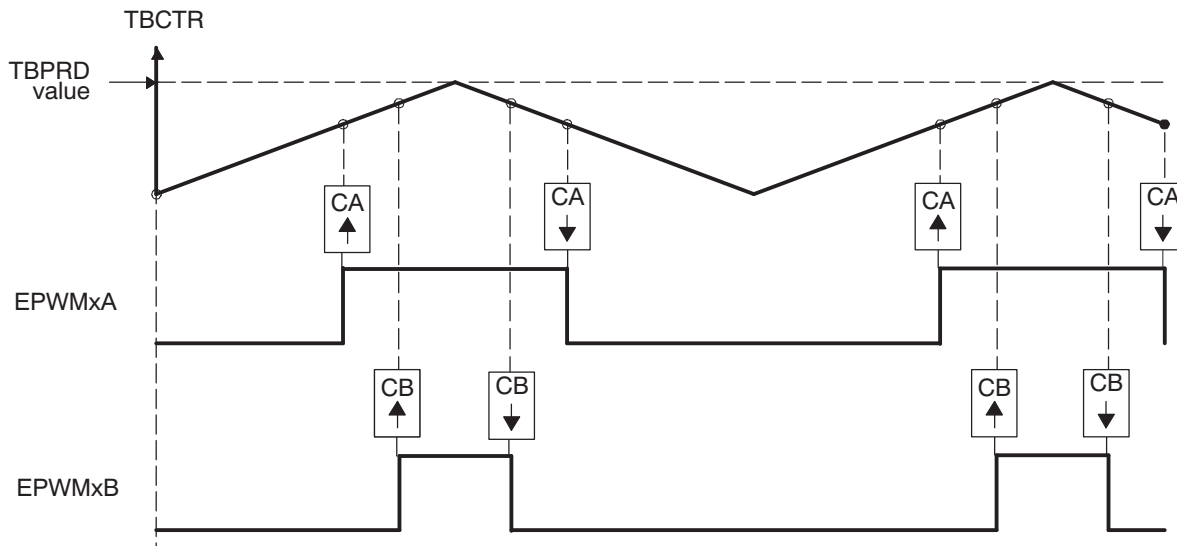
- A $PWM\ frequency = 1 / ((TBPRD + 1) \times T_{TBCLK})$
- B Pulse can be placed anywhere within the PWM cycle (0000 - TBPRD)
- C High time duty proportional to (CMPB - CMPA)
- D EPWMxB can be used to generate a 50% duty square wave with frequency = $1/2 \times ((TBPRD + 1) \times TBCLK)$

Example 19-3 contains a code sample showing initialization and run time for the waveforms in Figure 19-23.

Example 19-3. Code Sample for Figure 19-23

```
// Initialization Time
// =====
EPwm1Regs.TBPRD = 600;                // Period = 601 TBCLK counts
EPwm1Regs.CMPA.half.CMPA = 200;       // Compare A = 200 TBCLK counts
EPwm1Regs.CMPB = 400;                 // Compare B = 400 TBCLK counts
EPwm1Regs.TBPHS = 0;                  // Set Phase register to zero
EPwm1Regs.TBCTR = 0;                  // clear TB counter
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; // TBCLK = VCLK4
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on TBCTR = Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on TBCTR = Zero
EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;
EPwm1Regs.AQCTLA.bit.CBU = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.ZRO = AQ_TOGGLE;
//
// Run Time
// =====
EPwm1Regs.CMPA.half.CMPA = EdgePosA; // adjust duty for output EPWM1A only
EPwm1Regs.CMPB = EdgePosB;
```

Figure 19-24. Up-Down-Count, Dual Edge Symmetric Waveform, With Independent Modulation on EPWMxA and EPWMxB — Active Low



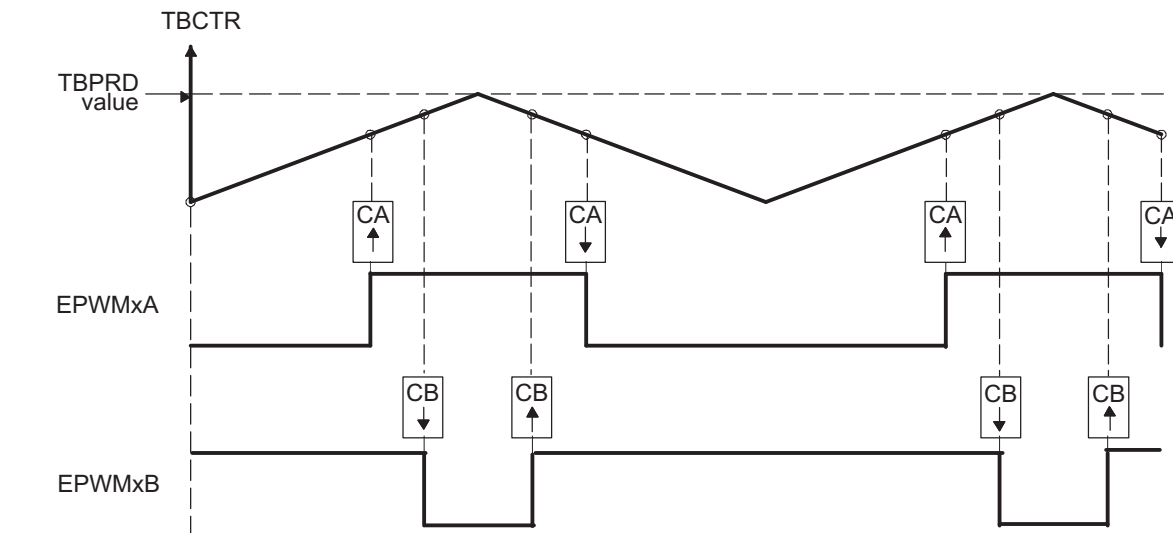
- A PWM period = $2 \times \text{TBPRD} \times T_{\text{TBCLK}}$
- B Duty modulation for EPWMxA is set by CMPA, and is active low (that is, the low time duty is proportional to CMPA).
- C Duty modulation for EPWMxB is set by CMPB and is active low (that is, the low time duty is proportional to CMPB).
- D Outputs EPWMxA and EPWMxB can drive independent power switches

Example 19-4 contains a code sample showing initialization and run time for the waveforms in Figure 19-24.

Example 19-4. Code Sample for Figure 19-24

```
// Initialization Time
// =====
EPwm1Regs.TBPRD = 600; // Period = 2*600 TBCLK counts
EPwm1Regs.CMPA.half.CMPA = 400; // Compare A = 400 TBCLK counts
EPwm1Regs.CMPB = 500; // Compare B = 500 TBCLK counts
EPwm1Regs.TBPHS = 0; // Set Phase register to zero
EPwm1Regs.TBCTR = 0; // clear TB counter
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetric
xEPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
xEPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; // TBCLK = VCLK4
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;
EPwm1Regs.AQCTLA.bit.CAD = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.CBU = AQ_SET;
EPwm1Regs.AQCTLB.bit.CBD = AQ_CLEAR;
//
// Run Time
// =====
EPwm1Regs.CMPA.half.CMPA = Duty1A; // adjust duty for output EPWM1A
EPwm1Regs.CMPB = Duty1B; // adjust duty for output EPWM1B
```

Figure 19-25. Up-Down-Count, Dual Edge Symmetric Waveform, With Independent Modulation on EPWMxA and EPWMxB — Complementary



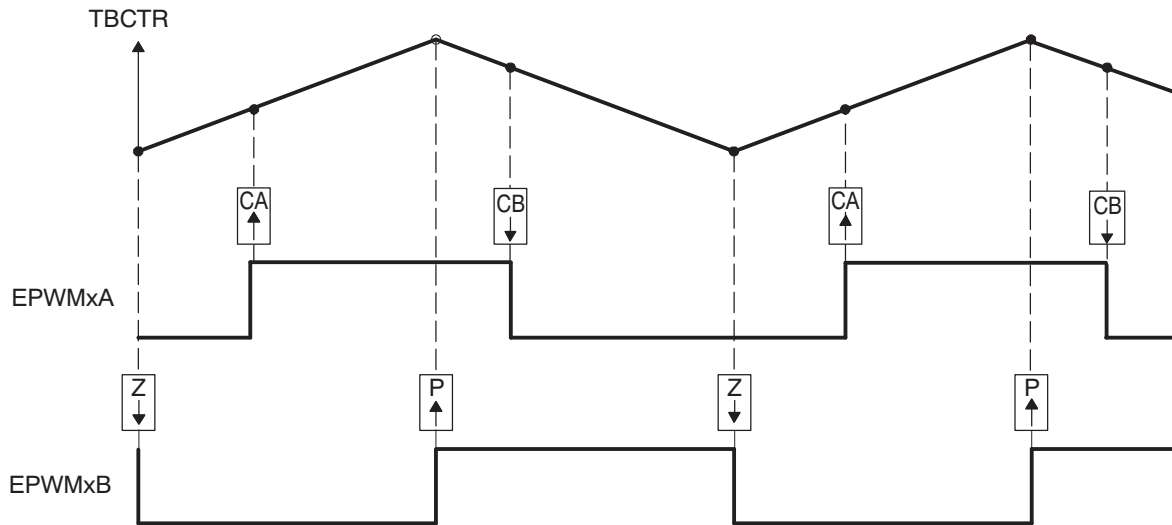
- A PWM period = $2 \times \text{TBPRD} \times T_{\text{TBCLK}}$
- B Duty modulation for EPWMxA is set by CMPA, and is active low, that is, low time duty proportional to CMPA
- C Duty modulation for EPWMxB is set by CMPB and is active high, that is, high time duty proportional to CMPB
- D Outputs EPWMx can drive upper/lower (complementary) power switches
- E Dead-band = CMPB - CMPA (fully programmable edge placement by software). Note the dead-band module is also available if the more classical edge delay method is required.

Example 19-5 contains a code sample showing initialization and run time for the waveforms in Figure 19-25.

Example 19-5. Code Sample for Figure 19-25

```
// Initialization Time
// =====
EPwm1Regs.TBPRD = 600;                // Period = 2*600 TBCLK counts
EPwm1Regs.CMPA.half.CMPA = 350;       // Compare A = 350 TBCLK counts
EPwm1Regs.CMPB = 400;                 // Compare B = 400 TBCLK counts
EPwm1Regs.TBPHS = 0;                  // Set Phase register to zero
EPwm1Regs.TBCTR = 0;                  // clear TB counter
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetric
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; // TBCLK = VCLK4
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;
EPwm1Regs.AQCTLA.bit.CAD = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.CBU = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.CBD = AQ_SET;
// Run Time
// =====
EPwm1Regs.CMPA.half.CMPA = Duty1A;    // adjust duty for output EPWM1A
EPwm1Regs.CMPB = Duty1B;              // adjust duty for output EPWM1B
```

Figure 19-26. Up-Down-Count, Dual Edge Asymmetric Waveform, With Independent Modulation on EPWMxA—Active Low



- A PWM period = $2 \times \text{TBPRD} \times \text{TBCLK}$
- B Rising edge and falling edge can be asymmetrically positioned within a PWM cycle. This allows for pulse placement techniques.
- C Duty modulation for EPWMxA is set by CMPA and CMPB.
- D Low time duty for EPWMxA is proportional to (CMPA + CMPB).
- E To change this example to active high, CMPA and CMPB actions need to be inverted (that is, Set ! Clear and Clear Set).
- F Duty modulation for EPWMxB is fixed at 50% (utilizes spare action resources for EPWMxB)

[Example 19-6](#) contains a code sample showing initialization and run time for the waveforms in [Figure 19-26](#).

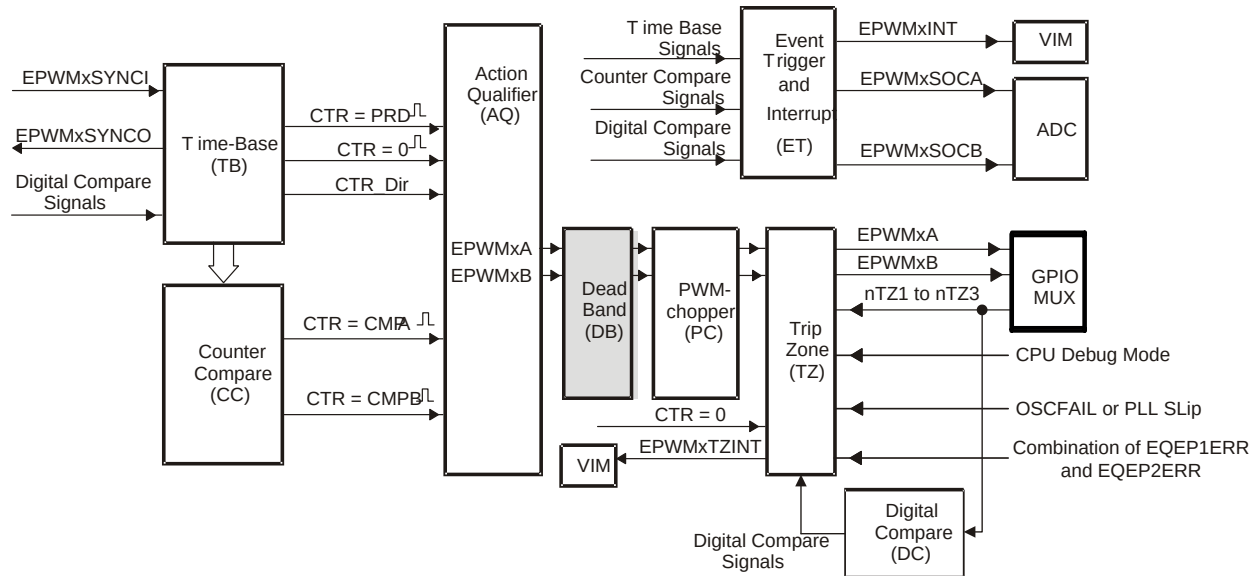
Example 19-6. Code Sample for [Figure 19-26](#)

```
// Initialization Time
// =====
EPwm1Regs.TBPRD = 600;                // Period = 2 * 600 TBCLK counts
EPwm1Regs.CMPA.half.CMPA = 250;      // Compare A = 250 TBCLK counts
EPwm1Regs.CMPB = 450;                // Compare B = 450 TBCLK counts
EPwm1Regs.TBPHS = 0;                 // Set Phase register to zero
EPwm1Regs.TBCTR = 0;                 // clear TB counter
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetric
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; // TBCLK = VCLK4
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR = Zero
EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;
EPwm1Regs.AQCTLA.bit.CBD = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.ZRO = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.PRDL = AQ_SET;
// Run Time
// =====
EPwm1Regs.CMPA.half.CMPA = EdgePosA; // adjust duty for output EPWM1A only
EPwm1Regs.CMPB = EdgePosB;
```

19.2.5 Dead-Band Generator (DB) Submodule

Figure 19-27 illustrates the dead-band submodule within the ePWM module.

Figure 19-27. Dead_Band Submodule



19.2.5.1 Purpose of the Dead-Band Submodule

Section 19.2.4 discussed how it is possible to generate the required dead-band by having full control over edge placement using both the CMPA and CMPB resources of the ePWM module. However, if the more classical edge delay-based dead-band with polarity control is required, then the dead-band submodule described here should be used.

The key functions of the dead-band module are:

- Generating appropriate signal pairs (EPWMxA and EPWMxB) with dead-band relationship from a single EPWMxA input
- Programming signal pairs for:
 - Active high (AH)
 - Active low (AL)
 - Active high complementary (AHC)
 - Active low complementary (ALC)
- Adding programmable delay to rising edges (RED)
- Adding programmable delay to falling edges (FED)
- Can be totally bypassed from the signal path (note dotted lines in diagram)

19.2.5.2 Controlling and Monitoring the Dead-Band Submodule

The dead-band submodule operation is controlled and monitored via the following registers:

Table 19-13. Dead-Band Generator Submodule Registers

Register Name	Address Offset	Shadowed	Description
DBCTL	0x001E	No	Dead-Band Control Register
DBRED	0x0020	No	Dead-Band Rising Edge Delay Count Register
DBFED	0x0022	No	Dead-Band Falling Edge Delay Count Register

19.2.5.3 Operational Highlights for the Dead-Band Submodule

The following sections provide the operational highlights.

The dead-band submodule has two groups of independent selection options as shown in [Figure 19-28](#).

- **Input Source Selection:**

The input signals to the dead-band module are the EPWMxA and EPWMxB output signals from the action-qualifier. In this section they will be referred to as EPWMxA In and EPWMxB In. Using the DBCTL[IN_MODE] control bits, the signal source for each delay, falling-edge or rising-edge, can be selected:

- EPWMxA In is the source for both falling-edge and rising-edge delay. This is the default mode.
- EPWMxA In is the source for falling-edge delay, EPWMxB In is the source for rising-edge delay.
- EPWMxA In is the source for rising edge delay, EPWMxB In is the source for falling-edge delay.
- EPWMxB In is the source for both falling-edge and rising-edge delay.

- **Half Cycle Clocking:**

The dead-band submodule can be clocked using half cycle clocking to double the resolution (that is, counter clocked at $2 \times \text{TBCLK}$)

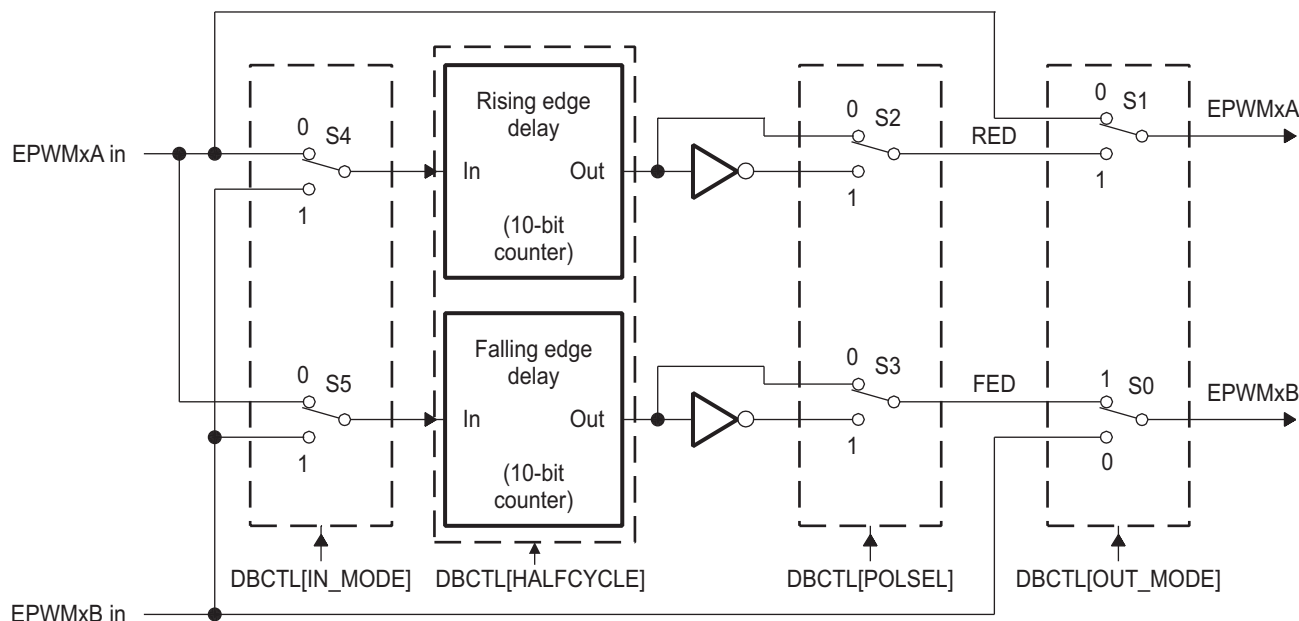
- **Output Mode Control:**

The output mode is configured by way of the DBCTL[OUT_MODE] bits. These bits determine if the falling-edge delay, rising-edge delay, neither, or both are applied to the input signals.

- **Polarity Control:**

The polarity control (DBCTL[POLSEL]) allows you to specify whether the rising-edge delayed signal and/or the falling-edge delayed signal is to be inverted before being sent out of the dead-band submodule.

Figure 19-28. Configuration Options for the Dead-Band Submodule



Although all combinations are supported, not all are typical usage modes. [Table 19-14](#) documents some classical dead-band configurations. These modes assume that the DBCTL[IN_MODE] is configured such that EPWMxA In is the source for both falling-edge and rising-edge delay. Enhanced, or non-traditional modes can be achieved by changing the input signal source. The modes shown in [Table 19-14](#) fall into the following categories:

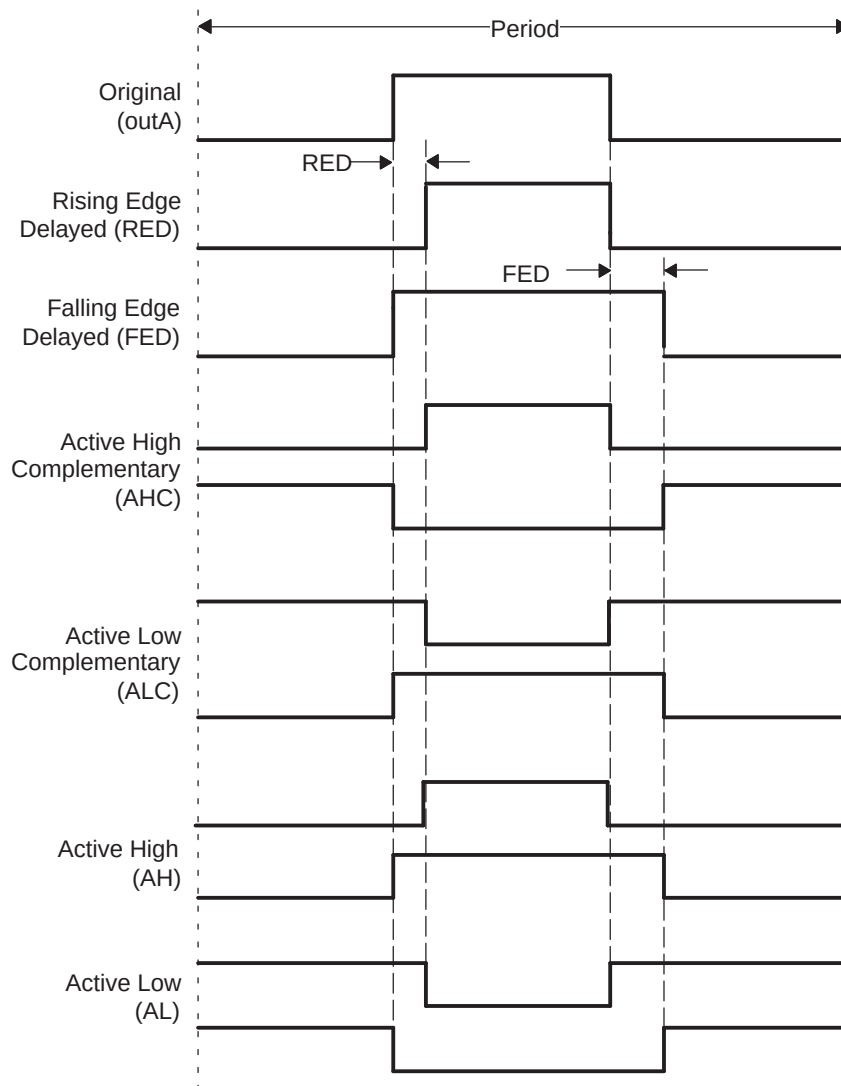
- **Mode 1: Bypass both falling-edge delay (FED) and rising-edge delay (RED)**
Allows you to fully disable the dead-band submodule from the PWM signal path.
- **Mode 2-5: Classical Dead-Band Polarity Settings:**
These represent typical polarity configurations that should address all the active high/low modes required by available industry power switch gate drivers. The waveforms for these typical cases are shown in [Figure 19-29](#). Note that to generate equivalent waveforms to [Figure 19-29](#), configure the action-qualifier submodule to generate the signal as shown for EPWMxA.
- **Mode 6: Bypass rising-edge-delay and Mode 7: Bypass falling-edge-delay**
Finally the last two entries in [Table 19-14](#) show combinations where either the falling-edge-delay (FED) or rising-edge-delay (RED) blocks are bypassed.

Table 19-14. Classical Dead-Band Operating Modes

Mode	Mode Description	DBCTL[POLSEL]		DBCTL[OUT_MODE]	
		S3	S2	S1	S0
1	EPWMxA and EPWMxB Passed Through (No Delay)	X	X	0	0
2	Active High Complementary (AHC)	1	0	1	1
3	Active Low Complementary (ALC)	0	1	1	1
4	Active High (AH)	0	0	1	1
5	Active Low (AL)	1	1	1	1
6	EPWMxA Out = EPWMxA In (No Delay) EPWMxB Out = EPWMxA In with Falling Edge Delay	0 or 1	0 or 1	0	1
7	EPWMxA Out = EPWMxA In with Rising Edge Delay EPWMxB Out = EPWMxB In with No Delay	0 or 1	0 or 1	1	0

Figure 19-29 shows waveforms for typical cases where $0\% < \text{duty} < 100\%$.

Figure 19-29. Dead-Band Waveforms for Typical Cases ($0\% < \text{Duty} < 100\%$)



The dead-band submodule supports independent values for rising-edge (RED) and falling-edge (FED) delays. The amount of delay is programmed using the DBRED and DBFED registers. These are 10-bit registers and their value represents the number of time-base clock, TBCLK, periods a signal edge is delayed by. For example, the formula to calculate falling-edge-delay and rising-edge-delay are:

$$\text{FED} = \text{DBFED} \times T_{\text{TBCLK}}$$

$$\text{RED} = \text{DBRED} \times T_{\text{TBCLK}}$$

Where T_{TBCLK} is the period of TBCLK, the prescaled version of VCLK4.

For convenience, delay values for various TBCLK options are shown in [Table 19-15](#).

Table 19-15. Dead-Band Delay Values in μS as a Function of DBFED and DBRED

Dead-Band Value	Dead-Band Delay in μS		
DBFED, DBRED	TBCLK = VCLK4/1	TBCLK = VCLK4 /2	TBCLK = VCLK4/4
1	0.02 μS	0.03 μS	0.07 μS
5	0.08 μS	0.17 μS	0.33 μS
10	0.17 μS	0.33 μS	0.67 μS
100	1.67 μS	3.33 μS	6.67 μS
200	3.33 μS	6.67 μS	13.33 μS
400	6.67 μS	13.33 μS	26.67 μS
500	8.33 μS	16.67 μS	33.33 μS
600	10.00 μS	20.00 μS	40.00 μS
700	11.67 μS	23.33 μS	46.67 μS
800	13.33 μS	26.67 μS	53.33 μS
900	15.00 μS	30.00 μS	60.00 μS
1000	16.67 μS	33.33 μS	66.67 μS

When half-cycle clocking is enabled, the formula to calculate the falling-edge-delay and rising-edge-delay becomes:

$$\text{FED} = \text{DBFED} \times T_{\text{TBCLK}}/2$$

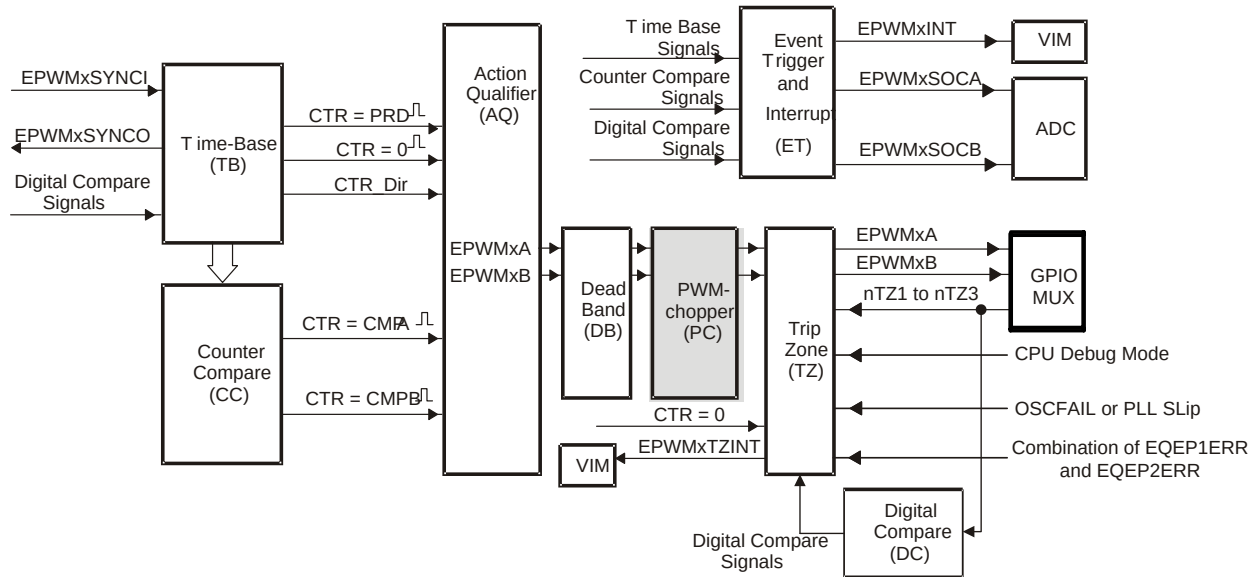
$$\text{RED} = \text{DBRED} \times T_{\text{TBCLK}}/2$$

19.2.6 PWM-Chopper (PC) Submodule

Figure 19-30 illustrates the PWM-chopper (PC) submodule within the ePWM module.

The PWM-chopper submodule allows a high-frequency carrier signal to modulate the PWM waveform generated by the action-qualifier and dead-band submodules. This capability is important if you need pulse transformer-based gate drivers to control the power switching elements.

Figure 19-30. PWM-Chopper Submodule



19.2.6.1 Purpose of the PWM-Chopper Submodule

The key functions of the PWM-chopper submodule are:

- Programmable chopping (carrier) frequency
- Programmable pulse width of first pulse
- Programmable duty cycle of second and subsequent pulses
- Can be fully bypassed if not required

19.2.6.2 Controlling the PWM-Chopper Submodule

The PWM-chopper submodule operation is controlled via the registers in Table 19-16.

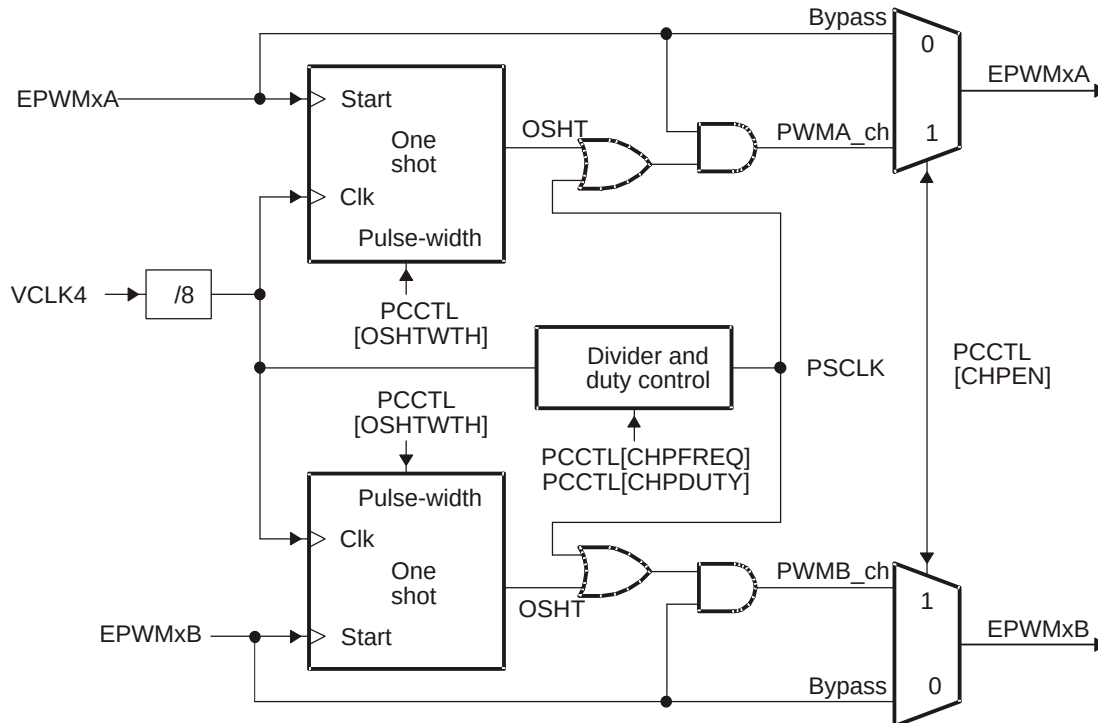
Table 19-16. PWM-Chopper Submodule Registers

Register Name	Address Offset	Shadowed	Description
PCCTL	0x003C	No	PWM-chopper Control Register

19.2.6.3 Operational Highlights for the PWM-Chopper Submodule

Figure 19-31 shows the operational details of the PWM-chopper submodule. The carrier clock is derived from VCLK4. Its frequency and duty cycle are controlled via the CHPFREQ and CHPDUTY bits in the PCCTL register. The one-shot block is a feature that provides a high energy first pulse to ensure hard and fast power switch turn on, while the subsequent pulses sustain pulses, ensuring the power switch remains on. The one-shot width is programmed via the OSHTWTH bits. The PWM-chopper submodule can be fully disabled (bypassed) via the CHPEN bit.

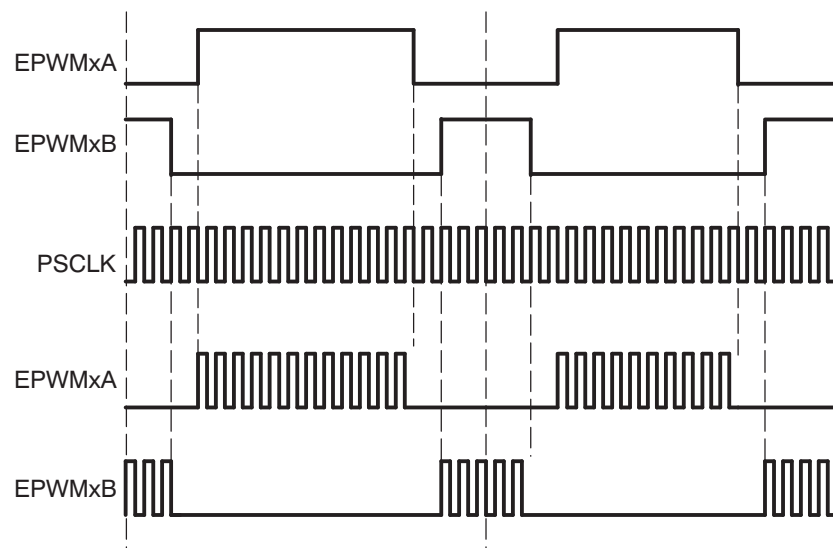
Figure 19-31. PWM-Chopper Submodule Operational Details



19.2.6.4 Waveforms

Figure 19-32 shows simplified waveforms of the chopping action only; one-shot and duty-cycle control are not shown. Details of the one-shot and duty-cycle control are discussed in the following sections.

Figure 19-32. Simple PWM-Chopper Submodule Waveforms Showing Chopping Action Only



19.2.6.4.1 One-Shot Pulse

The width of the first pulse can be programmed to any of 16 possible pulse width values. The width or period of the first pulse is given by:

$$T_{1stpulse} = T_{VCLK4} \times 8 \times OSHTWTH$$

Where T_{VCLK4} is the period of the system clock (VCLK4) and OSHTWTH is the four control bits (value from 1 to 16)

Figure 19-33 shows the first and subsequent sustaining pulses and Table 19-17 gives the possible pulse width values for a VCLK4 = 100 MHz.

Figure 19-33. PWM-Chopper Submodule Waveforms Showing the First Pulse and Subsequent Sustaining Pulses

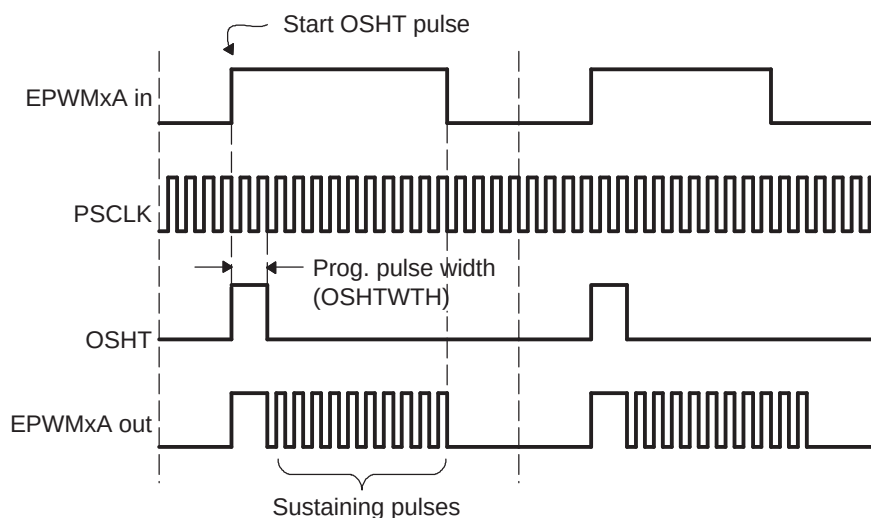


Table 19-17. Possible Pulse Width Values for VCLK4 = 100 MHz

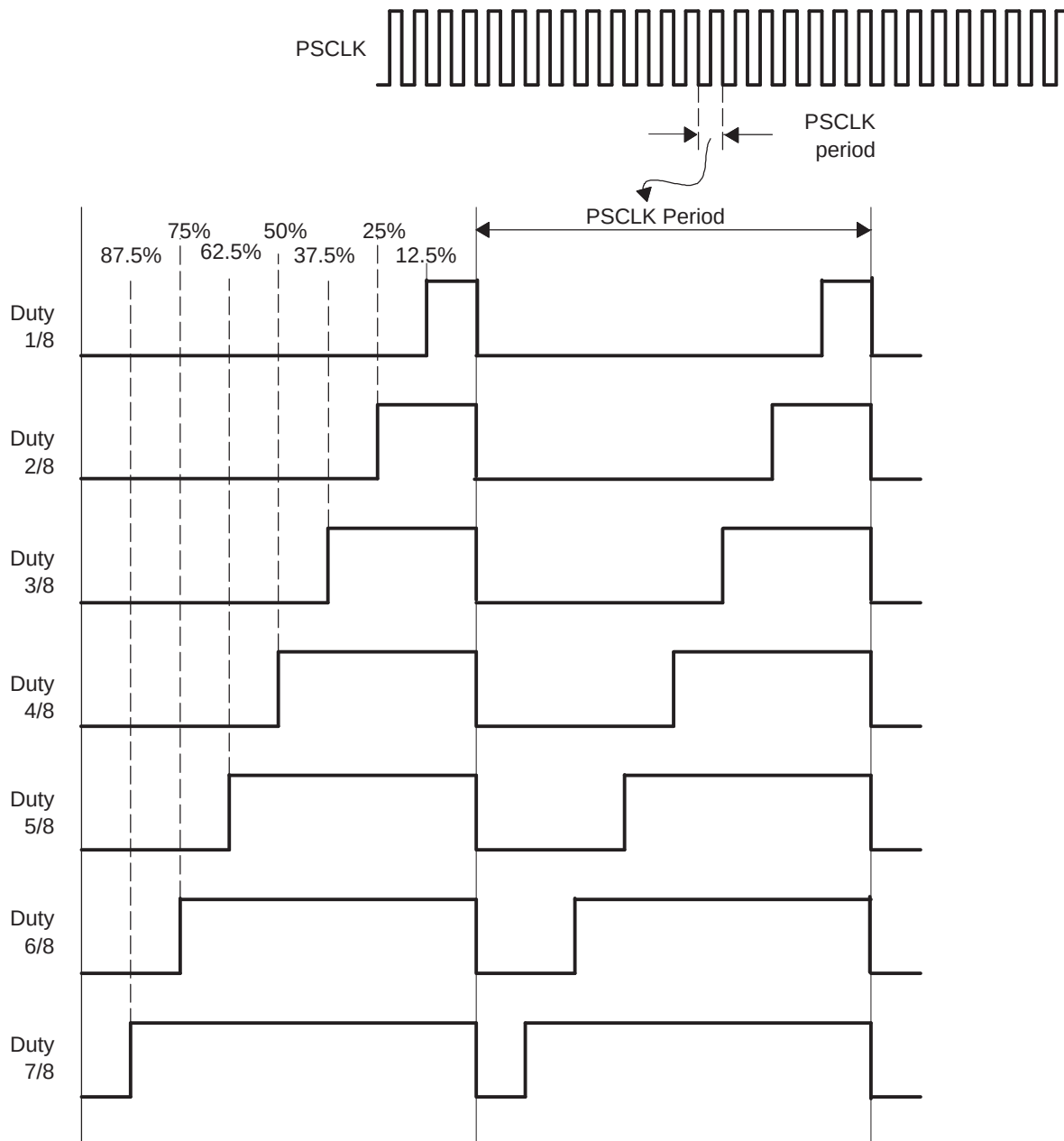
OSHTWTHz (hex)	Pulse Width (nS)
0	100
1	200
2	300
3	400
4	500
5	600
6	700
7	800
8	900
9	1000
A	1100
B	1200
C	1300
D	1400
E	1500
F	1600

19.2.6.4.2 Duty Cycle Control

Pulse transformer-based gate drive designs need to comprehend the magnetic properties or characteristics of the transformer and associated circuitry. Saturation is one such consideration. To assist the gate drive designer, the duty cycles of the second and subsequent pulses have been made programmable. These sustaining pulses ensure the correct drive strength and polarity is maintained on the power switch gate during the on period, and hence a programmable duty cycle allows a design to be tuned or optimized via software control.

Figure 19-34 shows the duty cycle control that is possible by programming the CHPDUTY bits. One of seven possible duty ratios can be selected ranging from 12.5% to 87.5%.

Figure 19-34. PWM-Chopper Submodule Waveforms Showing the Pulse Width (Duty Cycle) Control of Sustaining Pulses

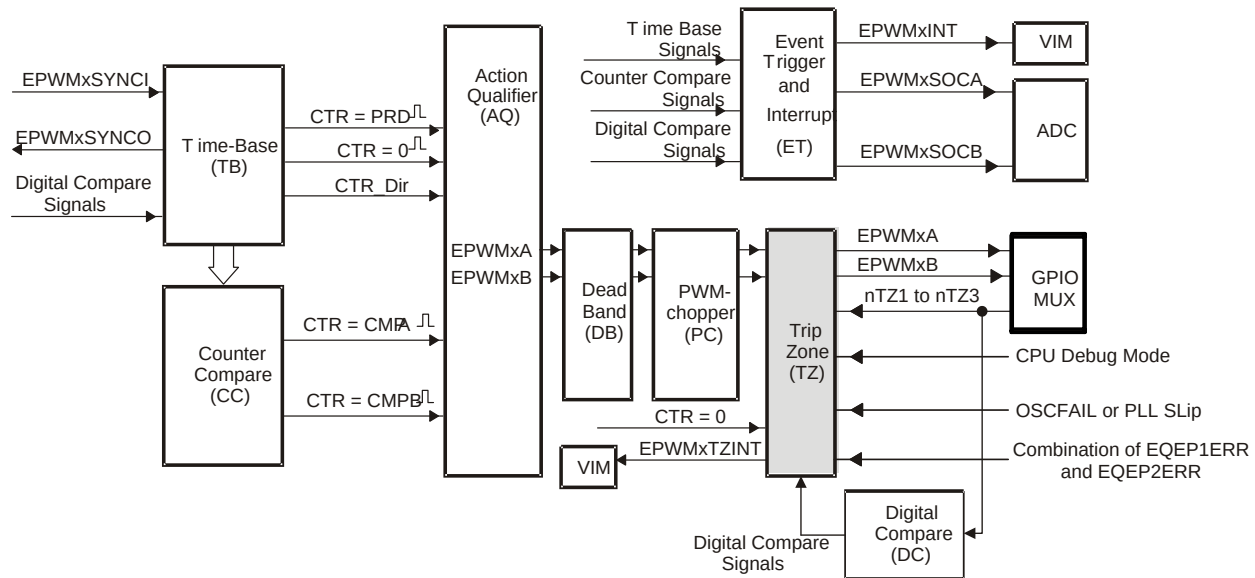


19.2.7 Trip-Zone (TZ) Submodule

Figure 19-35 shows how the trip-zone (TZ) submodule fits within the ePWM module.

Each ePWM module is connected to six $\overline{TZn}$ signals ($\overline{TZ1}$ to $\overline{TZ6}$). $\overline{TZ1}$ to $\overline{TZ3}$ are sourced from the GPIO mux. $\overline{TZ4}$ is sourced from a combination of EQEP1ERR and EQEP2ERR signals. $\overline{TZ5}$ is connected to the system oscillator or PLL clock fail logic, and $\overline{TZ6}$ is sourced from the debug mode halt indication output from the CPU. These signals indicate fault or trip conditions, and the ePWM outputs can be programmed to respond accordingly when faults occur.

Figure 19-35. Trip-Zone Submodule



19.2.7.1 Purpose of the Trip-Zone Submodule

The key functions of the Trip-Zone submodule are:

- Trip inputs $\overline{TZ1}$ to $\overline{TZ6}$ are mapped to all ePWM modules.
- Upon a fault indication, either no action is taken or the ePWM outputs $EPWMxA$ and $EPWMxB$ can be forced to one of the following:
 - High
 - Low
 - High-impedance
- Support for one-shot trip (OSHT) for major short circuits or over-current conditions.
- Support for cycle-by-cycle tripping (CBC) for current limiting operation.
- Support for digital compare tripping (DC) based on state of on-chip analog comparator module outputs and/or $\overline{TZ1}$ to $\overline{TZ3}$ signals.
- Each trip-zone input and digital compare (DC) submodule DCAEVT1/2 or DCBEVT1/2 force event can be allocated to either one-shot or cycle-by-cycle operation.
- Interrupt generation is possible on any trip-zone input.
- Software-forced tripping is also supported.
- The trip-zone submodule can be fully bypassed if it is not required.

19.2.7.2 Controlling and Monitoring the Trip-Zone Submodule

The trip-zone submodule operation is controlled and monitored through the following registers:

Table 19-18. Trip-Zone Submodule Registers

Register Name	Address Offset	Shadowed	Description ⁽¹⁾
TZSEL	0x0024	No	Trip-Zone Select Register
TZDCSEL	0x0026	No	Trip-zone Digital Compare Select Register ⁽²⁾
TZCTL	0x0028	No	Trip-Zone Control Register
TZEINT	0x002A	No	Trip-Zone Enable Interrupt Register
TZFLG	0x002C	No	Trip-Zone Flag Register
TZCLR	0x002E	No	Trip-Zone Clear Register
TZFRC	0x0030	No	Trip-Zone Force Register

⁽¹⁾ All trip-zone registers are writable only in privileged mode.

⁽²⁾ This register is discussed in more detail in [Section 19.2.9](#).

19.2.7.3 Operational Highlights for the Trip-Zone Submodule

The following sections describe the operational highlights and configuration options for the trip-zone submodule.

The trip-zone signals $\overline{TZ1}$ to $\overline{TZ6}$ (also collectively referred to as $\overline{TZn}$) are active low input signals. When one of these signals goes low, or when a DCAEVT1/2 or DCBEVT1/2 force happens based on the TZDCSEL register event selection, it indicates that a trip event has occurred. Each ePWM module can be individually configured to ignore or use each of the trip-zone signals or DC events. Which trip-zone signals or DC events are used by a particular ePWM module is determined by the TZSEL register for that specific ePWM module. The trip-zone signals may or may not be synchronized to the system clock (VCLK4) and digitally filtered within the GPIO MUX block. A minimum of 3*TBCLK low pulse width on $\overline{TZn}$ inputs is sufficient to trigger a fault condition on the ePWM module. If the pulse width is less than this, the trip condition may not be latched. The asynchronous trip makes sure that if clocks are missing for any reason, the outputs can still be tripped by a valid event present on $\overline{TZn}$ inputs. The GPIOs or peripherals must be appropriately configured. For more information, see the IOMM chapter of the device technical reference manual.

Each $\overline{TZn}$ input can be individually configured to provide either a cycle-by-cycle or one-shot trip event for an ePWM module. DCAEVT1 and DCBEVT1 events can be configured to directly trip an ePWM module or provide a one-shot trip event to the module. Likewise, DCAEVT2 and DCBEVT2 events can also be configured to directly trip an ePWM module or provide a cycle-by-cycle trip event to the module. This configuration is determined by the TZSEL[DCAEVT1/2], TZSEL[DCBEVT1/2], TZSEL[CBCn], and TZSEL[OSHTn] control bits (where n corresponds to the trip input) respectively.

- **Cycle-by-Cycle (CBC):**

When a cycle-by-cycle trip event occurs, the action specified in the TZCTL[TZA] and TZCTL[TZB] bits is carried out immediately on the EPWMxA and/or EPWMxB output. [Table 19-19](#) lists the possible actions. In addition, the cycle-by-cycle trip event flag (TZFLG[CBC]) is set and a EPWMx_TZINT interrupt is generated if it is enabled in the TZEINT register and VIM peripheral.

If the CBC interrupt is enabled via the TZEINT register, and DCAEVT2 or DCBEVT2 are selected as CBC trip sources via the TZSEL register, it is not necessary to also enable the DCAEVT2 or DCBEVT2 interrupts in the TZEINT register, as the DC events trigger interrupts through the CBC mechanism.

The specified condition on the inputs is automatically cleared when the ePWM time-base counter reaches zero (TBCTR = 0x0000) if the trip event is no longer present. Therefore, in this mode, the trip event is cleared or reset every PWM cycle. The TZFLG[CBC] flag bit will remain set until it is manually cleared by writing to the TZCLR[CBC] bit. If the cycle-by-cycle trip event is still present when the TZFLG[CBC] bit is cleared, then it will again be immediately set.

- **One-Shot (OSHT):**

When a one-shot trip event occurs, the action specified in the TZCTL[TZA] and TZCTL[TZB] bits is carried out immediately on the EPWMxA and/or EPWMxB output. [Table 19-19](#) lists the possible actions. In addition, the one-shot trip event flag (TZFLG[OST]) is set and a EPWMx_TZINT interrupt is generated if it is enabled in the TZEINT register and VIM peripheral. The one-shot trip condition must be cleared manually by writing to the TZCLR[OST] bit.

If the one-shot interrupt is enabled via the TZEINT register, and DCAEVT1 or DCBEVT1 are selected as OSHT trip sources via the TZSEL register, it is not necessary to also enable the DCAEVT1 or DCBEVT1 interrupts in the TZEINT register, as the DC events trigger interrupts through the OSHT mechanism.

- **Digital Compare Events (DCAEVT1/2 and DCBEVT1/2):**

A digital compare DCAEVT1/2 or DCBEVT1/2 event is generated based on a combination of the DCAH/DCAL and DCBH/DCBL signals as selected by the TZDCSEL register. The signals which source the DCAH/DCAL and DCBH/DCBL signals are selected via the DCTRIPSEL register and can be either trip zone input pins. For more information on the digital compare submodule signals, see [Section 19.2.9](#).

When a digital compare event occurs, the action specified in the TZCTL[DCAEVT1/2] and TZCTL[DCBEVT1/2] bits is carried out immediately on the EPWMxA and/or EPWMxB output. [Table 19-19](#) lists the possible actions. In addition, the relevant DC trip event flag (TZFLG[DCAEVT1/2] / TZFLG[DCBEVT1/2]) is set and a EPWMx_TZINT interrupt is generated if it is enabled in the TZEINT register and VIM peripheral.

The specified condition on the pins is automatically cleared when the DC trip event is no longer present. The TZFLG[DCAEVT1/2] or TZFLG[DCBEVT1/2] flag bit will remain set until it is manually cleared by writing to the TZCLR[DCAEVT1/2] or TZCLR[DCBEVT1/2] bit. If the DC trip event is still present when the TZFLG[DCAEVT1/2] or TZFLG[DCBEVT1/2] flag is cleared, then it will again be immediately set.

The action taken when a trip event occurs can be configured individually for each of the ePWM output pins by way of the TZCTL register bit fields. One of four possible actions, shown in [Table 19-19](#), can be taken on a trip event.

Table 19-19. Possible Actions On a Trip Event

TZCTL Register bit-field Settings	EPWMxA and/or EPWMxB	Comment
0,0	High-Impedance	Tripped
0,1	Force to High State	Tripped
1,0	Force to Low State	Tripped
1,1	No Change	Do Nothing. No change is made to the output.

Example 19-7. Trip-Zone Configurations

Scenario A:

A one-shot trip event on $\overline{TZ1}$ pulls both EPWM1A, EPWM1B low and also forces EPWM2A and EPWM2B high.

- Configure the ePWM1 registers as follows:
 - TZSEL[OSHT1] = 1: enables $\overline{TZ1}$ as a one-shot event source for ePWM1
 - TZCTL[TZA] = 2: EPWM1A will be forced low on a trip event.
 - TZCTL[TZB] = 2: EPWM1B will be forced low on a trip event.
- Configure the ePWM2 registers as follows:
 - TZSEL[OSHT1] = 1: enables $\overline{TZ1}$ as a one-shot event source for ePWM2
 - TZCTL[TZA] = 1: EPWM2A will be forced high on a trip event.
 - TZCTL[TZB] = 1: EPWM2B will be forced high on a trip event.

Scenario B:

A cycle-by-cycle event on $\overline{TZ5}$ pulls both EPWM1A, EPWM1B low.

A one-shot event on $\overline{TZ1}$ or $\overline{TZ6}$ puts EPWM2A into a high impedance state.

- Configure the ePWM1 registers as follows:
 - TZSEL[CBC5] = 1: enables $\overline{TZ5}$ as a one-shot event source for ePWM1
 - TZCTL[TZA] = 2: EPWM1A will be forced low on a trip event.
 - TZCTL[TZB] = 2: EPWM1B will be forced low on a trip event.
- Configure the ePWM2 registers as follows:
 - TZSEL[OSHT1] = 1: enables $\overline{TZ1}$ as a one-shot event source for ePWM2
 - TZSEL[OSHT6] = 1: enables $\overline{TZ6}$ as a one-shot event source for ePWM2
 - TZCTL[TZA] = 0: EPWM2A will be put into a high-impedance state on a trip event.
 - TZCTL[TZB] = 3: EPWM2B will ignore the trip event.

19.2.7.4 Generating Trip Event Interrupts

Figure 19-36 and Figure 19-37 illustrate the trip-zone submodule control and interrupt logic, respectively. DCAEVT1/2 and DCBEVT1/2 signals are described in further detail in Section 19.2.9.

Figure 19-36. Trip-Zone Submodule Mode Control Logic

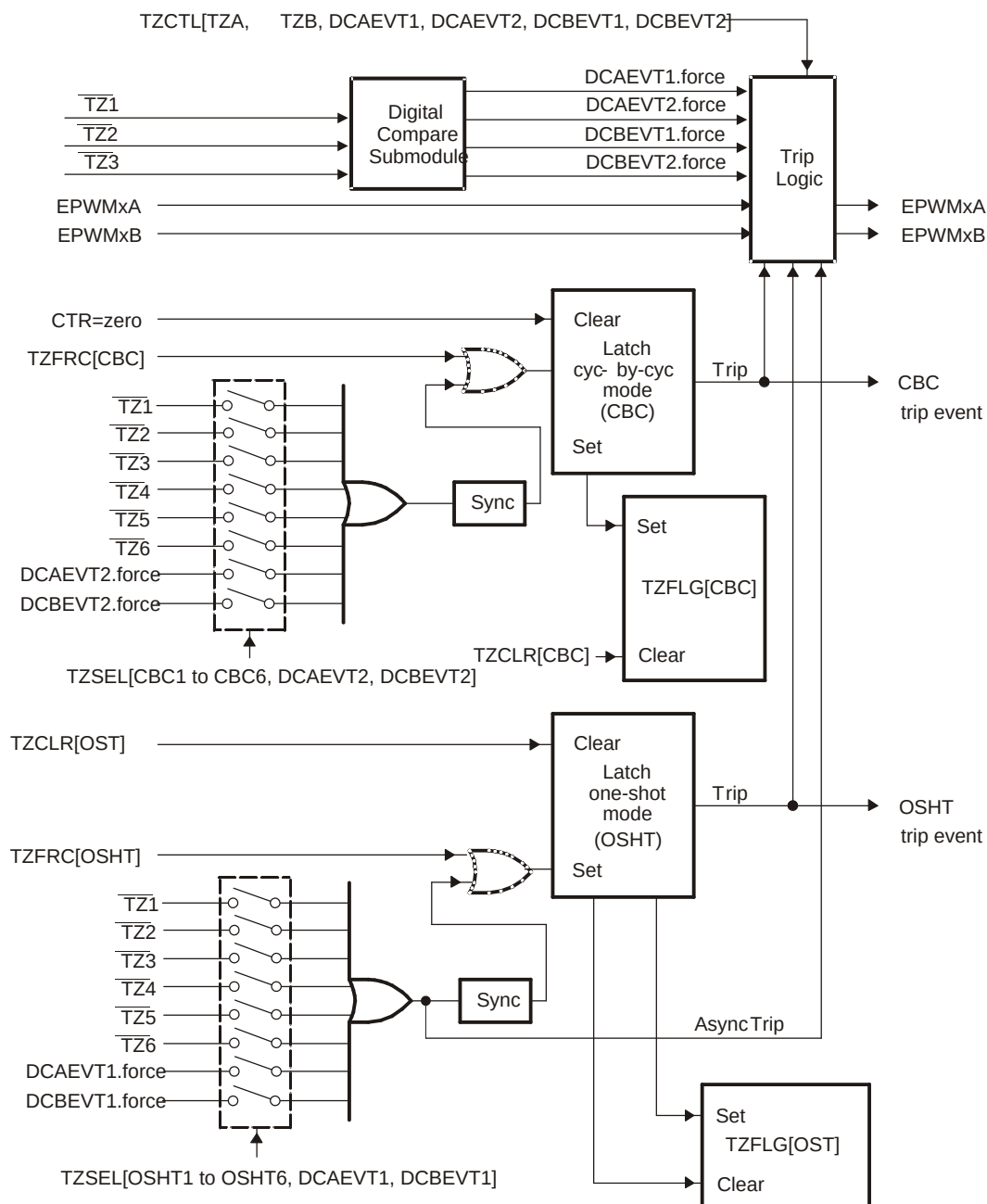
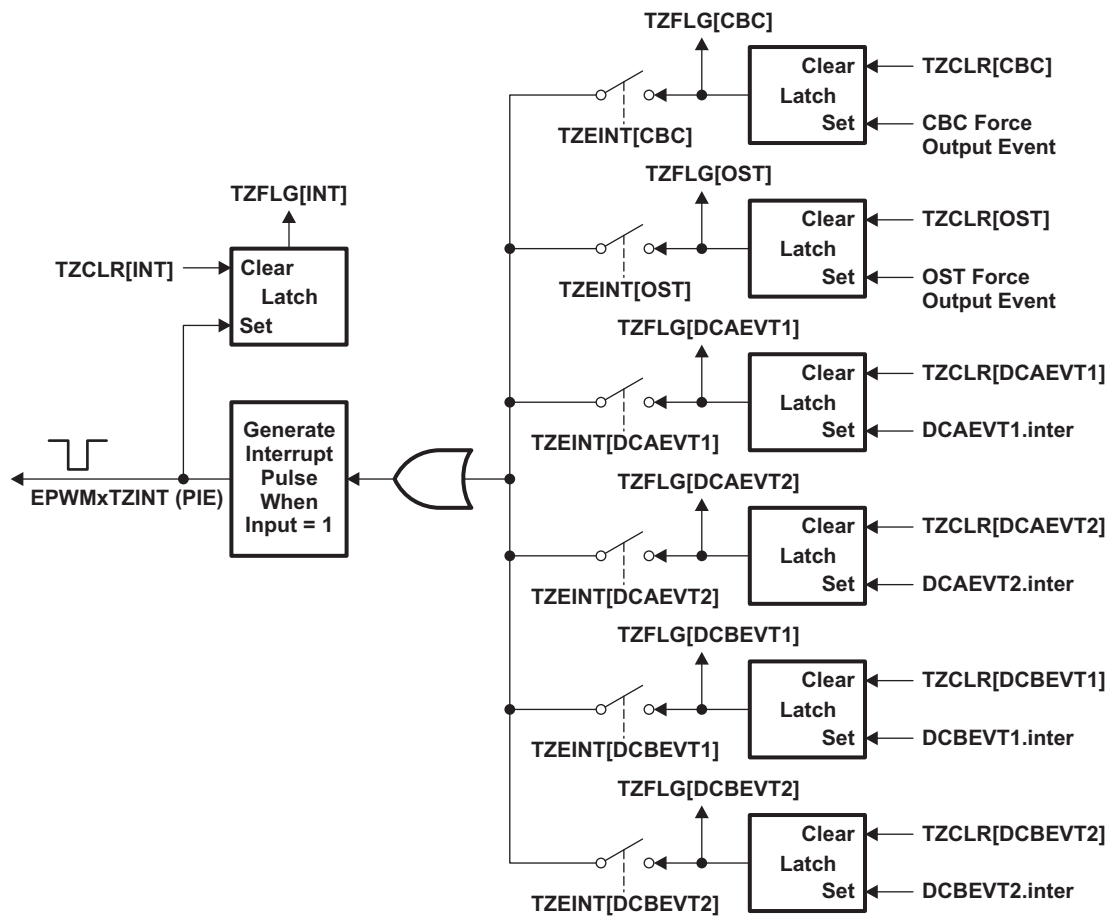


Figure 19-37. Trip-Zone Submodule Interrupt Logic



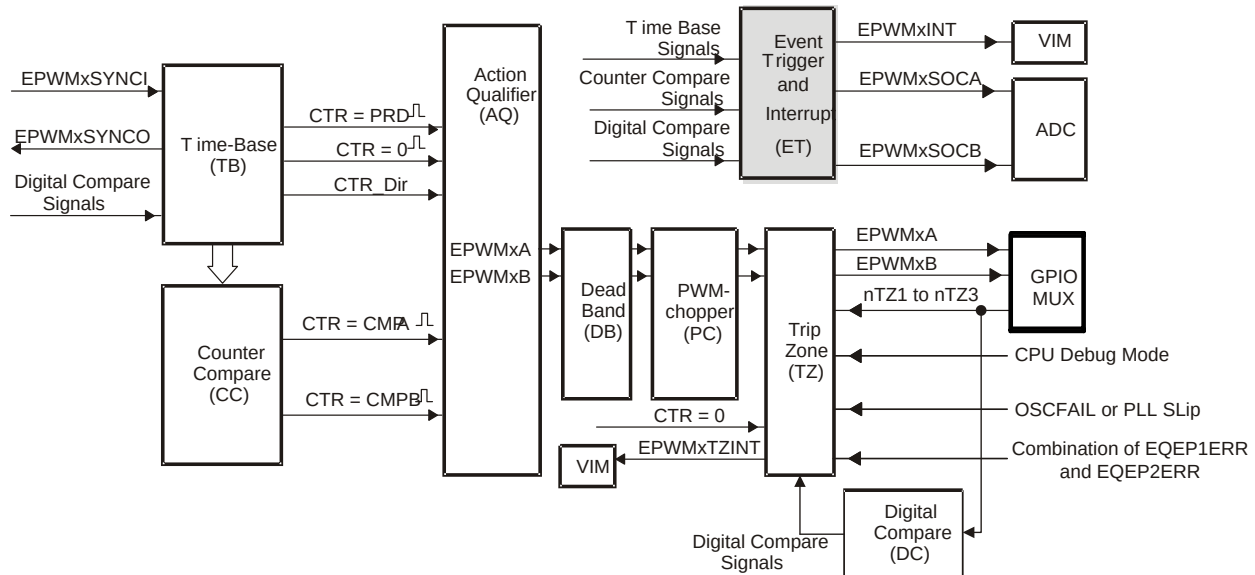
19.2.8 Event-Trigger (ET) Submodule

The key functions of the event-trigger submodule are:

- Receives event inputs generated by the time-base, counter-compare and digital-compare submodules
- Uses the time-base direction information for up/down event qualification
- Uses prescaling logic to issue interrupt requests and ADC start of conversion at:
 - Every event
 - Every second event
 - Every third event
- Provides full visibility of event generation via event counters and flags
- Allows software forcing of Interrupts and ADC start of conversion

The event-trigger submodule manages the events generated by the time-base submodule, the counter-compare submodule, and the digital-compare submodule to generate an interrupt to the CPU and/or a start of conversion pulse to the ADC when a selected event occurs. Figure 19-38 illustrates where the event-trigger submodule fits within the ePWM system.

Figure 19-38. Event-Trigger Submodule



19.2.8.1 Operational Overview of the Event-Trigger Submodule

The following sections describe the event-trigger submodule's operational highlights.

Each ePWM module has one interrupt request line connected to the VIM and two start of conversion signals connected to the ADC module. As shown in Figure 19-39, the EPWMxSOCA and EPWMxSOCB signals are combined to generate four special signals that can be used to trigger an ADC start of conversion, and hence multiple modules can initiate an ADC start of conversion via the ADC trigger inputs.

The event-trigger submodule monitors various event conditions (the left side inputs to event-trigger submodule shown in Figure 19-40) and can be configured to prescale these events before issuing an Interrupt request or an ADC start of conversion. The event-trigger prescaling logic can issue Interrupt requests and ADC start of conversion at:

- Every event
- Every second event
- Every third event

Figure 19-39. Event-Trigger Submodule Inter-Connectivity of ADC Start of Conversion

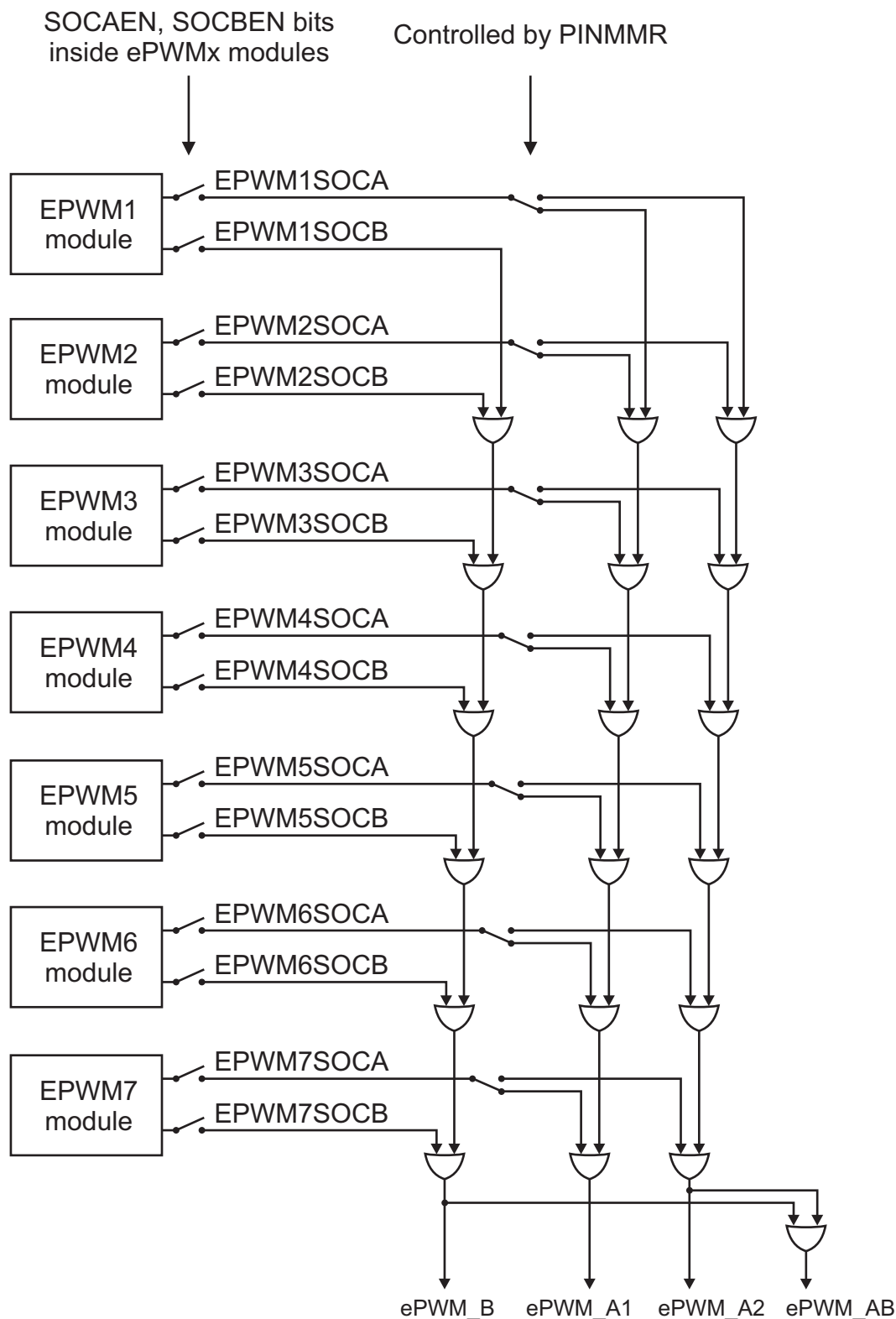
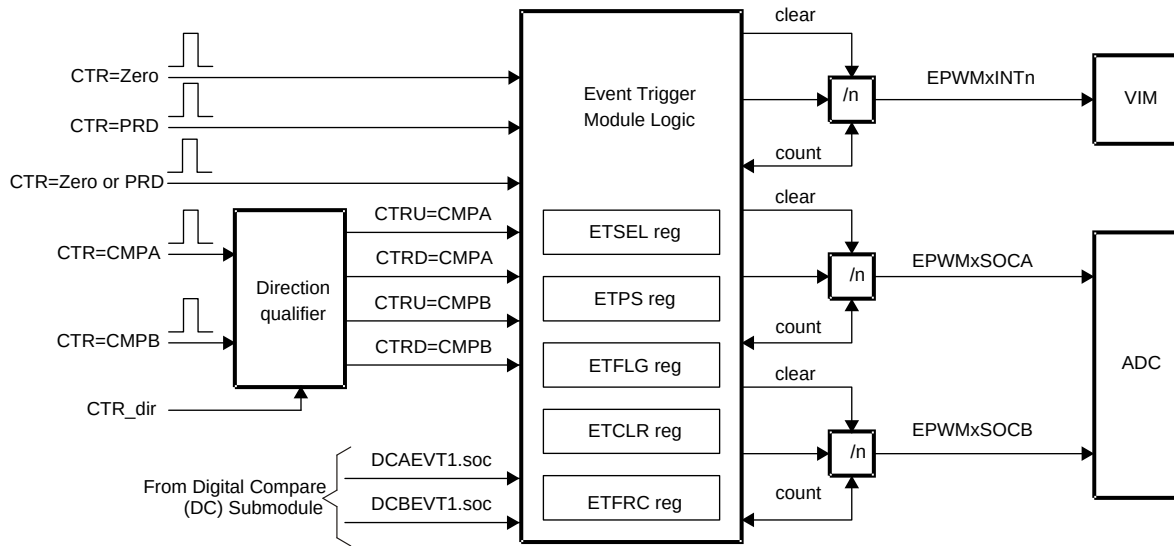


Figure 19-40. Event-Trigger Submodule Showing Event Inputs and Prescaled Outputs



The key registers used to configure the event-trigger submodule are shown in [Table 19-20](#).

Table 19-20. Event-Trigger Submodule Registers

Register Name	Address Offset	Shadowed	Description
ETSEL	0x0032	No	Event-trigger Selection Register
ETPS	0x0034	No	Event-trigger Prescale Register
ETFLG	0x0036	No	Event-trigger Flag Register
ETCLR	0x0038	No	Event-trigger Clear Register
ETFRC	0x003A	No	Event-trigger Force Register

- ETSEL—This selects which of the possible events will trigger an interrupt or start an ADC conversion
- ETPS—This programs the event prescaling options mentioned above.
- ETFLG—These are flag bits indicating status of the selected and prescaled events.
- ETCLR—These bits allow you to clear the flag bits in the ETFLG register via software.
- ETFRC—These bits allow software forcing of an event. Useful for debugging or s/w intervention.

A more detailed look at how the various register bits interact with the Interrupt and ADC start of conversion logic are shown in [Figure 19-41](#), [Figure 19-42](#), and [Figure 19-43](#).

[Figure 19-41](#) shows the event-trigger's interrupt generation logic. The interrupt-period (ETPS[INTPRD]) bits specify the number of events required to cause an interrupt pulse to be generated. The choices available are:

- Do not generate an interrupt.
- Generate an interrupt on every event
- Generate an interrupt on every second event
- Generate an interrupt on every third event

Which event can cause an interrupt is configured by the interrupt selection (ETSEL[INTSEL]) bits. The event can be one of the following:

- Time-base counter equal to zero (TBCTR = 0x0000).
- Time-base counter equal to period (TBCTR = TBPRD).
- Time-base counter equal to zero or period (TBCTR = 0x0000 || TBCTR = TBPRD)

- Time-base counter equal to the compare A register (CMPA) when the timer is incrementing.
- Time-base counter equal to the compare A register (CMPA) when the timer is decrementing.
- Time-base counter equal to the compare B register (CMPB) when the timer is incrementing.
- Time-base counter equal to the compare B register (CMPB) when the timer is decrementing.

The number of events that have occurred can be read from the interrupt event counter (ETPS[INTCNT]) register bits. That is, when the specified event occurs the ETPS[INTCNT] bits are incremented until they reach the value specified by ETPS[INTPRD]. When ETPS[INTCNT] = ETPS[INTPRD] the counter stops counting and its output is set. The counter is only cleared when an interrupt is sent to the VIM.

When ETPS[INTCNT] reaches ETPS[INTPRD] the following behaviors will occur:

- If interrupts are enabled, ETSEL[INTEN] = 1 and the interrupt flag is clear, ETFLG[INT] = 0, then an interrupt pulse is generated and the interrupt flag is set, ETFLG[INT] = 1, and the event counter is cleared ETPS[INTCNT] = 0. The counter will begin counting events again.
- If interrupts are disabled, ETSEL[INTEN] = 0, or the interrupt flag is set, ETFLG[INT] = 1, the counter stops counting events when it reaches the period value ETPS[INTCNT] = ETPS[INTPRD].
- If interrupts are enabled, but the interrupt flag is already set, then the counter will hold its output high until the ENTFLG[INT] flag is cleared. This allows for one interrupt to be pending while one is serviced.

Writing to the INTPRD bits will automatically clear the counter INTCNT = 0 and the counter output will be reset (so no interrupts are generated). Writing a 1 to the ETFRC[INT] bit will increment the event counter INTCNT. The counter will behave as described above when INTCNT = INTPRD. When INTPRD = 0, the counter is disabled and hence no events will be detected and the ETFRC[INT] bit is also ignored.

The above definition means that you can generate an interrupt on every event, on every second event, or on every third event. An interrupt cannot be generated on every fourth or more events.

Figure 19-41. Event-Trigger Interrupt Generator

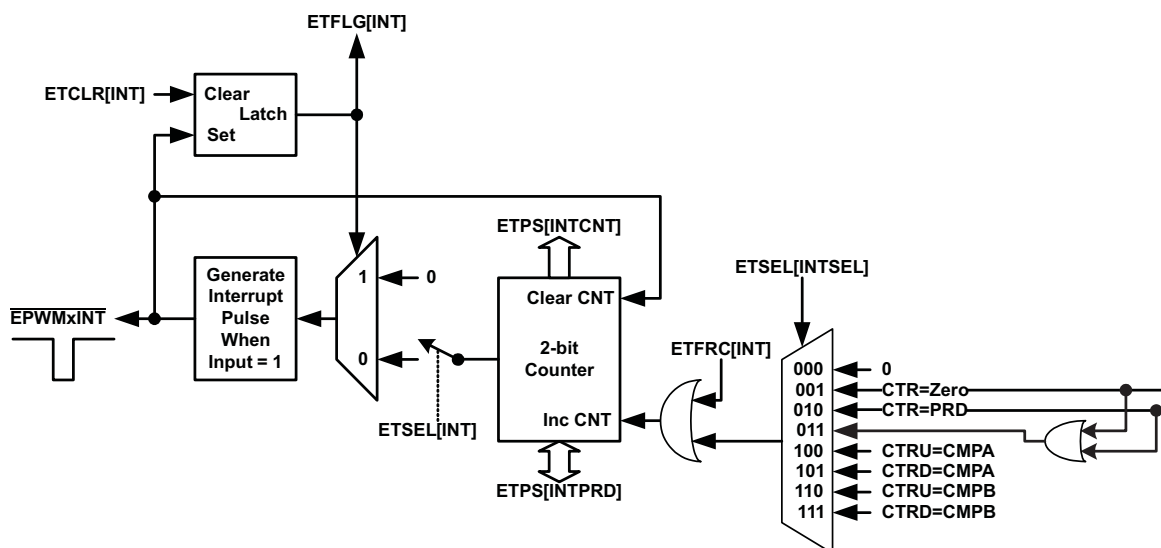
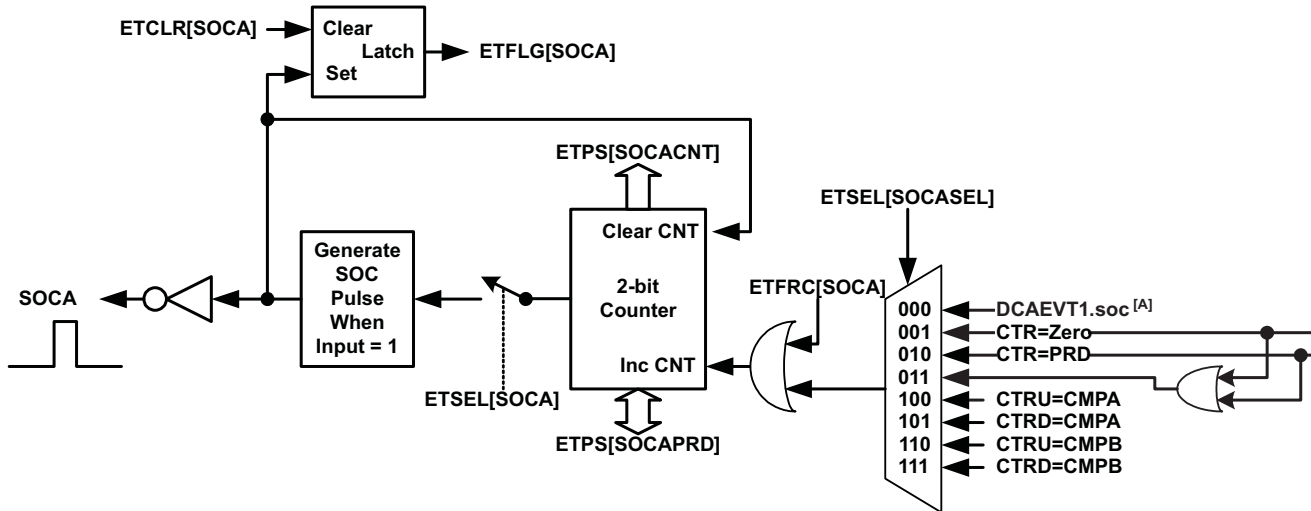


Figure 19-42 shows the operation of the event-trigger's start-of-conversion-A (SOCA) pulse generator. The ETPS[SOCACNT] counter and ETPS[SOCAPRD] period values behave similarly to the interrupt generator except that the pulses are continuously generated. That is, the pulse flag ETFLG[SOCA] is latched when a pulse is generated, but it does not stop further pulse generation. The enable/disable bit ETSEL[SOCAN] stops pulse generation, but input events can still be counted until the period value is reached as with the interrupt generation logic. The event that will trigger an SOCA and SOCB pulse can be configured separately in the ETSEL[SOCASEL] and ETSEL[SOCBSEL] bits. The possible events are the same events that can be specified for the interrupt generation logic with the addition of the DCAEVT1.soc and DCBEVT1.soc event signals from the digital compare (DC) submodule.

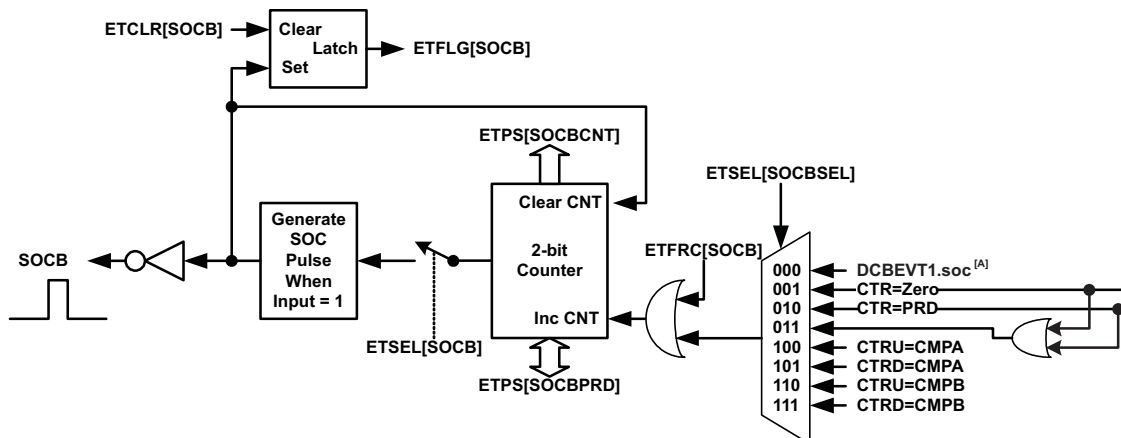
Figure 19-42. Event-Trigger SOCA Pulse Generator



- A The DCAEVT1.soc signals are signals generated by the Digital compare (DC) submodule described later in [Section 19.2.9](#).

Figure 19-43 shows the operation of the event-trigger's start-of-conversion-B (SOCB) pulse generator. The event-trigger's SOCB pulse generator operates the same way as the SOCA.

Figure 19-43. Event-Trigger SOCB Pulse Generator



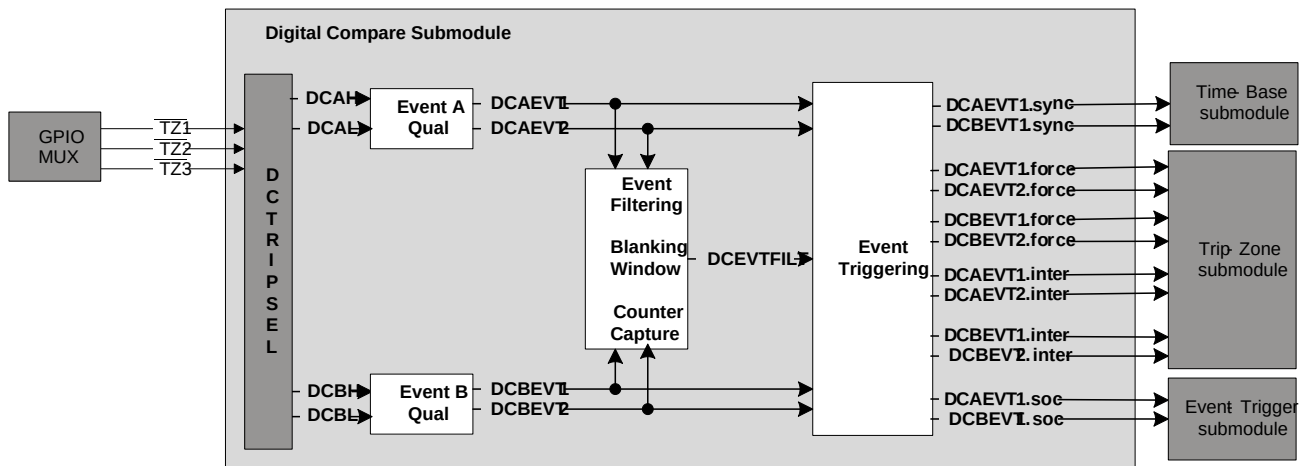
- A The DCBEVT1.soc signals are signals generated by the Digital compare (DC) submodule described later in [Section 19.2.9](#).

19.2.9 Digital Compare (DC) Submodule

Figure 19-44 illustrates where the digital compare (DC) submodule signals interface to other submodules in the ePWM system.

The digital compare (DC) submodule compares signals external to the ePWM module to directly generate PWM events/actions that then feed to the event-trigger, trip-zone, and time-base submodules. Additionally, blanking window functionality is supported to filter noise or unwanted pulses from the DC event signals.

Figure 19-44. Digital-Compare Submodule High-Level Block Diagram



19.2.9.1 Purpose of the Digital Compare Submodule

The key functions of the digital compare submodule are:

- TZ1, TZ2, and TZ3 inputs generate Digital Compare A High/Low (DCAH, DCAL) and Digital Compare B High/Low (DCBH, DCBL) signals.
- DCAH/L and DCBH/L signals trigger events which can then either be filtered or fed directly to the trip-zone, event-trigger, and time-base submodules to:
 - generate a trip zone interrupt
 - generate an ADC start of conversion
 - force an event
 - generate a synchronization event for synchronizing the ePWM module TBCTR.
- Event filtering (blanking window logic) can optionally blank the input signal to remove noise.

19.2.9.2 Controlling and Monitoring the Digital Compare Submodule

The digital compare submodule operation is controlled and monitored through the following registers:

Table 19-21. Digital Compare Submodule Registers

Register Name	Address Offset	Shadowed	Description
TZDSEL ⁽¹⁾ ⁽²⁾	0x0026	No	Trip Zone Digital Compare Select Register
DCTRIPSEL ⁽¹⁾	0x0060	No	Digital Compare Trip Select Register
DCACTL ⁽¹⁾	0x0062	No	Digital Compare A Control Register
DCBCTL ⁽¹⁾	0x0064	No	Digital Compare B Control Register
DCFCTL ⁽¹⁾	0x0066	No	Digital Compare Filter Control Register
DCCAPCTL ⁽¹⁾	0x0068	No	Digital Compare Capture Control Register
DCFOFFSET	0x006A	Writes	Digital Compare Filter Offset Register
DCFOFFSETCNT	0x006C	No	Digital Compare Filter Offset Counter Register
DCFWINDOW	0x006E	No	Digital Compare Filter Window Register
DCFWINDOWCNT	0x0070	No	Digital Compare Filter Window Counter Register
DCCAP	0x0072	Yes	Digital Compare Counter Capture Register

⁽¹⁾ These registers are writable only in privileged mode.

⁽²⁾ The TZDSEL register is part of the trip-zone submodule but is mentioned again here because of its functional significance to the digital compare submodule.

19.2.9.3 Operation Highlights of the Digital Compare Submodule

The following sections describe the operational highlights and configuration options for the digital compare submodule.

19.2.9.3.1 Digital Compare Events

As illustrated in [Figure 19-44](#), trip zone inputs ($\overline{TZ1}$, $\overline{TZ2}$, and $\overline{TZ3}$) can be selected via the DCTRIPSEL bits to generate the Digital Compare A High and Low (DCAH/L) and Digital Compare B High and Low (DCBH/L) signals. Then, the configuration of the TZDSEL register qualifies the actions on the selected DCAH/L and DCBH/L signals, which generate the DCAEVT1/2 and DCBEVT1/2 events (Event Qualification A and B).

NOTE: The $\overline{TZn}$ signals, when used as a DCEVT tripping functions, are treated as a normal input signal and can be defined to be active high or active low inputs. EPWM outputs are asynchronously tripped when either the $\overline{TZn}$, DCAEVTx.force, or DCBEVTx.force signals are active. For the condition to remain latched, a minimum of $3 \times \text{TBCLK}$ sync pulse width is required. If pulse width is $< 3 \times \text{TBCLK}$ sync pulse width, the trip condition may or may not get latched by CBC or OST latches.

The DCAEVT1/2 and DCBEVT1/2 events can then be filtered to provide a filtered version of the event signals (DCEVTFILT) or the filtering can be bypassed. Filtering is discussed further in [Section 19.2.9.3.2](#). Either the DCAEVT1/2 and DCBEVT1/2 event signals or the filtered DCEVTFILT event signals can generate a force to the trip zone module, a TZ interrupt, an ADC SOC, or a PWM sync signal.

- **force signal:**

DCAEVT1/2.force signals force trip zone conditions which either directly influence the output on the EPWMxA pin (via TZCTL[DCAEVT1 or DCAEVT2] configurations) or, if the DCAEVT1/2 signals are selected as one-shot or cycle-by-cycle trip sources (via the TZSEL register), the DCAEVT1/2.force signals can effect the trip action via the TZCTL[TZA] configuration. The DCBEVT1/2.force signals behaves similarly, but affect the EPWMxB output pin instead of the EPWMxA output pin.

The priority of conflicting actions on the TZCTL register is as follows (highest priority overrides lower priority):

Output EPWMxA: TZA (highest) -> DCAEVT1 -> DCAEVT2 (lowest)

Output EPWMxB: TZB (highest) -> DCBEVT1 -> DCBEVT2 (lowest)

- **interrupt signal:**

DCAEVT1/2.interrupt signals generate trip zone interrupts to the VIM. To enable the interrupt, the user must set the DCAEVT1, DCAEVT2, DCBEVT1, or DCBEVT2 bits in the TZEINT register. Once one of these events occurs, an EPWMxTZINT interrupt is triggered, and the corresponding bit in the TZCLR register must be set in order to clear the interrupt.

- **soc signal:**

The DCAEVT1.soc signal interfaces with the event-trigger submodule and can be selected as an event which generates an ADC start-of-conversion-A (SOCA) pulse via the ETSEL[SOCASEL] bit. Likewise, the DCBEVT1.soc signal can be selected as an event which generates an ADC start-of-conversion-B (SOCB) pulse via the ETSEL[SOCBSEL] bit.

- **sync signal:**

The DCAEVT1.sync and DCBEVT1.sync events are ORed with the EPWMxSYNCl input signal and the TBCTL[SWFSYNC] signal to generate a synchronization pulse to the time-base counter.

[Figure 19-45](#) and [Figure 19-46](#) show how the DCAEVT1, DCAEVT2, or DCEVTFILT signals are processed to generate the digital compare A event force, interrupt, soc and sync signals.

Figure 19-45. DCAEVT1 Event Triggering

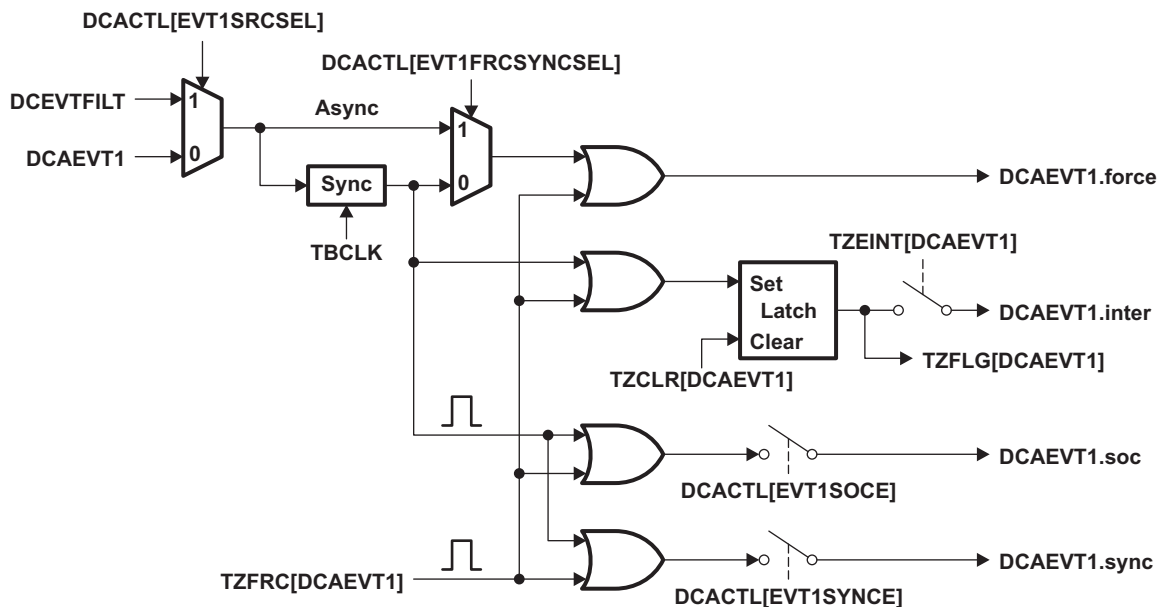


Figure 19-46. DCAEVT2 Event Triggering

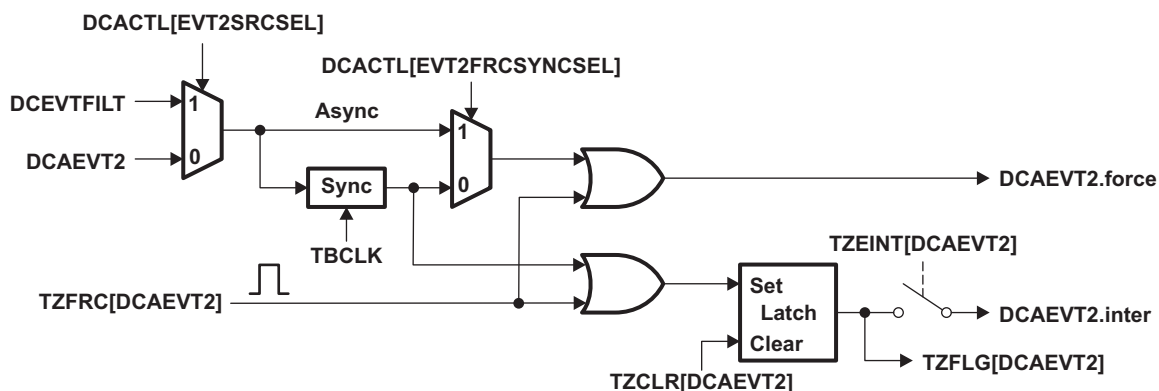


Figure 19-47 and Figure 19-48 show how the DCBEVT1, DCBEVT2, or DCEVTFILT signals are processed to generate the digital compare B event force, interrupt, soc, and sync signals.

Figure 19-47. DCBEVT1 Event Triggering

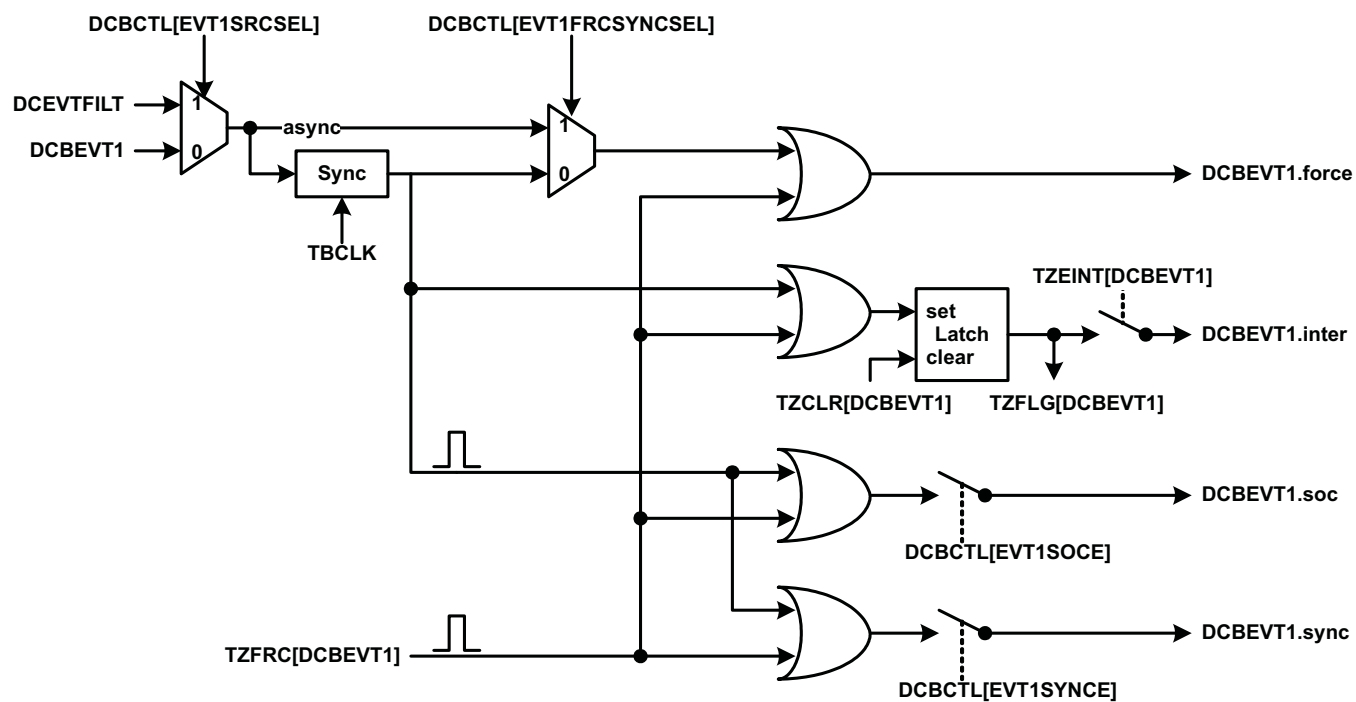
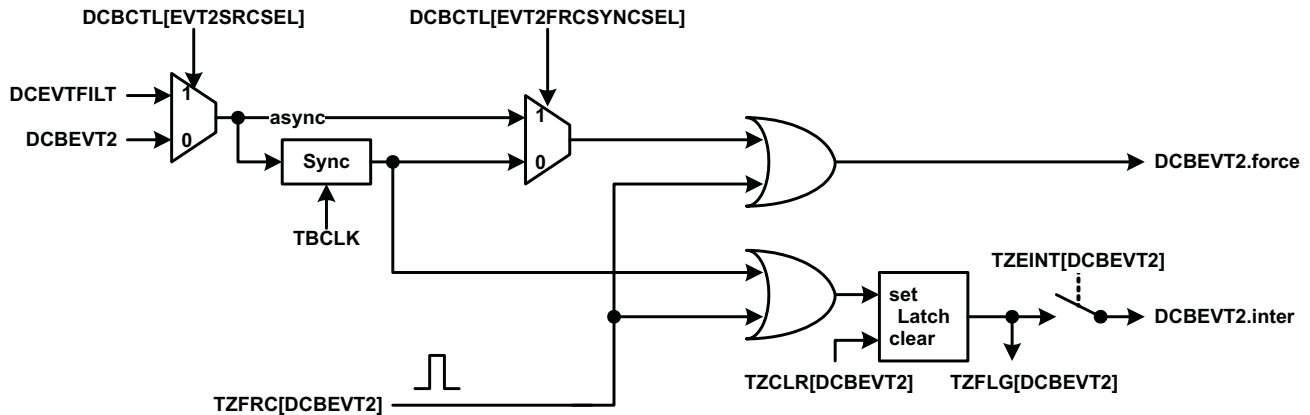


Figure 19-48. DCBEVT2 Event Triggering



19.2.9.3.2 Event Filtering

The DCAEVT1/2 and DCBEVT1/2 events can be filtered via event filtering logic to remove noise by optionally blanking events for a certain period of time. This is useful for cases where the analog comparator outputs may be selected to trigger DCAEVT1/2 and DCBEVT1/2 events, and the blanking logic is used to filter out potential noise on the signal prior to tripping the PWM outputs or generating an interrupt or ADC start-of-conversion. The event filtering can also capture the TBCTR value of the trip event. Figure 19-49 shows the details of the event filtering logic.

If the blanking logic is enabled, one of the digital compare events – DCAEVT1, DCAEVT2, DCBEVT1, DCBEVT2 – is selected for filtering. The blanking window, which filters out all event occurrences on the signal while it is active, will be aligned to either a CTR = PRD pulse or a CTR = 0 pulse (configured by the DCFCTL[PULSESEL] bits). An offset value in TBCLK counts is programmed into the DCFOFFSET register, which determines at what point after the CTR = PRD or CTR = 0 pulse the blanking window starts. The duration of the blanking window, in number of TBCLK counts after the offset counter expires, is written to the DCFWINDOW register by the application. During the blanking window, all events are ignored. Before and after the blanking window ends, events can generate soc, sync, interrupt, and force signals as before.

Figure 19-50 illustrates several timing conditions for the offset and blanking window within an ePWM period. Notice that if the blanking window crosses the CTR = 0 or CTR = PRD boundary, the next window still starts at the same offset value after the CTR = 0 or CTR = PRD pulse.

Figure 19-49. Event Filtering

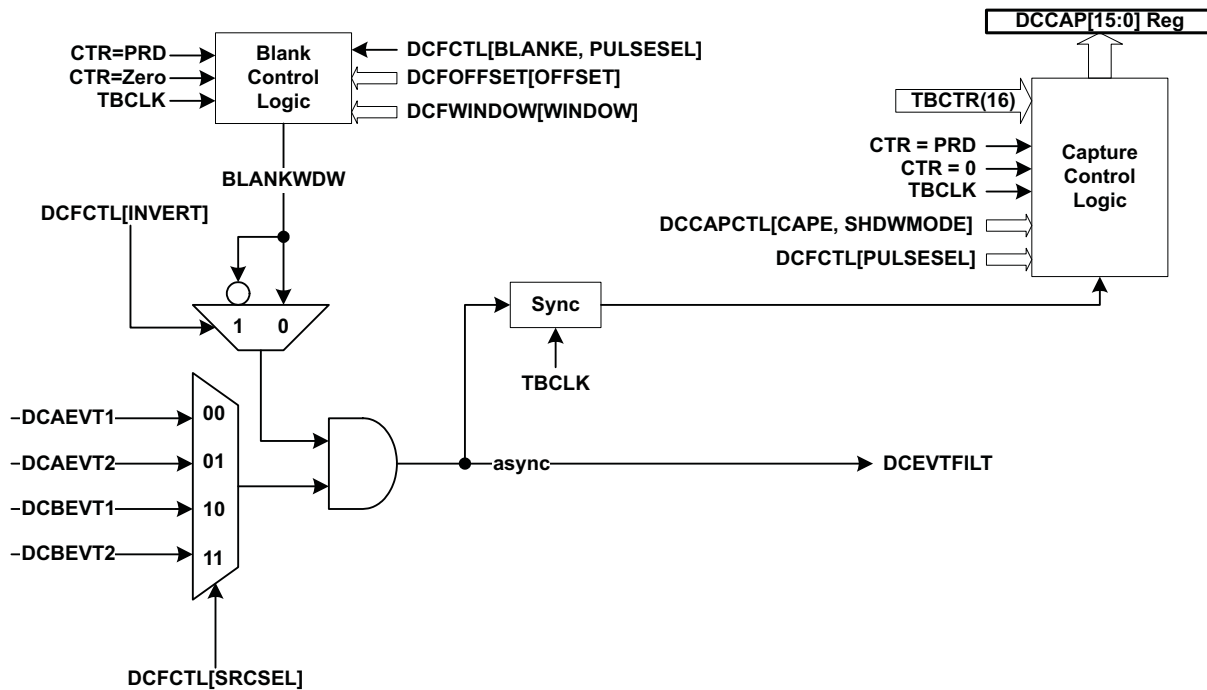
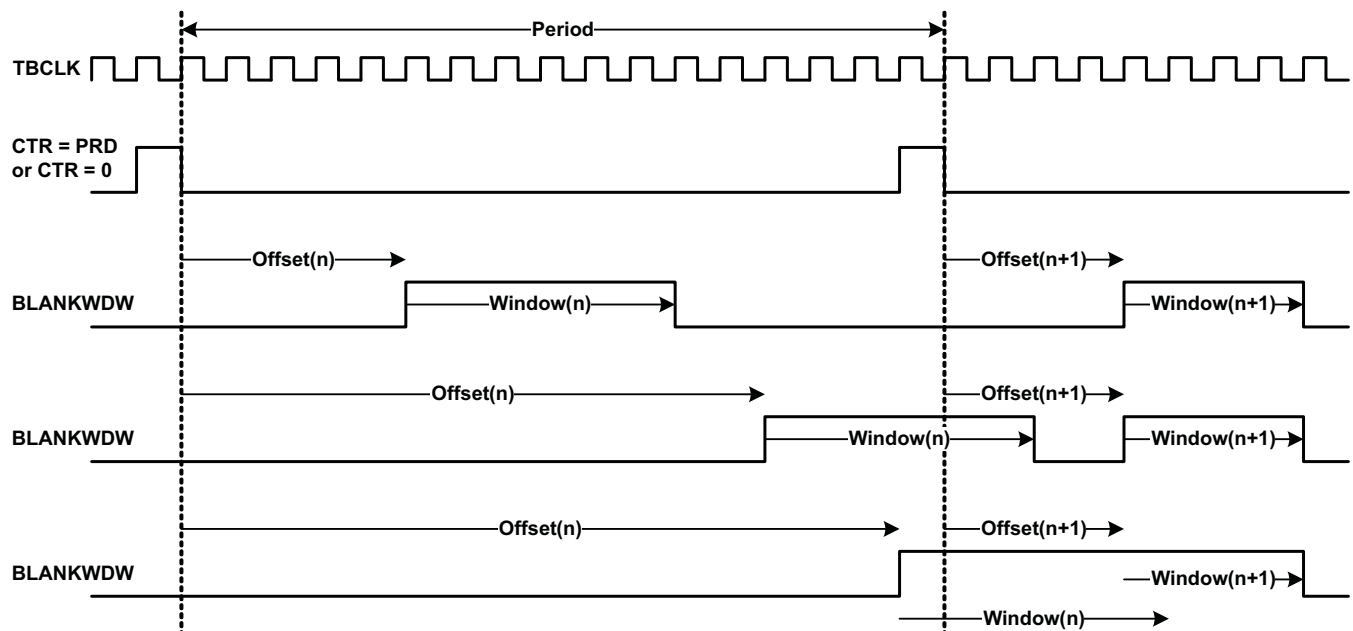


Figure 19-50. Blanking Window Timing Diagram



19.2.10 Proper Interrupt Initialization Procedure

When the ePWM peripheral clock is enabled it may be possible that interrupt flags may be set due to spurious events due to the ePWM registers not being properly initialized. The proper procedure for initializing the ePWM peripheral is as follows:

1. Disable global interrupts (CPU INTM flag)
2. Disable ePWM interrupts
3. Set TBCLKSYNC = 0
4. Initialize peripheral registers
5. Set TBCLKSYNC = 1
6. Clear any spurious ePWM flags (including interrupt flags)
7. Enable ePWM interrupts
8. Enable global interrupts

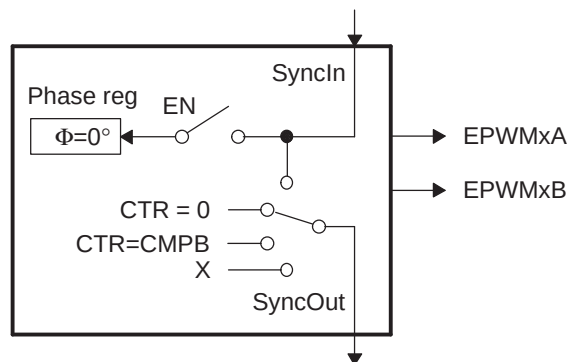
19.3 Application Examples

An ePWM module has all the local resources necessary to operate completely as a standalone module or to operate in synchronization with other identical ePWM modules.

19.3.1 Overview of Multiple Modules

Previously in this chapter, all discussions have described the operation of a single module. To facilitate the understanding of multiple modules working together in a system, the ePWM module described in reference is represented by the more simplified block diagram shown in [Figure 19-51](#). This simplified ePWM block shows only the key resources needed to explain how a multiswitch power topology is controlled with multiple ePWM modules working together.

Figure 19-51. Simplified ePWM Module



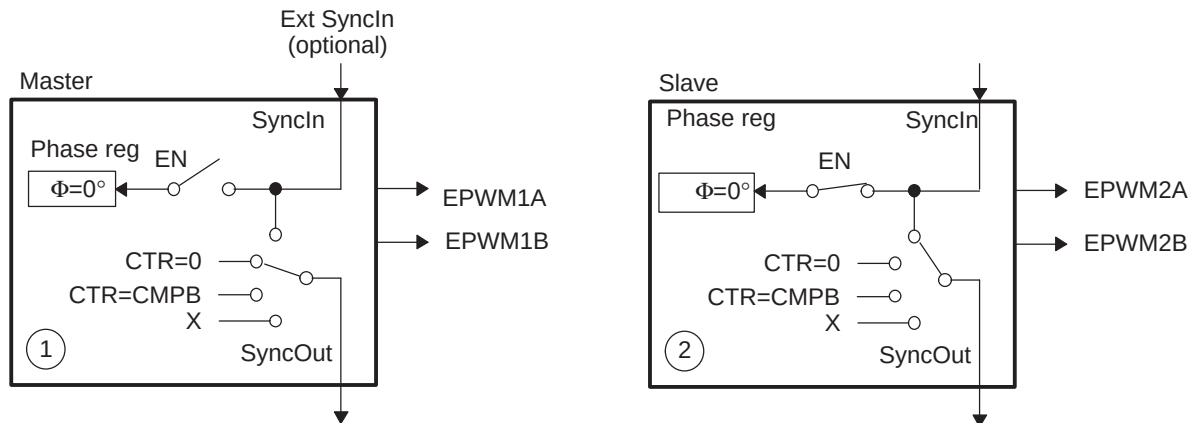
19.3.2 Key Configuration Capabilities

The key configuration choices available to each module are as follows:

- Options for SyncIn
 - Load own counter with phase register on an incoming sync strobe—enable (EN) switch closed
 - Do nothing or ignore incoming sync strobe—enable switch open
 - Sync flow-through - SyncOut connected to SyncIn
 - Master mode, provides a sync at PWM boundaries—SyncOut connected to CTR = PRD
 - Master mode, provides a sync at any programmable point in time—SyncOut connected to CTR = CMPB
 - Module is in standalone mode and provides No sync to other modules—SyncOut connected to X (disabled)
- Options for SyncOut
 - Sync flow-through - SyncOut connected to SyncIn
 - Master mode, provides a sync at PWM boundaries—SyncOut connected to CTR = PRD
 - Master mode, provides a sync at any programmable point in time—SyncOut connected to CTR = CMPB
 - Module is in standalone mode and provides No sync to other modules—SyncOut connected to X (disabled)

For each choice of SyncOut, a module may also choose to load its own counter with a new phase value on a SyncIn strobe input or choose to ignore it, that is, via the enable switch. Although various combinations are possible, the two most common—master module and slave module modes—are shown in [Figure 19-52](#).

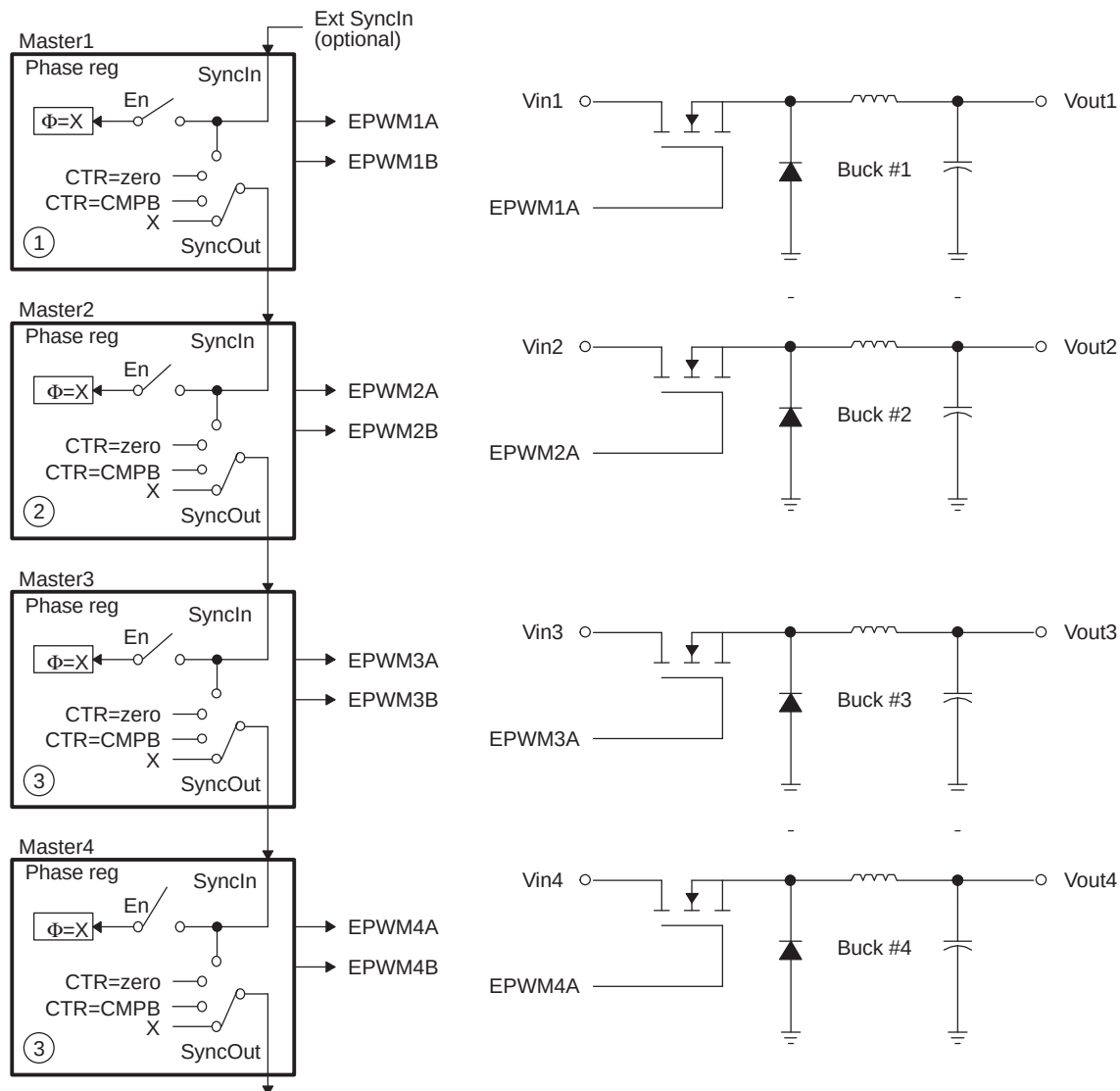
Figure 19-52. EPWM1 Configured as a Typical Master, EPWM2 Configured as a Slave



19.3.3 Controlling Multiple Buck Converters With Independent Frequencies

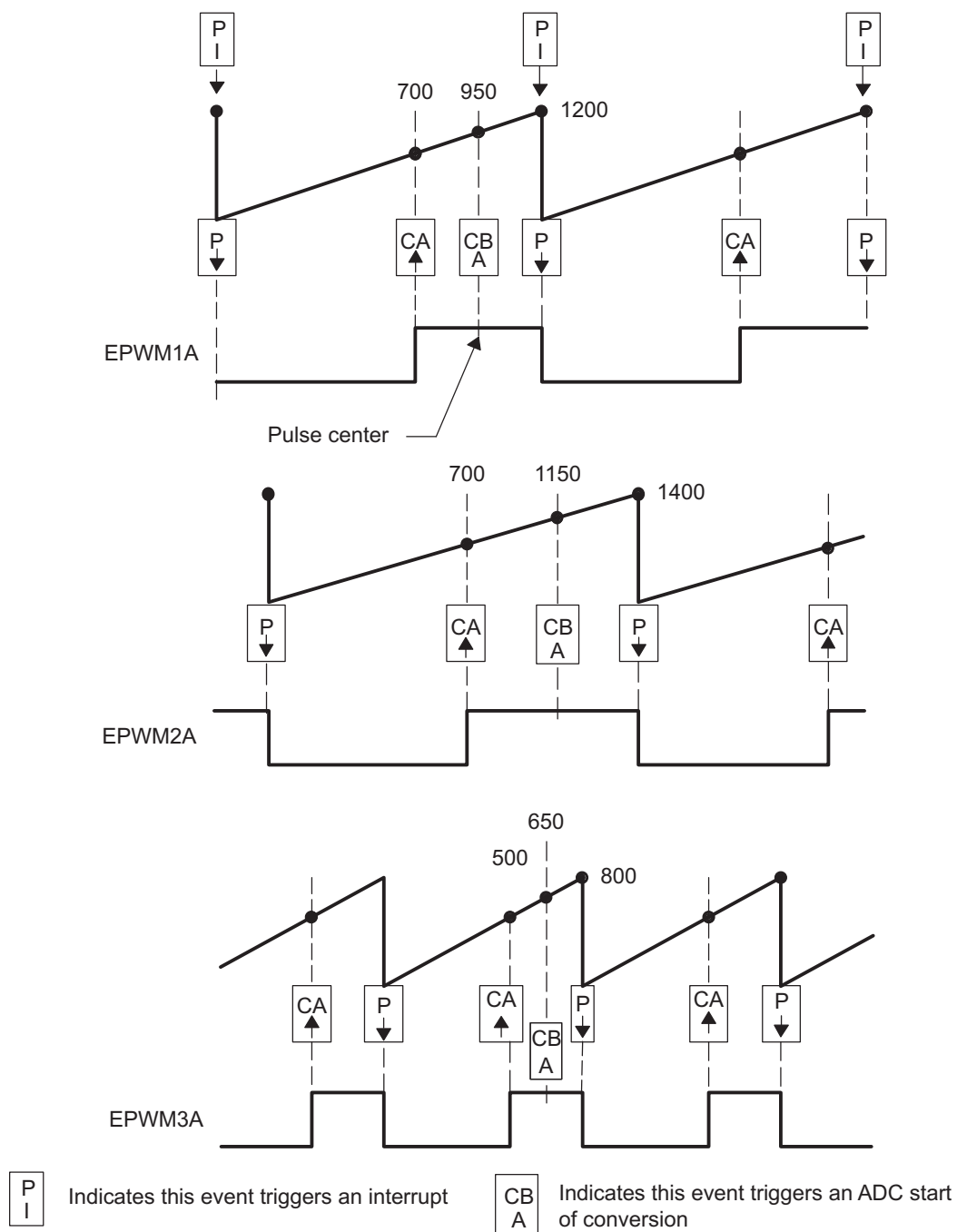
One of the simplest power converter topologies is the buck. A single ePWM module configured as a master can control two buck stages with the same PWM frequency. If independent frequency control is required for each buck converter, then one ePWM module must be allocated for each converter stage. [Figure 19-53](#) shows four buck stages, each running at independent frequencies. In this case, all four ePWM modules are configured as Masters and no synchronization is used. [Figure 19-54](#) shows the waveforms generated by the setup shown in [Figure 19-53](#); note that only three waveforms are shown, although there are four stages.

Figure 19-53. Control of Four Buck Stages. Here $F_{PWM1} \neq F_{PWM2} \neq F_{PWM3} \neq F_{PWM4}$



NOTE: $\Phi = X$ indicates value in phase register is a "don't care"

Figure 19-54. Buck Waveforms for Figure 19-53 (Note: Only three bucks shown here)



Example 19-8. Configuration for Example in Figure 19-54

```
//=====
// (Note: code for only 3 modules shown)
// Initialization Time
//=====
// EPWM Module 1 config
EPwm1Regs.TBPRD = 1200; // Period = 1201 TBCLK counts
EPwm1Regs.TBPHS.half.TBPHS = 0; // Set Phase register to zero
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP; // Asymmetrical mode
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm1Regs.AQCTLA.bit.PRDL = AQ_CLEAR;
EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;
// EPWM Module 2 config
EPwm2Regs.TBPRD = 1400; // Period = 1401 TBCLK counts
EPwm2Regs.TBPHS.half.TBPHS = 0; // Set Phase register to zero
EPwm2Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP; // Asymmetrical mode
EPwm2Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm2Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm2Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm2Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm2Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm2Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm2Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm2Regs.AQCTLA.bit.PRDL = AQ_CLEAR;
EPwm2Regs.AQCTLA.bit.CAU = AQ_SET;
// EPWM Module 3 config
EPwm3Regs.TBPRD = 800; // Period = 801 TBCLK counts
EPwm3Regs.TBPHS.half.TBPHS = 0; // Set Phase register to zero
EPwm3Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
EPwm3Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Phase loading disabled
EPwm3Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm3Regs.TBCTL.bit.SYNCSEL = TB_SYNC_DISABLE;
EPwm3Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm3Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm3Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm3Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm3Regs.AQCTLA.bit.PRDL = AQ_CLEAR;
EPwm3Regs.AQCTLA.bit.CAU = AQ_SET;
//
// Run Time (Note: Example execution of one run-time instant)
//=====
EPwm1Regs.CMPA.half.CMPA = 700; // adjust duty for output EPWM1A
EPwm2Regs.CMPA.half.CMPA = 700; // adjust duty for output EPWM2A
EPwm3Regs.CMPA.half.CMPA = 500; // adjust duty for output EPWM3A
```

19.3.4 Controlling Multiple Buck Converters With Same Frequencies

If synchronization is a requirement, ePWM module 2 can be configured as a slave and can operate at integer multiple (N) frequencies of module 1. The sync signal from master to slave ensures these modules remain locked. Figure 19-55 shows such a configuration; Figure 19-56 shows the waveforms generated by the configuration.

Figure 19-55. Control of Four Buck Stages. (Note: $F_{PWM2} = N \times F_{PWM1}$)

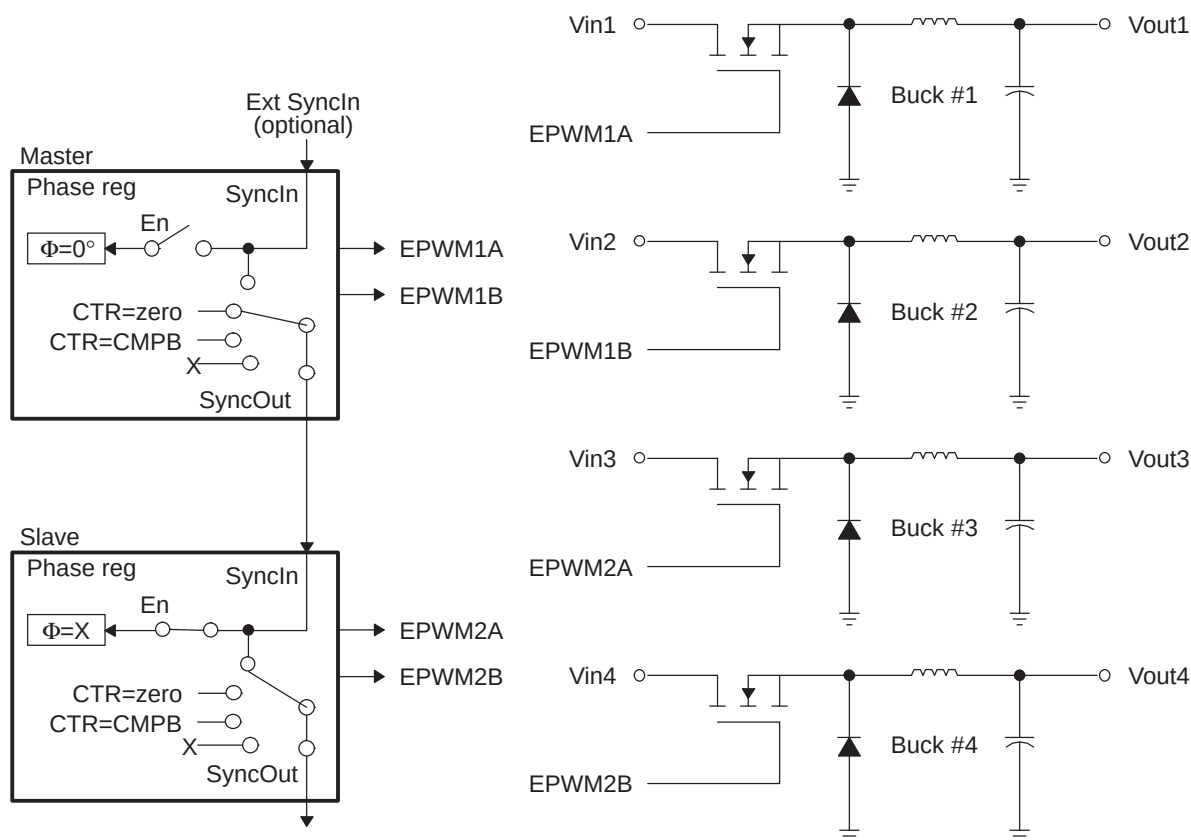
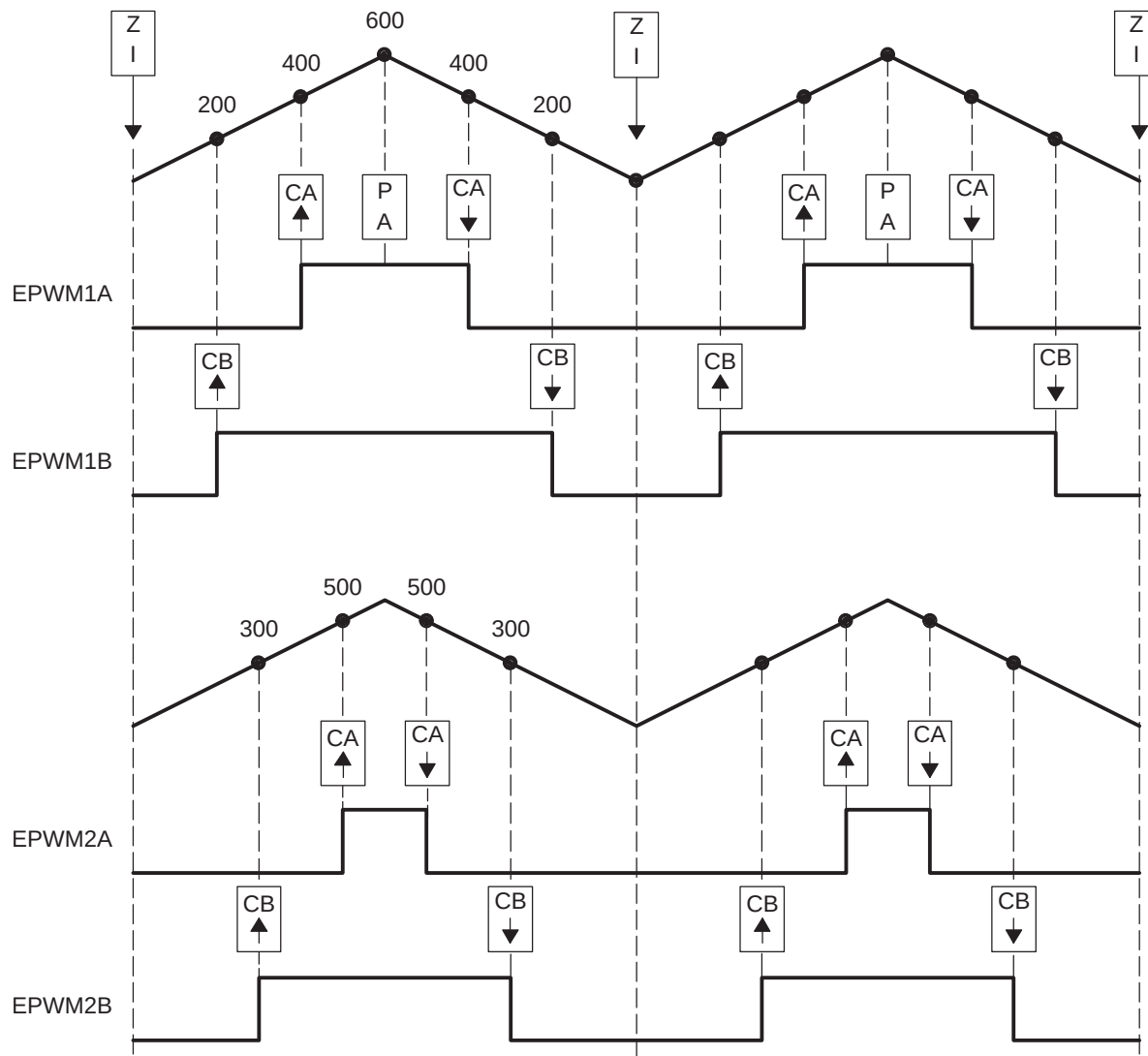
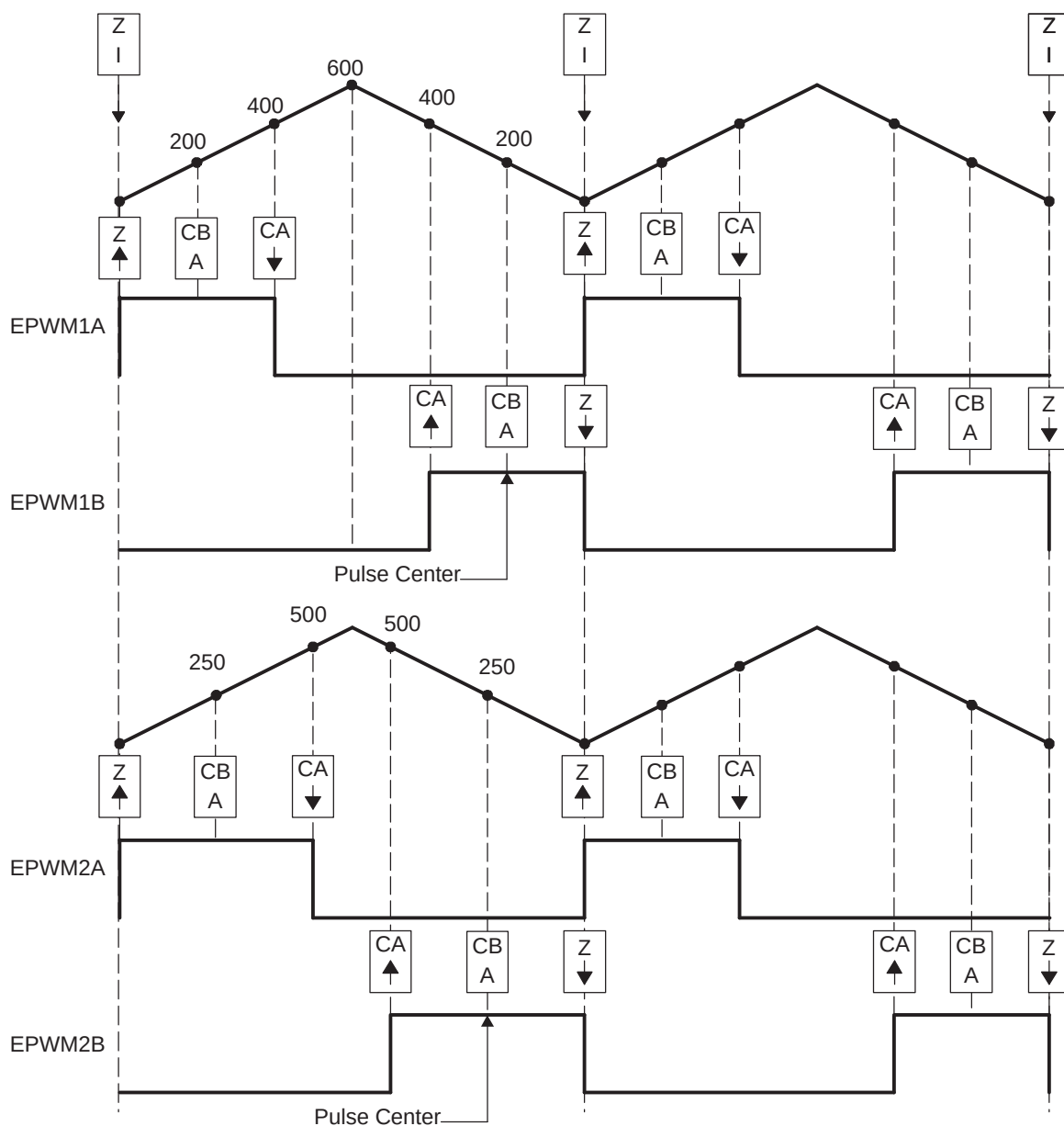


Figure 19-56. Buck Waveforms for Figure 19-55 (Note: $F_{PWM2} = F_{PWM1}$)


Example 19-9. Code Snippet for Configuration in Figure 19-55

```
//=====
// EPWM Module 1 config
    EPwm1Regs.TBPRD = 600;                // Period = 1200 TBCLK counts
    EPwm1Regs.TBPHS.half.TBPHS = 0;      // Set Phase register to zero
    EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetrical mode
    EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Master module
    EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
    EPwm1Regs.TBCTL.bit.SYNCSEL = TB_CTR_ZERO; // Sync down-stream module
    EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
    EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
    EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
    EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
    EPwm1Regs.AQCTLA.bit.CAU = AQ_SET;    // set actions for EPWM1A
    EPwm1Regs.AQCTLA.bit.CAD = AQ_CLEAR;
    EPwm1Regs.AQCTLB.bit.CBU = AQ_SET;    // set actions for EPWM1B
    EPwm1Regs.AQCTLB.bit.CBD = AQ_CLEAR;
// EPWM Module 2 config
    EPwm2Regs.TBPRD = 600;                // Period = 1200 TBCLK counts
    EPwm2Regs.TBPHS.half.TBPHS = 0;      // Set Phase register to zero
    EPwm2Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetrical mode
    EPwm2Regs.TBCTL.bit.PHSEN = TB_ENABLE; // Slave module
    EPwm2Regs.TBCTL.bit.PRDL = TB_SHADOW;
    EPwm2Regs.TBCTL.bit.SYNCSEL = TB_SYNC_IN; // sync flow-through
    EPwm2Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
    EPwm2Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
    EPwm2Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
    EPwm2Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
    EPwm2Regs.AQCTLA.bit.CAU = AQ_SET;    // set actions for EPWM2A
    EPwm2Regs.AQCTLA.bit.CAD = AQ_CLEAR;
    EPwm2Regs.AQCTLB.bit.CBU = AQ_SET;    // set actions for EPWM2B
    EPwm2Regs.AQCTLB.bit.CBD = AQ_CLEAR;
//
// Run Time (Note: Example execution of one run-time instance)
//=====
    EPwm1Regs.CMPA.half.CMPA = 400;      // adjust duty for output EPWM1A
    EPwm1Regs.CMPB = 200;                // adjust duty for output EPWM1B
    EPwm2Regs.CMPA.half.CMPA = 500;      // adjust duty for output EPWM2A
    EPwm2Regs.CMPB = 300;                // adjust duty for output EPWM2B
```


Figure 19-58. Half-H Bridge Waveforms for Figure 19-57 (Note: Here $F_{PWM2} = F_{PWM1}$)



Example 19-10. Code Snippet for Configuration in Figure 19-57

```
//=====
// Config
//=====
// Initialization Time
//=====
// EPWM Module 1 config
EPwm1Regs.TBPRD = 600; // Period = 1200 TBCLK counts
EPwm1Regs.TBPHS.half.TBPHS = 0; // Set Phase register to zero
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetrical mode
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE; // Master module
EPwm1Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm1Regs.TBCTL.bit.SYNCSEL = TB_CTR_ZERO; // Sync down-stream module
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm1Regs.AQCTLA.bit.ZR0 = AQ_SET; // set actions for EPWM1A
EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.ZR0 = AQ_CLEAR; // set actions for EPWM1B
EPwm1Regs.AQCTLB.bit.CAD = AQ_SET;
// EPWM Module 2 config
EPwm2Regs.TBPRD = 600; // Period = 1200 TBCLK counts
EPwm2Regs.TBPHS.half.TBPHS = 0; // Set Phase register to zero
EPwm2Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; // Symmetrical mode
EPwm2Regs.TBCTL.bit.PHSEN = TB_ENABLE; // Slave module
EPwm2Regs.TBCTL.bit.PRDL = TB_SHADOW;
EPwm2Regs.TBCTL.bit.SYNCSEL = TB_SYNC_IN; // sync flow-through
EPwm2Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm2Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm2Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm2Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO; // load on CTR=Zero
EPwm2Regs.AQCTLA.bit.ZR0 = AQ_SET; // set actions for EPWM1A
EPwm2Regs.AQCTLA.bit.CAU = AQ_CLEAR;
EPwm2Regs.AQCTLB.bit.ZR0 = AQ_CLEAR; // set actions for EPWM1B
EPwm2Regs.AQCTLB.bit.CAD = AQ_SET;
//=====
EPwm1Regs.CMPA.half.CMPA = 400; // adjust duty for output EPWM1A & EPWM1B

EPwm1Regs.CMPB = 200; // adjust point-in-time for ADCSOC trigger
EPwm2Regs.CMPA.half.CMPA = 500; // adjust duty for output EPWM2A & EPWM2B
EPwm2Regs.CMPB = 250; // adjust point-in-time for ADCSOC trigger
```

19.3.6 Controlling Dual 3-Phase Inverters for Motors (ACI and PMSM)

The idea of multiple modules controlling a single power stage can be extended to the 3-phase Inverter case. In such a case, six switching elements can be controlled using three PWM modules, one for each leg of the inverter. Each leg must switch at the same frequency and all legs must be synchronized. A master + two slaves configuration can easily address this requirement. [Figure 19-59](#) shows how six PWM modules can control two independent 3-phase inverters; each running a motor.

As in the cases shown in the previous sections, we have a choice of running each inverter at a different frequency (module 1 and module 4 are masters as in [Figure 19-59](#)), or both inverters can be synchronized by using one master (module 1) and five slaves. In this case, the frequency of modules 4, 5, and 6 (all equal) can be integer multiples of the frequency for modules 1, 2, 3 (also all equal).

Figure 19-59. Control of Dual 3-Phase Inverter Stages as Is Commonly Used in Motor Control

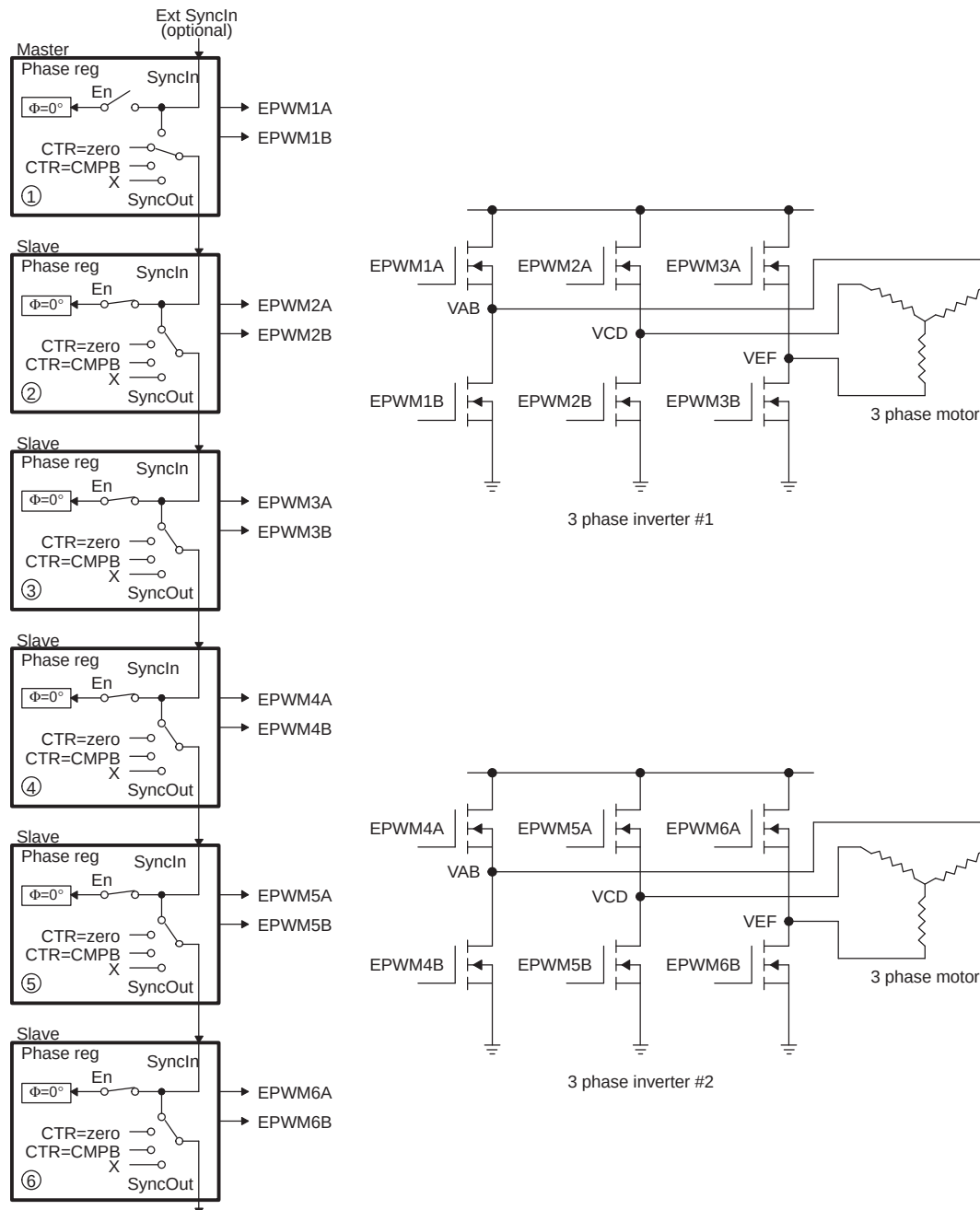
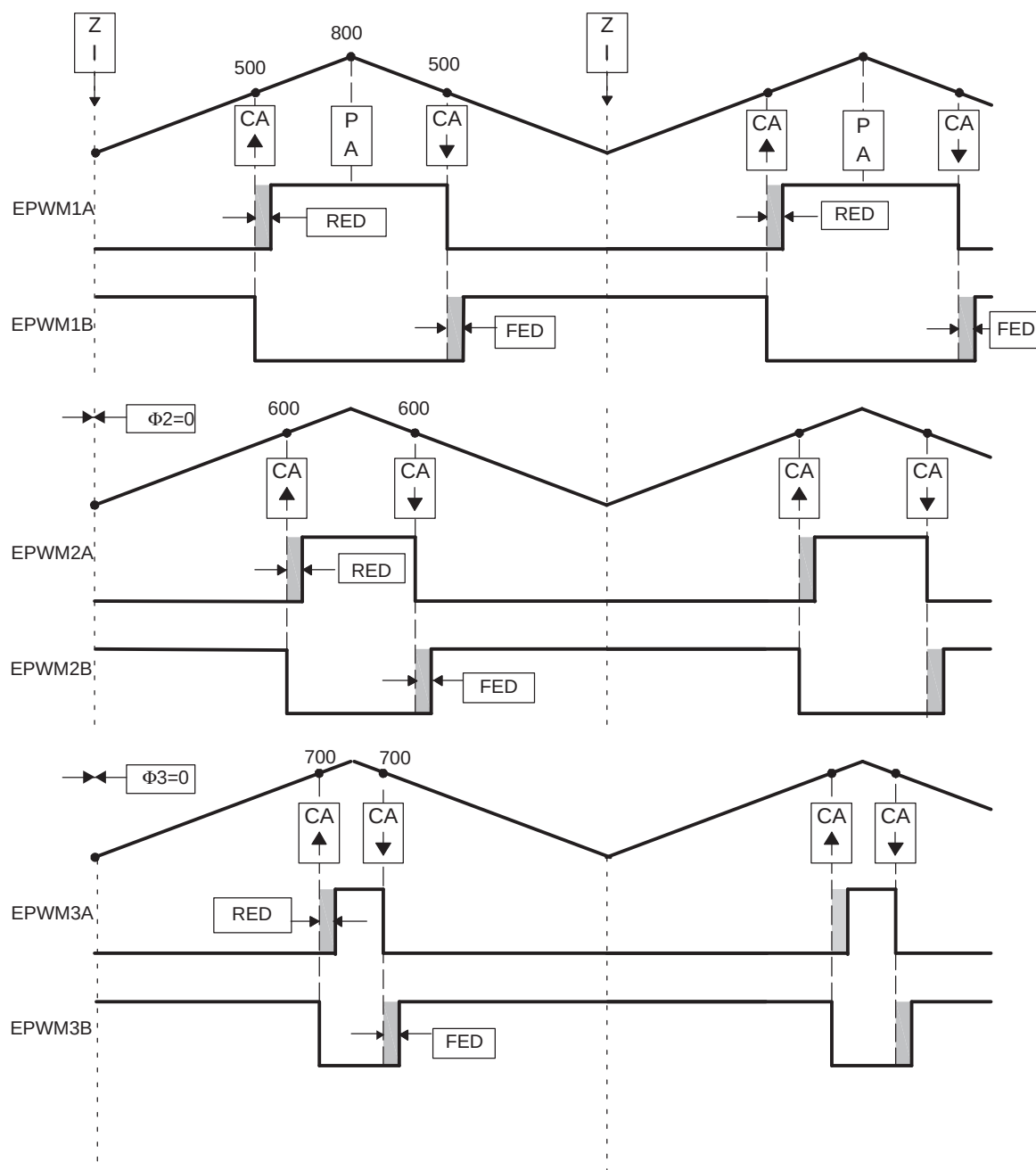


Figure 19-60. 3-Phase Inverter Waveforms for Figure 19-59 (Only One Inverter Shown)



19.3.7 Practical Applications Using Phase Control Between PWM Modules

So far, none of the examples have made use of the phase register (TBPHS). It has either been set to zero or its value has been a don't care. However, by programming appropriate values into TBPHS, multiple PWM modules can address another class of applications that rely on phase relationship between stages for correct operation. As described in the TB module section, a PWM module can be configured to allow a SyncIn pulse to cause the TBPHS register to be loaded into the TBCTR register. To illustrate this concept, [Figure 19-61](#) shows a master and slave module with a phase relationship of 120°, that is, the slave leads the master.

Figure 19-61. Configuring Two PWM Modules for Phase Control

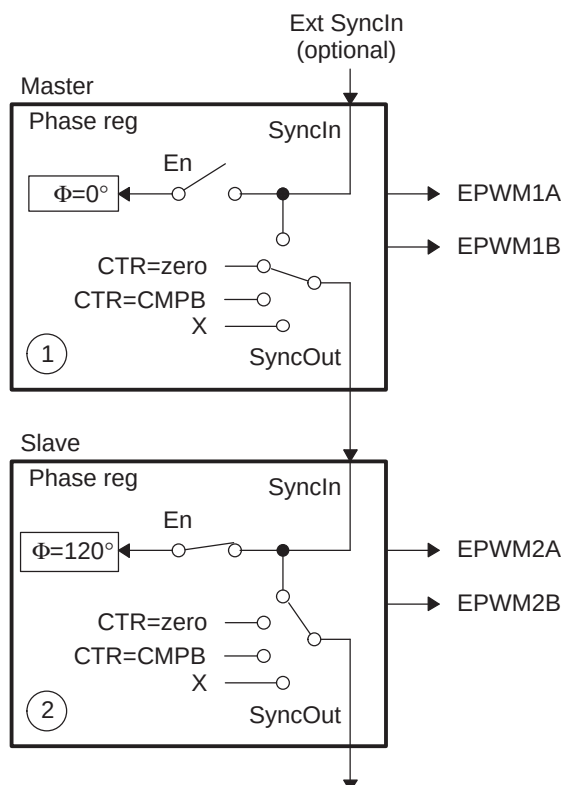
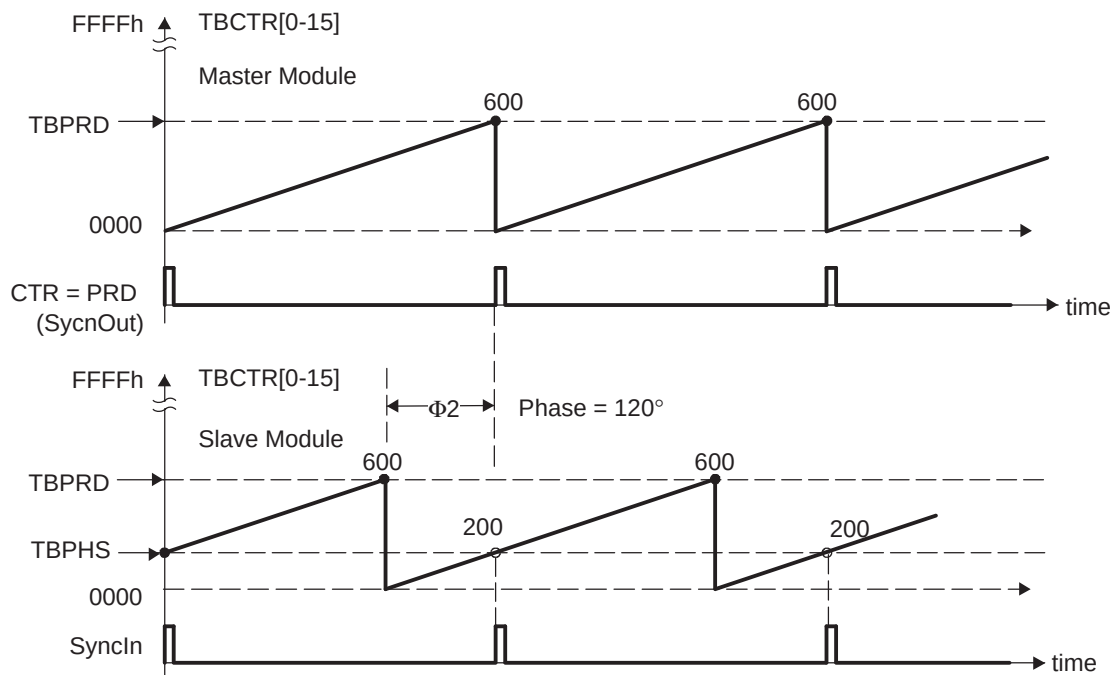


Figure 19-62 shows the associated timing waveforms for this configuration. Here, TBPRD = 600 for both master and slave. For the slave, TBPHS = 200 (that is, $200/600 \times 360^\circ = 120^\circ$). Whenever the master generates a SyncIn pulse (CTR = PRD), the value of TBPHS = 200 is loaded into the slave TBCTR register so the slave time-base is always leading the master's time-base by 120° .

Figure 19-62. Timing Waveforms Associated With Phase Control Between 2 Modules



19.4 EPWM_MODULE_CONTROL_AND_STATUS_REGISTER Registers

Table 19-22 lists the memory-mapped registers for the EPWM_MODULE_CONTROL_AND_STATUS_REGISTER. All register offset addresses not listed in Table 19-22 should be considered as reserved locations and the register contents should not be modified.

The base address for the control registers is FCF7 8C00h for ePWM1, FCF7 8D00h for ePWM2, FCF7 8E00h for ePWM3, FCF7 8F00h for ePWM4, FCF7 9000h for ePWM5, FCF7 9100h for ePWM6, and FCF7 9200h for ePWM7.

Table 19-22. EPWM_MODULE_CONTROL_AND_STATUS_REGISTER Registers

Offset	Acronym	Register Name	Section
0h	TBCTL	Time-Base Control Register	Section 19.4.1
2h	TBSTS	Time-Base Status Register	Section 19.4.2
6h	TBPHS	Time-Base Phase Register	Section 19.4.3
8h	TBCTR	Time-Base Counter Register	Section 19.4.4
Ah	TBPRD	Time-Base Period Register	Section 19.4.5
Eh	CMPCTL	Counter-Compare Control Register	Section 19.4.6
12h	CPMA	Counter-Compare A Register	Section 19.4.7
14h	CMPB	Counter-Compare B Register	Section 19.4.8
16h	AQCTLA	Action-Qualifier Control Register for Output A (EPWMxA)	Section 19.4.9
18h	AQCTLB	Action-Qualifier Control Register for Output B (EPWMxB)	Section 19.4.10
1Ah	AQSFRC	Action-Qualifier Software Force Register	Section 19.4.11
1Ch	AQCSFRC	Action-Qualifier Continuous S/W Force Register Set	Section 19.4.12
1Eh	DBCTL	Dead-Band Generator Control Register	Section 19.4.13
20h	DBRED	Dead-Band Generator Rising Edge Delay Count Register	Section 19.4.14
22h	DBFED	Dead-Band Generator Falling Edge Delay Count Register	Section 19.4.15
24h	TZSEL	Trip-Zone Select Register	Section 19.4.16
26h	TZDCSEL	Trip Zone Digital Compare Event Select Register	Section 19.4.17
28h	TZCTL	Trip-Zone Control Register	Section 19.4.18
2Ah	TZEINT	Trip-Zone Enable Interrupt Register	Section 19.4.19
2Ch	TZFLG	Trip-Zone Flag Register	Section 19.4.20
2Eh	TZCLR	Trip-Zone Clear Register	Section 19.4.21
30h	TZFRC	Trip-Zone Force Register	Section 19.4.22
32h	ETSEL	Event-Trigger Selection Register	Section 19.4.23
34h	ETPS	Event-Trigger Pre-Scale Register	Section 19.4.24
36h	ETFLG	Event-Trigger Flag Register	Section 19.4.25
38h	ETCLR	Event-Trigger Clear Register	Section 19.4.26
3Ah	ETFRC	Event-Trigger Force Register	Section 19.4.27
3Ch	PCCTL	PWM-Chopper Control Register	Section 19.4.28
60h	DCTRIPSEL	Digital Compare Trip Select Register	Section 19.4.29
62h	DCACTL	Digital Compare A Control Register	Section 19.4.30
64h	DCBCTL	Digital Compare B Control Register	Section 19.4.31
66h	DCFCTL	Digital Compare Filter Control Register	Section 19.4.32
68h	DCCAPCTL	Digital Compare Capture Control Register	Section 19.4.33
6Ah	DCFOFFSET	Digital Compare Filter Offset Register	Section 19.4.34
6Ch	DCFOFFSETCNT	Digital Compare Filter Offset Counter Register	Section 19.4.35
6Eh	DCFWINDOW	Digital Compare Filter Window Register	Section 19.4.36
70h	DCFWINDOWCNT	Digital Compare Filter Window Counter Register	Section 19.4.37
72h	DCCAP	Digital Compare Counter Capture Register	Section 19.4.38

Complex bit access types are encoded to fit into small table cells. [Table 19-23](#) shows the codes that are used for access types in this section.

**Table 19-
23.
EPWM_MODULE_CONTROL_AND_STATUS_REGISTER
R Access Type Codes**

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W	W	Write
W1C	1C W	1 to clear Write
Reset or Default Value		
-n		Value after reset or the default value

19.4.1 TBCTL Register (Offset = 0h) [reset = 83h]

TBCTL is shown in [Figure 19-63](#) and described in [Table 19-24](#).

Return to [Summary Table](#).

Figure 19-63. TBCTL Register

15	14	13	12	11	10	9	8
FREE_SOFT		PHSDIR	CLKDIV		HSPCLKDIV		
R/W-0h		R/W-0h	R/W-0h		R/W-1h		
7	6	5	4	3	2	1	0
HSPCLKDIV	SWFSYNC	SYNCOSSEL		PRDLD	PHSEN	CTRMODE	
R/W-1h	R/W-0h	R/W-0h		R/W-0h	R/W-0h	R/W-3h	

Table 19-24. TBCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	FREE_SOFT	R/W	0h	Emulation Mode Bits. These bits select the behavior of the ePWM time-base counter during emulation events. 0h = Stop after the next time-base counter increment or decrement. 1h = Stop when counter completes a whole cycle. In up-count mode, stop when the time-base counter = period (TBCTR = TBPRD). In down-count mode, stop when the time-base counter = 0x0000 (TBCTR = 0x0000). In up-down-count mode, stop when the time-base counter = 0x0000 (TBCTR = 0x0000). 2h = Free run 3h = Free run
13	PHSDIR	R/W	0h	Phase Direction Bit. This bit is only used when the time-base counter is configured in the up-down-count mode. The PHSDIR bit indicates the direction the time-base counter (TBCTR) counts after a synchronization event occurs and a new phase value is loaded from the phase (TBPHS) register. This is irrespective of the direction of the counter before the synchronization event. In the up-count and down-count modes this bit is ignored. 0h = Count down after the synchronization event. 1h = Count up after the synchronization event.
12-10	CLKDIV	R/W	0h	Time-base Clock Prescale Bits. These bits determine part of the time-base clock prescale value. $TBCLK = VCLK4 / (HSPCLKDIV \times CLKDIV)$ 0h = /1 (default on reset) 1h = /2 2h = /4 3h = /8 4h = /16 5h = /32 6h = /64 7h = /128

Table 19-24. TBCTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
9-7	HSPCLKDIV	R/W	1h	High Speed Time-base Clock Prescale Bits. These bits determine part of the time-base clock prescale value. $TBCLK = VCLK4 / (HSPCLKDIV \times CLKDIV)$ 0h = /1 1h = /2 (default on reset) 2h = /4 3h = /6 4h = /8 5h = /10 6h = /12 7h = /14
6	SWFSYNC	R/W	0h	Software Forced Synchronization Pulse. This event is ORed with the EPWMxSYNCl input of the ePWM module. SWFSYNC is valid (operates) only when EPWMxSYNCl is selected by SYNCOSSEL = 0. 0h = Writing a 0 has no effect and reads always return a 0. 1h = Writing a 1 forces a one-time synchronization pulse to be generated.
5-4	SYNCOSSEL	R/W	0h	Synchronization Output Select. These bits select the source of the EPWMxSYNCO signal. 0h = EPWMxSYNC 1h = CTR = zero: Time-base counter equal to zero (TBCTR = 0x0000) 2h = CTR = CMPB : Time-base counter equal to counter-compare B (TBCTR = CMPB) 3h = Disable EPWMxSYNCO signal
3	PRDLD	R/W	0h	Active Period Register Load From Shadow Register Select. 0h = The period register (TBPRD) is loaded from its shadow register when the time-base counter, TBCTR, is equal to 0. A write or read to the TBPRD register accesses the shadow register. 1h = Load the TBPRD register immediately without using a shadow register. A write or read to the TBPRD register directly accesses the active register.
2	PHSEN	R/W	0h	Counter Register Load From Phase Register Enable. 0h = Do not load the time-base counter (TBCTR) from the time-base phase register (TBPHS) 1h = Load the time-base counter with the phase register when an EPWMxSYNCl input signal occurs or when a software synchronization is forced by the SWFSYNC bit, or when a digital compare sync event occurs.
1-0	CTRMODE	R/W	3h	Counter Mode. The time-base counter mode is normally configured once and not changed during normal operation. If you change the mode of the counter, the change takes effect at the next TBCLK edge and the current counter value shall increment or decrement from the value before the mode change. These bits set the time-base counter mode of operation as follows: 0h = Up-count mode 1h = Down-count mode 2h = Up-down-count mode 3h = Stop-freeze counter operation (default on reset)

19.4.2 TBSTS Register (Offset = 2h) [reset = 1h]

TBSTS is shown in [Figure 19-64](#) and described in [Table 19-25](#).

Return to [Summary Table](#).

Figure 19-64. TBSTS Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED					CTRMX	SYNCl	CTDRlR
R-0h					R-0h	R/W-0h	R-1h

Table 19-25. TBSTS Register Field Descriptions

Bit	Field	Type	Reset	Description
15-3	RESERVED	R	0h	Reserved
2	CTRMX	R	0h	Time-Base Counter Max Latched Status Bit. 0h = Reading a 0 indicates the time-base counter never reached its maximum value. Writing a 0 has no effect. 1h = Reading a 1 on this bit indicates that the time-base counter reached the max value 0xFFFF. Writing a 1 to this bit clears the latched event.
1	SYNCl	R/W	0h	Input Synchronization Latched Status Bit. 0h = Writing a 0 has no effect. Reading a 0 indicates no external synchronization event has occurred. 1h = Reading a 1 on this bit indicates that an external synchronization event has occurred (EPWMxSYNCl). Writing a 1 to this bit clears the latched event.
0	CTDRlR	R	1h	Time-Base Counter Direction Status Bit. At reset, the counter is frozen, therefore, this bit has no meaning. To make this bit meaningful, you must first set the appropriate mode via TBCTL[CTRMODE]. 0h = Time-Base Counter is currently counting down. 1h = Time-Base Counter is currently counting up.

19.4.3 TBPHS Register (Offset = 6h) [reset = 0h]

TBPHS is shown in [Figure 19-65](#) and described in [Table 19-26](#).

Return to [Summary Table](#).

Figure 19-65. TBPHS Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBPHS															
R/W-0h															

Table 19-26. TBPHS Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	TBPHS	R/W	0h	These bits set time-base counter phase of the selected ePWM relative to the time-base that is supplying the synchronization input signal. Valid values: 0-FFFFh. If TBCTL[PHSEN] = 0, then the synchronization event is ignored and the time-base counter is not loaded with the phase. If TBCTL[PHSEN] = 1, then the time-base counter (TBCTR) is loaded with the phase (TBPHS) when a synchronization event occurs. The synchronization event is initiated by the input synchronization signal (EPWMxSYNCI) or by a software forced synchronization.

19.4.4 TBCTR Register (Offset = 8h) [reset = 0h]

TBCTR is shown in [Figure 19-66](#) and described in [Table 19-27](#).

Return to [Summary Table](#).

Figure 19-66. TBCTR Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBCTR															
R/W-0h															

Table 19-27. TBCTR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	TBCTR	R/W	0h	Reading these bits gives the current time-base counter value. Valid values: 0-FFFFh. Writing to these bits sets the current time-base counter value. The update happens as soon as the write occurs. The write is NOT synchronized to the time-base clock (TBCLK) and the register is not shadowed.

19.4.5 TBPRD Register (Offset = Ah) [reset = 0h]

TBPRD is shown in [Figure 19-67](#) and described in [Table 19-28](#).

Return to [Summary Table](#).

Figure 19-67. TBPRD Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBPRD															
R/W-0h															

Table 19-28. TBPRD Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	TBPRD	R/W	0h	These bits determine the period of the time-base counter. This sets the PWM frequency. Valid values: 0-FFFFh. Shadowing of this register is enabled and disabled by the TBCTL[PRDLD] bit. By default this register is shadowed. If TBCTL[PRDLD] = 0, then the shadow is enabled and any write or read automatically goes to the shadow register. In this case, the active register is loaded from the shadow register when the time-base counter equals 0. If TBCTL[PRDLD] = 1, then the shadow is disabled and any write or read goes directly to the active register, that is the register actively controlling the hardware. The active and shadow registers share the same memory map address.

19.4.6 CMPCTL Register (Offset = Eh) [reset = 0h]

CMPCTL is shown in [Figure 19-68](#) and described in [Table 19-29](#).

Return to [Summary Table](#).

Figure 19-68. CMPCTL Register

15	14	13	12	11	10	9	8
RESERVED						SHDWBFULL	SHDWAFULL
R-0h						R-0h	R-0h
7	6	5	4	3	2	1	0
RESERVED	SHDWBMODE	RESERVED	SHDWAMODE	LOADBMODE		LOADAMODE	
R-0h	R/W-0h	R-0h	R/W-0h	R/W-0h		R/W-0h	

Table 19-29. CMPCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9	SHDWBFULL	R	0h	Counter-compare B (CMPB) Shadow Register Full Status Flag. This bit self clears once a load-strobe occurs. 0h = CMPB shadow FIFO not full yet. 1h = Indicates the CMPB shadow FIFO is full, a CPU write overwrites current shadow value.
8	SHDWAFULL	R	0h	Counter-compare A (CMPA) Shadow Register Full Status Flag. The flag bit is set when a 32-bit write to CMPA:CMPAHR register or a 16-bit write to CMPAHR register does not affect the flag. This bit self clears once a load-strobe occurs. 0h = CMPA shadow FIFO not full yet. 1h = Indicates the CMPA shadow FIFO is full, a CPU write overwrites the current shadow value.
7	RESERVED	R	0h	Reserved
6	SHDWBMODE	R/W	0h	Counter-compare B (CMPB) Register Operating Mode. 0h = Shadow mode. Operates as a double buffer. All writes via the CPU access the shadow register. 1h = Immediate mode. Only the active compare B register is used. All writes and reads directly access the active register for immediate compare action.
5	RESERVED	R	0h	Reserved
4	SHDWAMODE	R/W	0h	Counter-compare A (CMPA) Register Operating Mode. 0h = Shadow mode. Operates as a double buffer. All writes via the CPU access the shadow register. 1h = Immediate mode. Only the active compare register is used. All writes and reads directly access the active register for immediate compare action.
3-2	LOADBMODE	R/W	0h	Active Counter-Compare B (CMPB) Load From Shadow Select Mode. This bit has no effect in immediate mode (CMPCTL[SHDWBMODE] = 1). 0h = Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000) 1h = Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD) 2h = Load on either CTR = Zero or CTR = PRD 3h = Freeze (no loads possible)

Table 19-29. CMPCTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1-0	LOADAMODE	R/W	0h	Active Counter-Compare A (CMPA) Load From Shadow Select Mode. This bit has no effect in immediate mode (CMPCTL[SHDWAMODE] = 1). 0h = Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000) 1h = Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD) 2h = Load on either CTR = Zero or CTR = PRD 3h = Freeze (no loads possible)

19.4.7 CMPA Register (Offset = 12h) [reset = 0h]

CMPA is shown in [Figure 19-69](#) and described in [Table 19-30](#).

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Figure 19-69. CMPA Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMPA															
R/W-0h															

Table 19-30. CMPA Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	CMPA	R/W	0h	The value in the active CMPA register is continuously compared to the time-base counter (TBCTR). When the values are equal, the counter-compare module generates a "time-base counter equal to counter compare A" event. This event is sent to the action-qualifier where it is qualified and converted it into one or more actions. These actions are applied to either the EPWMxA or the EPWMxB output depending on the configuration of the AQCTLA and AQCTLB registers. The actions that are defined in the AQCTLA and AQCTLB registers include: -Do nothing, the event is ignored. -Clear: Pull the EPWMxA and/or EPWMxB signal low. -Set: Pull the EPWMxA and/or EPWMxB signal high. -Toggle the EPWMxA and/or EPWMxB signal. Shadowing of this register is enabled and disabled by the CMPCTL[SHDWAMODE] bit. By default this register is shadowed. If CMPCTL[SHDWAMODE] = 0, then the shadow is enabled and any write or read automatically goes to the shadow register. In this case, the CMPCTL[LOADAMODE] bit field determines which event loads the active register from the shadow register. Before a write, the CMPCTL[SHDWAFULL] bit is read to determine if the shadow register is currently full. If CMPCTL[SHDWAMODE] = 1, then the shadow register is disabled and any write or read goes directly to the active register, that is the register actively controlling the hardware. In either mode, the active and shadow registers share the same memory map address.

19.4.8 CMPB Register (Offset = 14h) [reset = 0h]

CMPB is shown in [Figure 19-70](#) and described in [Table 19-31](#).

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Figure 19-70. CMPB Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMPB															
R/W-0h															

Table 19-31. CMPB Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	CMPB	R/W	0h	The value in the active CMPB register is continuously compared to the time-base counter (TBCTR). When the values are equal, the counter-compare module generates a "time-base counter equal to counter compare B" event. This event is sent to the action-qualifier where it is qualified and converted it into one or more actions. These actions are applied to either the EPWMxA or the EPWMxB output depending on the configuration of the AQCTLA and AQCTLB registers. The actions that are defined in the AQCTLA and AQCTLB registers include: -Do nothing, event is ignored. -Clear: Pull the EPWMxA and/or EPWMxB signal low. -Set: Pull the EPWMxA and/or EPWMxB signal high. -Toggle the EPWMxA and/or EPWMxB signal. Shadowing of this register is enabled and disabled by the CMPCTL[SHDWBMODE] bit. By default this register is shadowed. If CMPCTL[SHDWBMODE] = 0, then the shadow is enabled and any write or read automatically goes to the shadow register. In this case, the CMPCTL[LOADBMODE] bit field determines which event loads the active register from the shadow register. Before a write, the CMPCTL[SHDWBFULL] bit is read to determine if the shadow register is currently full. If CMPCTL[SHDWBMODE] = 1, then the shadow register is disabled and any write or read goes directly to the active register, that is the register actively controlling the hardware. In either mode, the active and shadow registers share the same memory map address.

19.4.9 AQCTLA Register (Offset = 16h) [reset = 0h]

AQCTLA is shown in [Figure 19-71](#) and described in [Table 19-32](#).

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Figure 19-71. AQCTLA Register

15	14	13	12	11	10	9	8
RESERVED				CBD		CBU	
R-0h				R/W-0h		R/W-0h	
7	6	5	4	3	2	1	0
CAD		CAU		PRD		ZRO	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	

Table 19-32. AQCTLA Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-10	CBD	R/W	0h	Action when the time-base counter equals the active CMPB register and the counter is decrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxA output low. 2h = Set: force EPWMxA output high. 3h = Toggle EPWMxA output: low output signal is forced high and a high signal is forced low.
9-8	CBU	R/W	0h	Action when the counter equals the active CMPB register and the counter is incrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxA output low. 2h = Set: force EPWMxA output high. 3h = Toggle EPWMxA output: low output signal is forced high and a high signal is forced low.
7-6	CAD	R/W	0h	Action when the counter equals the active CMPA register and the counter is decrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxA output low. 2h = Set: force EPWMxA output high. 3h = Toggle EPWMxA output: low output signal is forced high and a high signal is forced low.
5-4	CAU	R/W	0h	Action when the counter equals the active CMPA register and the counter is incrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxA output low. 2h = Set: force EPWMxA output high. 3h = Toggle EPWMxA output: low output signal is forced high and a high signal is forced low.
3-2	PRD	R/W	0h	Action when the counter equals the period. By definition, in count up-down mode when the counter equals period the direction is defined as 0 or counting down. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxA output low. 2h = Set: force EPWMxA output high. 3h = Toggle EPWMxA output: low output signal is forced high and a high signal is forced low.

Table 19-32. AQCTLA Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1-0	ZRO	R/W	0h	Action when counter equals zero. By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxA output low. 2h = Set: force EPWMxA output high. 3h = Toggle EPWMxA output: low output signal is forced high and a high signal is forced low.

19.4.10 AQCTLB Register (Offset = 18h) [reset = 0h]

AQCTLB is shown in [Figure 19-72](#) and described in [Table 19-33](#).

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Figure 19-72. AQCTLB Register

15	14	13	12	11	10	9	8
RESERVED				CBD		CBU	
R-0h				R/W-0h		R/W-0h	
7	6	5	4	3	2	1	0
CAD		CAU		PRD		ZRO	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	

Table 19-33. AQCTLB Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-10	CBD	R/W	0h	Action when the counter equals the active CMPB register and the counter is decrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxB output low. 2h = Set: force EPWMxB output high. 3h = Toggle EPWMxB output: low output signal is forced high and a high signal is forced low.
9-8	CBU	R/W	0h	Action when the counter equals the active CMPB register and the counter is incrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxB output low. 2h = Set: force EPWMxB output high. 3h = Toggle EPWMxB output: low output signal is forced high and a high signal is forced low.
7-6	CAD	R/W	0h	Action when the counter equals the active CMPA register and the counter is decrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxB output low. 2h = Set: force EPWMxB output high. 3h = Toggle EPWMxB output: low output signal is forced high and a high signal is forced low.
5-4	CAU	R/W	0h	Action when the counter equals the active CMPA register and the counter is incrementing. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxB output low. 2h = Set: force EPWMxB output high. 3h = Toggle EPWMxB output: low output signal is forced high and a high signal is forced low.
3-2	PRD	R/W	0h	Action when the counter equals the period. Note: By definition, in count up-down mode when the counter equals period the direction is defined as 0 or counting down. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxB output low. 2h = Set: force EPWMxB output high. 3h = Toggle EPWMxB output: low output signal is forced high and a high signal is forced low.

Table 19-33. AQCTLB Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1-0	ZRO	R/W	0h	Action when counter equals zero. Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up. 0h = Do nothing (action is disabled). 1h = Clear: force EPWMxB output low. 2h = Set: force EPWMxB output high. 3h = Toggle EPWMxB output: low output signal is forced high and a high signal is forced low.

19.4.11 AQSFR Register (Offset = 1Ah) [reset = 0h]

AQSFR is shown in [Figure 19-73](#) and described in [Table 19-34](#).

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Figure 19-73. AQSFR Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RLDCSF		OTSFB	ACTSFB		OTSFA	ACTSFA	
R/W-0h		R/W-0h	R/W-0h		R/W-0h	R/W-0h	

Table 19-34. AQSFR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Reserved
7-6	RLDCSF	R/W	0h	AQCSFRC Active Register Reload From Shadow Options. 0h = Load on event counter equals zero. 1h = Load on event counter equals period. 2h = Load on event counter equals zero or counter equals period. 3h = Load immediately (the active register is directly accessed by the CPU and is not loaded from the shadow register).
5	OTSFB	R/W	0h	One-Time Software Forced Event on Output B. 0h = Writing a 0 has no effect. Always reads back a 0. This bit is auto cleared once a write to this register is complete, that is, a forced event is initiated.) This is a one-shot forced event. It is overridden by another subsequent event on output B. 1h = Initiates a single software forced event.
4-3	ACTSFB	R/W	0h	Action when One-Time Software Force B Is Invoked. 0h = Does nothing (action is disabled) 1h = Clear (low) 2h = Set (high) 3h = Toggle (low to high, high to low). This action is not qualified by counter direction (CNT_dir).
2	OTSFA	R/W	0h	One-Time Software Forced Event on Output A. 0h = Writing a 0 has no effect. Always reads back a 0. This bit is auto cleared once a write to this register is complete (that is, a forced event is initiated). 1h = Initiates a single software forced event.
1-0	ACTSFA	R/W	0h	Action When One-Time Software Force A Is Invoked. 0h = Does nothing (action is disabled) 1h = Clear (low) 2h = Set (high) 3h = Toggle (low to high, high to low). This action is not qualified by counter direction (CNT_dir).

19.4.12 AQCSFRC Register (Offset = 1Ch) [reset = 0h]

AQCSFRC is shown in [Figure 19-74](#) and described in [Table 19-35](#).

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Figure 19-74. AQCSFRC Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				CSFB		CSFA	
R-0h				R/W-0h		R/W-0h	

Table 19-35. AQCSFRC Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3-2	CSFB	R/W	0h	Continuous Software Force on Output B. In immediate mode, a continuous force takes effect on the next TBCLK edge. In shadow mode, a continuous force takes effect on the next TBCLK edge after a shadow load into the active register. To configure shadow mode, use AQSFRC[RLDCSF]. 0h = Forcing is disabled and has no effect. 1h = Forces a continuous low on output B. 2h = Forces a continuous high on output B. 3h = Software forcing is disabled and has no effect.
1-0	CSFA	R/W	0h	Continuous Software Force on Output A. In immediate mode, a continuous force takes effect on the next TBCLK edge. In shadow mode, a continuous force takes effect on the next TBCLK edge after a shadow load into the active register. 0h = Forcing is disabled and has no effect. 1h = Forces a continuous low on output A . 2h = Forces a continuous high on output A. 3h = Software forcing is disabled and has no effect.

19.4.13 DBCTL Register (Offset = 1Eh) [reset = 0h]

DBCTL is shown in [Figure 19-75](#) and described in [Table 19-36](#).

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Figure 19-75. DBCTL Register

15	14	13	12	11	10	9	8
HALFCYCLE	RESERVED						
R/W-0h	R-0h						
7	6	5	4	3	2	1	0
RESERVED		IN_MODE		POLSEL		OUT_MODE	
R-0h		R/W-0h		R/W-0h		R/W-0h	

Table 19-36. DBCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	HALFCYCLE	R/W	0h	Half Cycle Clocking Enable Bit. 0h = Full cycle clocking is enabled. The dead-band counters are clocked at the TBCLK rate. 1h = Half cycle clocking is enabled. The dead-band counters are clocked at TBCLK x 2.
14-6	RESERVED	R	0h	Reserved
5-4	IN_MODE	R/W	0h	Dead Band Input Mode Control. This bit selects the input source to the falling-edge and rising-edge delay. To produce classical dead-band waveforms, the default is EPWMxA In. 0h = EPWMxA In (from the action-qualifier) is the source for both falling-edge and rising-edge delay. 1h = EPWMxB In (from the action-qualifier) is the source for rising-edge delayed signal. EPWMxA In (from the action-qualifier) is the source for falling-edge delayed signal. 2h = EPWMxA In (from the action-qualifier) is the source for rising-edge delayed signal. EPWMxB In (from the action-qualifier) is the source for falling-edge delayed signal. 3h = EPWMxB In (from the action-qualifier) is the source for both rising-edge delay and falling-edge delayed signal.
3-2	POLSEL	R/W	0h	Polarity Select Control. This bit selectively inverts one of the delayed signals before it is sent out of the dead-band submodule. The following descriptions correspond to classical upper and lower switch control, as found in one leg of a digital motor control inverter. These assume that DBCTL[OUT_MODE] = 1,1 and DBCTL[IN_MODE] = 0,0. Other enhanced modes are also possible, but not regarded as typical usage modes. 0h = Active high (AH) mode. Neither EPWMxA nor EPWMxB is inverted (default). 1h = Active low complementary (ALC) mode. EPWMxA is inverted. 2h = Active high complementary (AHC). EPWMxB is inverted. 3h = Active low (AL) mode. Both EPWMxA and EPWMxB are inverted.

Table 19-36. DBCTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1-0	OUT_MODE	R/W	0h	Dead-band Output Mode Control. This bit enables or bypasses the dead-band generation for the falling-edge and rising-edge delay. 0h = Dead-band generation is bypassed for both output signals. In this mode, both the EPWMxA and EPWMxB output signals from the action-qualifier are passed directly to the PWM-chopper submodule. In this mode, the POLSEL and IN_MODE bits have no effect. 1h = Disable rising-edge delay. The EPWMxA signal from the action-qualifier is passed straight through to the EPWMxA input of the PWM-chopper submodule. The falling-edge delayed signal is seen on output EPWMxB. The input signal for the delay is determined by DBCTL[IN_MODE]. 2h = The rising-edge delayed signal is seen on output EPWMxA. The input signal for the delay is determined by DBCTL[IN_MODE]. Disable falling-edge delay. The EPWMxB signal from the action-qualifier is passed straight through to the EPWMxB input of the PWM-chopper submodule. 3h = Dead-band is fully enabled for both rising-edge delay on output EPWMxA and falling-edge delay on output EPWMxB. The input signal for the delay is determined by DBCTL[IN_MODE].

19.4.14 DBRED Register (Offset = 20h) [reset = 0h]

DBRED is shown in [Figure 19-76](#) and described in [Table 19-37](#).

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Figure 19-76. DBRED Register

15	14	13	12	11	10	9	8
RESERVED						DEL	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
DEL							
R/W-0h							

Table 19-37. DBRED Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-0	DEL	R/W	0h	Rising Edge Delay Count. 10-bit counter.

19.4.15 DBFED Register (Offset = 22h) [reset = 0h]

DBFED is shown in [Figure 19-77](#) and described in [Table 19-38](#).

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Figure 19-77. DBFED Register

15	14	13	12	11	10	9	8
RESERVED						DEL	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
DEL							
R/W-0h							

Table 19-38. DBFED Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9-0	DEL	R/W	0h	Falling Edge Delay Count. 10-bit counter.

19.4.16 TZSEL Register (Offset = 24h) [reset = 0h]

TZSEL is shown in Figure 19-78 and described in Table 19-39.

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One-Shot (OSHT) Trip-zone enable/disable (bits 15-8). When any of the enabled pins go low, a one-shot trip event occurs for this ePWM module. When the event occurs, the action defined in the TZCTL register is taken on the EPWMxA and EPWMxB outputs. The one-shot trip condition remains latched until the user clears the condition via the TZCLR register. Cycle-by-Cycle (CBC) Trip-zone enable/disable (bits 7-0). When any of the enabled pins go low, a cycle-by-cycle trip event occurs for this ePWM module. When the event occurs, the action defined in the TZCTL register is taken on the EPWMxA and EPWMxB outputs. A cycle-by-cycle trip condition is automatically cleared when the time-base counter reaches zero.

Figure 19-78. TZSEL Register

15	14	13	12	11	10	9	8
DCBEVT1	DCAEVT1	OSHT6	OSHT5	OSHT4	OSHT3	OSHT2	OSHT1
R-0h	R-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
DCBEVT2	DCAEVT2	CBC6	CBC5	CBC4	CBC3	CBC2	CBC1
R-0h	R-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 19-39. TZSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	DCBEVT1	R	0h	Digital Compare Output B Event 1 Select. 0h = Disable DCBEVT1 as one-shot-trip source for this ePWM module. 1h = Enable DCBEVT1 as one-shot-trip source for this ePWM module.
14	DCAEVT1	R	0h	Digital Compare Output A Event 1 Select. 0h = Disable DCAEVT1 as one-shot-trip source for this ePWM module. 1h = Enable DCAEVT1 as one-shot-trip source for this ePWM module.
13	OSHT6	R/W	0h	Trip-zone 6 (/TZ6) Select. 0h = Disable /TZ6 as a one-shot trip source for this ePWM module. 1h = Enable /TZ6 as a one-shot trip source for this ePWM module.
12	OSHT5	R/W	0h	Trip-zone 5 (/TZ5) Select. 0h = Disable /TZ5 as a one-shot trip source for this ePWM module. 1h = Enable /TZ5 as a one-shot trip source for this ePWM module.
11	OSHT4	R/W	0h	Trip-zone 4 (/TZ4) Select. 0h = Disable /TZ4 as a one-shot trip source for this ePWM module. 1h = Enable /TZ4 as a one-shot trip source for this ePWM module.
10	OSHT3	R/W	0h	Trip-zone 3 (/TZ3) Select. 0h = Disable /TZ3 as a one-shot trip source for this ePWM module. 1h = Enable /TZ3 as a one-shot trip source for this ePWM module.
9	OSHT2	R/W	0h	Trip-zone 2 (/TZ2) Select. 0h = Disable /TZ2 as a one-shot trip source for this ePWM module. 1h = Enable /TZ2 as a one-shot trip source for this ePWM module.
8	OSHT1	R/W	0h	Trip-zone 1 (/TZ1) Select. 0h = Disable /TZ1 as a one-shot trip source for this ePWM module. 1h = Enable /TZ1 as a one-shot trip source for this ePWM module.
7	DCBEVT2	R	0h	Digital Compare Output B Event 2 Select. 0h = Disable DCBEVT2 as a CBC trip source for this ePWM module. 1h = Enable DCBEVT2 as a CBC trip source for this ePWM module.

Table 19-39. TZSEL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
6	DCAEVT2	R	0h	Digital Compare Output A Event 2 Select. 0h = Disable DCAEVT2 as a CBC trip source for this ePWM module. 1h = Enable DCAEVT2 as a CBC trip source for this ePWM module.
5	CBC6	R/W	0h	Trip-zone 6 (/TZ6) Select. 0h = Disable /TZ6 as a CBC trip source for this ePWM module. 1h = Enable /TZ6 as a CBC trip source for this ePWM module.
4	CBC5	R/W	0h	Trip-zone 5 (/TZ5) Select. 0h = Disable /TZ5 as a CBC trip source for this ePWM module. 1h = Enable /TZ5 as a CBC trip source for this ePWM module.
3	CBC4	R/W	0h	Trip-zone 4 (/TZ4) Select. 0h = Disable /TZ4 as a CBC trip source for this ePWM module. 1h = Enable /TZ4 as a CBC trip source for this ePWM module.
2	CBC3	R/W	0h	Trip-zone 3 (/TZ3) Select. 0h = Disable /TZ3 as a CBC trip source for this ePWM module. 1h = Enable /TZ3 as a CBC trip source for this ePWM module.
1	CBC2	R/W	0h	Trip-zone 2 (/TZ2) Select. 0h = Disable /TZ2 as a CBC trip source for this ePWM module. 1h = Enable /TZ2 as a CBC trip source for this ePWM module.
0	CBC1	R/W	0h	Trip-zone 1 (/TZ1) Select. 0h = Disable /TZ1 as a CBC trip source for this ePWM module. 1h = Enable /TZ1 as a CBC trip source for this ePWM module.

19.4.17 TZDCSEL Register (Offset = 26h) [reset = 0h]

TZDCSEL is shown in [Figure 19-79](#) and described in [Table 19-40](#).

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Figure 19-79. TZDCSEL Register

15	14	13	12	11	10	9	8
RESERVED				DCBEVT2		DCBEVT1	
R-0h				R/W-0h		R/W-0h	
7	6	5	4	3	2	1	0
DCBEVT1		DCAEVT2			DCAEVT1		
R/W-0h		R/W-0h			R/W-0h		

Table 19-40. TZDCSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-9	DCBEVT2	R/W	0h	Digital Compare Output B Event 2 Selection. 0h = Event is disabled. 1h = DCBH = low, DCBL = don't care 2h = DCBH = high, DCBL = don't care 3h = DCBL = low, DCBH = don't care 4h = DCBL = high, DCBH = don't care 5h = DCBL = high, DCBH = low 6h = Reserved 7h = Reserved
8-6	DCBEVT1	R/W	0h	Digital Compare Output B Event 1 Selection. 0h = Event is disabled. 1h = DCBH = low, DCBL = don't care 2h = DCBH = high, DCBL = don't care 3h = DCBL = low, DCBH = don't care 4h = DCBL = high, DCBH = don't care 5h = DCBL = high, DCBH = low 6h = Reserved 7h = Reserved
5-3	DCAEVT2	R/W	0h	Digital Compare Output A Event 2 Selection. 0h = Event is disabled. 1h = DCAH = low, DCAL = don't care 2h = DCAH = high, DCAL = don't care 3h = DCAL = low, DCAH = don't care 4h = DCAL = high, DCAH = don't care 5h = DCAL = high, DCAH = low 6h = Reserved 7h = Reserved
2-0	DCAEVT1	R/W	0h	Digital Compare Output A Event 1 Selection. 0h = Event is disabled. 1h = DCAH = low, DCAL = don't care 2h = DCAH = high, DCAL = don't care 3h = DCAL = low, DCAH = don't care 4h = DCAL = high, DCAH = don't care 5h = DCAL = high, DCAH = low 6h = Reserved 7h = Reserved

19.4.18 TZCTL Register (Offset = 28h) [reset = 0h]

TZCTL is shown in Figure 19-80 and described in Table 19-41.

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Figure 19-80. TZCTL Register

15	14	13	12	11	10	9	8
RESERVED				DCBEVT2		DCBEVT1	
R-0h				R/W-0h		R/W-0h	
7	6	5	4	3	2	1	0
DCAEVT2		DCAEVT1		TZB		TZA	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	

Table 19-41. TZCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Reserved
11-10	DCBEVT2	R/W	0h	Digital Compare Output B Event 2 Action On EPWMxB. 0h = High-impedance (EPWMxB = High-impedance state). 1h = Force EPWMxB to a high state. 2h = Force EPWMxB to a low state. 3h = Do nothing, trip action is disabled.
9-8	DCBEVT1	R/W	0h	Digital Compare Output B Event 1 Action On EPWMxB. 0h = High-impedance (EPWMxB = High-impedance state). 1h = Force EPWMxB to a high state. 2h = Force EPWMxB to a low state. 3h = Do nothing, trip action is disabled.
7-6	DCAEVT2	R/W	0h	Digital Compare Output A Event 2 Action On EPWMxA. 0h = High-impedance (EPWMxA = High-impedance state). 1h = Force EPWMxA to a high state. 2h = Force EPWMxA to a low state. 3h = Do nothing, trip action is disabled.
5-4	DCAEVT1	R/W	0h	Digital Compare Output A Event 1 Action On EPWMxA. 0h = High-impedance (EPWMxA = High-impedance state). 1h = Force EPWMxA to a high state. 2h = Force EPWMxA to a low state. 3h = Do nothing, trip action is disabled.
3-2	TZB	R/W	0h	When a trip event occurs the following action is taken on output EPWMxB. Which trip-zone pins can cause an event is defined in the TZSEL register. 0h = High-impedance (EPWMxB = High-impedance state). 1h = Force EPWMxB to a high state. 2h = Force EPWMxB to a low state. 3h = Do nothing, no action is taken on EPWMxB.
1-0	TZA	R/W	0h	When a trip event occurs the following action is taken on output EPWMxA. Which trip-zone pins can cause an event is defined in the TZSEL register. 0h = High-impedance (EPWMxA = High-impedance state). 1h = Force EPWMxA to a high state. 2h = Force EPWMxA to a low state. 3h = Do nothing, no action is taken on EPWMxA.

19.4.19 TZEINT Register (Offset = 2Ah) [reset = 0h]

TZEINT is shown in [Figure 19-81](#) and described in [Table 19-42](#).

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Figure 19-81. TZEINT Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED	DCBEVT2	DCBEVT1	DCAEVT2	DCAEVT1	OST	CBC	RESERVED
R-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R-0h

Table 19-42. TZEINT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-7	RESERVED	R	0h	Reserved
6	DCBEVT2	R/W	0h	Digital Comparator Output B Event 2 Interrupt Enable. 0h = Disabled 1h = Enabled
5	DCBEVT1	R/W	0h	Digital Comparator Output B Event 1 Interrupt Enable. 0h = Disabled 1h = Enabled
4	DCAEVT2	R/W	0h	Digital Comparator Output A Event 2 Interrupt Enable. 0h = Disabled 1h = Enabled
3	DCAEVT1	R/W	0h	Digital Comparator Output A Event 1 Interrupt Enable. 0h = Disabled 1h = Enabled
2	OST	R/W	0h	Trip-zone One-Shot Interrupt Enable. 0h = Disable one-shot interrupt generation. 1h = Enable Interrupt generation. A one-shot trip event causes a EPWMx_TZINT VIM interrupt.
1	CBC	R/W	0h	Trip-zone Cycle-by-Cycle Interrupt Enable. 0h = Disable cycle-by-cycle interrupt generation. 1h = Enable interrupt generation. A cycle-by-cycle trip event causes an EPWMx_TZINT VIM interrupt.
0	RESERVED	R	0h	Reserved

19.4.20 TZFLG Register (Offset = 2Ch) [reset = 0h]

TZFLG is shown in [Figure 19-82](#) and described in [Table 19-43](#).

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Figure 19-82. TZFLG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED	DCBEVT2	DCBEVT1	DCAEVT2	DCAEVT1	OST	CBC	INT
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 19-43. TZFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-7	RESERVED	R	0h	Reserved
6	DCBEVT2	R	0h	Latched Status Flag for Digital Compare Output B Event 2. 0h = No trip event has occurred on DCBEVT2. 1h = A trip event has occurred for the event defined for DCBEVT2.
5	DCBEVT1	R	0h	Latched Status Flag for Digital Compare Output B Event 1. 0h = No trip event has occurred on DCBEVT1. 1h = A trip event has occurred for the event defined for DCBEVT1.
4	DCAEVT2	R	0h	Latched Status Flag for Digital Compare Output A Event 2. 0h = No trip event has occurred on DCAEVT2. 1h = A trip event has occurred for the event defined for DCAEVT2.
3	DCAEVT1	R	0h	Latched Status Flag for Digital Compare Output A Event 1. 0h = No trip event has occurred on DCAEVT1. 1h = A trip event has occurred for the event defined for DCAEVT1.
2	OST	R	0h	Latched Status Flag for A One-Shot Trip Event. This bit is cleared by writing the appropriate value to the TZCLR register. 0h = No one-shot trip event has occurred. 1h = A trip event has occurred on a pin selected as a one-shot trip source.
1	CBC	R	0h	Latched Status Flag for Cycle-By-Cycle Trip Event. This bit is cleared by writing the appropriate value to the TZCLR register. 0h = No cycle-by-cycle trip event has occurred. 1h = A trip event has occurred on a signal selected as a cycle-by-cycle trip source. The TZFLG.CBC bit remains set until it is manually cleared by the user. If the cycle-by-cycle trip event is still present when the CBC bit is cleared, then CBC is immediately set again. The specified condition on the signal is automatically cleared when the ePWM time-base counter reaches zero (TBCTR = 0x0000), if the trip condition is no longer present. The condition on the signal is only cleared when the TBCTR = 0x0000, no matter where in the cycle the CBC flag is cleared.
0	INT	R	0h	Latched Trip Interrupt Status Flag. No further EPWMx_TZINT VIM interrupts are generated until this flag is cleared. If the interrupt flag is cleared when either CBC or OST is set, then another interrupt pulse is generated. Clearing all flag bits prevents further interrupts. This bit is cleared by writing the appropriate value to the TZCLR register. 0h = No interrupt is generated. 1h = An EPWMx_TZINT VIM interrupt is generated because of a trip condition.

19.4.21 TZCLR Register (Offset = 2Eh) [reset = 0h]

TZCLR is shown in [Figure 19-83](#) and described in [Table 19-44](#).

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Figure 19-83. TZCLR Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED	DCBEVT2	DCBEVT1	DCAEVT2	DCAEVT1	OST	CBC	INT
R-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W-0h	R/W-0h	R/W-0h

Table 19-44. TZCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-7	RESERVED	R	0h	Reserved
6	DCBEVT2	R/W1C	0h	Clear Flag for Digital Compare Output B Event 2. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 clears the DCBEVT2 event trip condition.
5	DCBEVT1	R/W1C	0h	Clear Flag for Digital Compare Output B Event 1. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 clears the DCBEVT1 event trip condition.
4	DCAEVT2	R/W1C	0h	Clear Flag for Digital Compare Output A Event 2. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 clears the DCAEVT2 event trip condition.
3	DCAEVT1	R/W1C	0h	Clear Flag for Digital Compare Output A Event 1. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 clears the DCAEVT1 event trip condition.
2	OST	R/W	0h	Clear Flag for One-Shot Trip (OST) Latch. 0h = Has no effect. Always reads back a 0. 1h = Writing 1 clears the Trip (set) condition.
1	CBC	R/W	0h	Clear Flag for Cycle-By-Cycle (CBC) Trip Latch. 0h = Has no effect. Always reads back a 0. 1h = Writing 1 clears the Trip (set) condition.
0	INT	R/W	0h	Global Interrupt Clear Flag. NOTE: No further EPWMx_TZINT VIM interrupts are generated until the flag is cleared. If the TZFLG.INT bit is cleared and any of the other flag bits are set, then another interrupt pulse is generated. Clearing all flag bits prevents further interrupts. 0h = Has no effect. Always reads back a 0. 1h = Writing 1 clears the trip-interrupt flag for this ePWM module, TZFLG.INT.

19.4.22 TZFRC Register (Offset = 30h) [reset = 0h]

TZFRC is shown in [Figure 19-84](#) and described in [Table 19-45](#).

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Figure 19-84. TZFRC Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED	DCBEVT2	DCBEVT1	DCAEVT2	DCAEVT1	OST	CBC	RESERVED
R-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R-0h

Table 19-45. TZFRC Register Field Descriptions

Bit	Field	Type	Reset	Description
15-7	RESERVED	R	0h	Reserved
6	DCBEVT2	R/W	0h	Force Flag for Digital Compare Output B Event 2. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 forces the DCBEVT2 event trip condition and sets the TZFLG[DCBEVT2] bit.
5	DCBEVT1	R/W	0h	Force Flag for Digital Compare Output B Event 1. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 forces the DCBEVT1 event trip condition and sets the TZFLG[DCBEVT1] bit.
4	DCAEVT2	R/W	0h	Force Flag for Digital Compare Output A Event 2. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 forces the DCAEVT2 event trip condition and sets the TZFLG[DCAEVT2] bit.
3	DCAEVT1	R/W	0h	Force Flag for Digital Compare Output A Event 1. 0h = Writing 0 has no effect. This bit always reads back 0. 1h = Writing 1 forces the DCAEVT1 event trip condition and sets the TZFLG[DCAEVT1] bit.
2	OST	R/W	0h	Force a One-Shot Trip Event via Software. 0h = Writing of 0 is ignored. Always reads back a 0. 1h = Writing 1 forces a one-shot trip event and sets the TZFLG[OST] bit.
1	CBC	R/W	0h	Force a Cycle-by-Cycle Trip Event via Software. 0h = Writing of 0 is ignored. Always reads back a 0. 1h = Writing 1 forces a cycle-by-cycle trip event and sets the TZFLG[CBC] bit.
0	RESERVED	R	0h	Reserved

19.4.23 ETSEL Register (Offset = 32h) [reset = 0h]

ETSEL is shown in [Figure 19-85](#) and described in [Table 19-46](#).

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Figure 19-85. ETSEL Register

15	14	13	12	11	10	9	8
SOCBEN	SOCBSEL			SOCAEN	SOCASEL		
R/W-0h	R/W-0h			R/W-0h	R/W-0h		
7	6	5	4	3	2	1	0
RESERVED				INTEN	INTSEL		
R-0h				R/W-0h	R/W-0h		

Table 19-46. ETSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	SOCBEN	R/W	0h	Enable the ADC Start of Conversion B (EPWMxSOCB) Pulse. 0h = Disable EPWMxSOCB. 1h = Enable EPWMxSOCB pulse.
14-12	SOCBSEL	R/W	0h	EPWMxSOCB Selection Options. These bits determine when a EPWMxSOCB pulse is generated. 0h = Enable DCBEVT1.soc event. 1h = Enable event time-base counter equal to zero. (TBCTR = 0x0000). 2h = Enable event time-base counter equal to period (TBCTR = TBPRD). 3h = Enable event time-base counter equal to zero or period (TBCTR = 0x0000 or TBCTR = TBPRD). This mode is useful in up-down count mode. 4h = Enable event time-base counter equal to CMPA when the timer is incrementing. 5h = Enable event time-base counter equal to CMPA when the timer is decrementing. 6h = Enable event: time-base counter equal to CMPB when the timer is incrementing. 7h = Enable event: time-base counter equal to CMPB when the timer is decrementing.
11	SOCAEN	R/W	0h	Enable the ADC Start of Conversion A (EPWMxSOCA) Pulse. 0h = Disable EPWMxSOCA. 1h = Enable EPWMxSOCA pulse.
10-8	SOCASEL	R/W	0h	EPWMxSOCA Selection Options. These bits determine when a EPWMxSOCA pulse is generated. 0h = Enable DCAEVT1.soc event. 1h = Enable event time-base counter equal to zero. (TBCTR = 0x0000). 2h = Enable event time-base counter equal to period (TBCTR = TBPRD). 3h = Enable event time-base counter equal to zero or period (TBCTR = 0x0000 or TBCTR = TBPRD). This mode is useful in up-down count mode. 4h = Enable event time-base counter equal to CMPA when the timer is incrementing. 5h = Enable event time-base counter equal to CMPA when the timer is decrementing. 6h = Enable event: time-base counter equal to CMPB when the timer is incrementing. 7h = Enable event: time-base counter equal to CMPB when the timer is decrementing.
7-4	RESERVED	R	0h	Reserved

Table 19-46. ETSEL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	INTEN	R/W	0h	Enable ePWM Interrupt (EPWMx_INT) Generation. 0h = Disable EPWMx_INT generation. 1h = Enable EPWMx_INT generation.
2-0	INTSEL	R/W	0h	ePWM Interrupt (EPWMx_INT) Selection Options. 0h = Reserved 1h = Enable event time-base counter equal to zero. (TBCTR = 0x0000). 2h = Enable event time-base counter equal to period (TBCTR = TBPRD). 3h = Enable event time-base counter equal to zero or period (TBCTR = 0x0000 or TBCTR = TBPRD). This mode is useful in up-down count mode. 4h = Enable event time-base counter equal to CMPA when the timer is incrementing. 5h = Enable event time-base counter equal to CMPA when the timer is decrementing. 6h = Enable event: time-base counter equal to CMPB when the timer is incrementing. 7h = Enable event: time-base counter equal to CMPB when the timer is decrementing.

19.4.24 ETPS Register (Offset = 34h) [reset = 0h]

ETPS is shown in [Figure 19-86](#) and described in [Table 19-47](#).

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Figure 19-86. ETPS Register

15	14	13	12	11	10	9	8
SOCBCNT		SOCBPRD		SOCACNT		SOCAPRD	
R-0h		R/W-0h		R-0h		R/W-0h	
7	6	5	4	3	2	1	0
RESERVED				INTCNT		INTPRD	
R-0h				R-0h		R/W-0h	

Table 19-47. ETPS Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	SOCBCNT	R	0h	ePWM ADC Start-of-Conversion B Event (EPWMxSOCB) Counter Register. These bits indicate how many selected ETSEL[SOCBSEL] events have occurred. 0h = No events have occurred. 1h = 1 event has occurred. 2h = 2 events have occurred. 3h = 3 events have occurred.
13-12	SOCBPRD	R/W	0h	ePWM ADC Start-of-Conversion B Event (EPWMxSOCB) Period Select. These bits determine how many selected ETSEL[SOCBSEL] events need to occur before an EPWMxSOCB pulse is generated. To be generated, the pulse must be enabled (ETSEL[SOCBEN] = 1). The SOCB pulse is generated even if the status flag is set from a previous start of conversion (ETFLG[SOCB] = 1). Once the SOCB pulse is generated, the ETPS[SOCBCNT] bits is automatically cleared. 0h = Disable the SOCB event counter. No EPWMxSOCB pulse is generated. 1h = Generate the EPWMxSOCB pulse on the first event: ETPS[SOCBCNT] = 0,1. 2h = Generate the EPWMxSOCB pulse on the second event: ETPS[SOCBCNT] = 1,0. 3h = Generate the EPWMxSOCB pulse on the third event: ETPS[SOCBCNT] = 1,1.
11-10	SOCACNT	R	0h	ePWM ADC Start-of-Conversion A Event (EPWMxSOCA) Counter Register. These bits indicate how many selected ETSEL[SOCASEL] events have occurred. 0h = No events have occurred. 1h = 1 event has occurred. 2h = 2 events have occurred. 3h = 3 events have occurred.

Table 19-47. ETPS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
9-8	SOCAPRD	R/W	0h	ePWM ADC Start-of-Conversion A Event (EPWMxSOCA) Period Select. These bits determine how many selected ETSSEL[SOCASEL] events need to occur before an EPWMxSOCA pulse is generated. To be generated, the pulse must be enabled (ETSSEL[SOCASEN] = 1). The SOCA pulse is generated even if the status flag is set from a previous start of conversion (ETFLG[SOCA] = 1). Once the SOCA pulse is generated, the ETPS[SOCACNT] bits are automatically cleared. 0h = Disable the SOCA event counter. No EPWMxSOCA pulse is generated. 1h = Generate the EPWMxSOCA pulse on the first event: ETPS[SOCACNT] = 0,1. 2h = Generate the EPWMxSOCA pulse on the second event: ETPS[SOCACNT] = 1,0. 3h = Generate the EPWMxSOCA pulse on the third event: ETPS[SOCACNT] = 1,1.
7-4	RESERVED	R	0h	Reserved
3-2	INTCNT	R	0h	ePWM Interrupt Event (EPWMx_INT) Counter Register. These bits indicate how many selected ETSSEL[INTSEL] events have occurred. These bits are automatically cleared when an interrupt pulse is generated. If interrupts are disabled, ETSSEL[INT] = 0 or the interrupt flag is set, ETFLG[INT] = 1, the counter stops counting events when it reaches the period value ETPS[INTCNT] = ETPS[INTPRD]. 0h = No events have occurred. 1h = 1 event has occurred. 2h = 2 events have occurred. 3h = 3 events have occurred.
1-0	INTPRD	R/W	0h	ePWM Interrupt (EPWMx_INT) Period Select. These bits determine how many selected ETSSEL[INTSEL] events need to occur before an interrupt is generated. To be generated, the interrupt must be enabled (ETSSEL[INT] = 1). If the interrupt status flag is set from a previous interrupt (ETFLG[INT] = 1) then no interrupt is generated until the flag is cleared via the ETCLR[INT] bit. This allows for one interrupt to be pending while another is still being serviced. Once the interrupt is generated, the ETPS[INTCNT] bits are automatically cleared. Writing a INTPRD value that is the same as the current counter value triggers an interrupt if it is enabled and the status flag is clear. Writing a INTPRD value that is less than the current counter value results in an undefined state. If a counter event occurs at the same instant as a new zero or non-zero INTPRD value is written, the counter is incremented. 0h = Disable the interrupt event counter. No interrupt is generated and ETFRC[INT] is ignored. 1h = Generate an interrupt on the first event INTCNT = 01 (first event). 2h = Generate interrupt on ETPS[INTCNT] = 1,0 (second event). 3h = Generate interrupt on ETPS[INTCNT] = 1,1 (third event).

19.4.25 ETFLG Register (Offset = 36h) [reset = 0h]

ETFLG is shown in [Figure 19-87](#) and described in [Table 19-48](#).

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Figure 19-87. ETFLG Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				SOCB	SOCA	RESERVED	INT
R-0h				R-0h	R-0h	R-0h	R-0h

Table 19-48. ETFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	SOCB	R	0h	Latched ePWM ADC Start-of-Conversion B (EPWMxSOCB) Status Flag. 0h = No EPWMxSOCB event occurred. 1h = A start of conversion pulse is generated on EPWMxSOCB. The EPWMxSOCB output continues to be generated even if the flag bit is set.
2	SOCA	R	0h	Latched ePWM ADC Start-of-Conversion A (EPWMxSOCA) Status Flag. Unlike the ETFLG[INT] flag, the EPWMxSOCA output continues to pulse even if the flag bit is set. 0h = No event occurred. 1h = A start of conversion pulse is generated on EPWMxSOCA. The EPWMxSOCA output continues to be generated even if the flag bit is set.
1	RESERVED	R	0h	Reserved
0	INT	R	0h	Latched ePWM Interrupt (EPWMx_INT) Status Flag. 0h = No event occurred. 1h = An ePWMx interrupt (EPWMx_INT) is generated. No further interrupts are generated until the flag bit is cleared. Up to one interrupt can be pending while the ETFLG[INT] bit is still set. If an interrupt is pending, it is not generated until after the ETFLG[INT] bit is cleared.

19.4.26 ETCLR Register (Offset = 38h) [reset = 0h]

ETCLR is shown in [Figure 19-88](#) and described in [Table 19-49](#).

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Figure 19-88. ETCLR Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				SOCB	SOCA	RESERVED	INT
R-0h				R/W-0h	R/W-0h	R-0h	R/W-0h

Table 19-49. ETCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	SOCB	R/W	0h	ePWM ADC Start-of-Conversion B (EPWMxSOCB) Flag Clear Bit. 0h = Writing a 0 has no effect. Always reads back a 0. 1h = Clears the ETFLG[SOCB] flag bit.
2	SOCA	R/W	0h	ePWM ADC Start-of-Conversion A (EPWMxSOCA) Flag Clear Bit. 0h = Writing a 0 has no effect. Always reads back a 0. 1h = Clears the ETFLG[SOCA] flag bit.
1	RESERVED	R	0h	Reserved
0	INT	R/W	0h	ePWM Interrupt (EPWMx_INT) Flag Clear Bit. 0h = Writing a 0 has no effect. Always reads back a 0. 1h = Clears the ETFLG[INT] flag bit and enables further interrupts pulses to be generated.

19.4.27 ETFRC Register (Offset = 3Ah) [reset = 0h]

ETFRC is shown in [Figure 19-89](#) and described in [Table 19-50](#).

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Figure 19-89. ETFRC Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				SOCB	SOCA	RESERVED	INT
R-0h				R/W-0h	R/W-0h	R-0h	R/W-0h

Table 19-50. ETFRC Register Field Descriptions

Bit	Field	Type	Reset	Description
15-4	RESERVED	R	0h	Reserved
3	SOCB	R/W	0h	SOCB Force Bit. The SOCB pulse is only generated if the event is enabled in the ETSEL register. The ETFLG[SOCB] flag bit is set regardless. 0h = Writing 0 to this bit is ignored. Always reads back a 0. 1h = Generates a pulse on EPWMxSOCB and sets the SOCBFLG bit. This bit is used for test purposes.
2	SOCA	R/W	0h	SOCA Force Bit. The SOCA pulse is only generated if the event is enabled in the ETSEL register. The ETFLG[SOCA] flag bit is set regardless. 0h = Writing 0 to this bit is ignored. Always reads back a 0. 1h = Generates a pulse on EPWMxSOCB and set the SOCAFLG bit. This bit is used for test purposes.
1	RESERVED	R	0h	Reserved
0	INT	R/W	0h	INT Force Bit. The interrupt is only generated if the event is enabled in the ETSEL register. The INT flag bit is set regardless. 0h = Writing 0 to this bit is ignored. Always reads back a 0. 1h = Generates an interrupt on /EPWMxINT and set the INT flag bit. This bit is used for test purposes.

19.4.28 PCCTL Register (Offset = 3Ch) [reset = 0h]

PCCTL is shown in [Figure 19-90](#) and described in [Table 19-51](#).

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Figure 19-90. PCCTL Register

15	14	13	12	11	10	9	8
RESERVED						CHPDUTY	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
CHPFREQ			OSHTWTH				CHPEN
R/W-0h			R/W-0h				R/W-0h

Table 19-51. PCCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-11	RESERVED	R	0h	Reserved
10-8	CHPDUTY	R/W	0h	Chopping Clock Duty Cycle. 0h = Duty = 1/8 (12.5%) 1h = Duty = 2/8 (25.0%) 2h = Duty = 3/8 (37.5%) 3h = Duty = 4/8 (50.0%) 4h = Duty = 5/8 (62.5%) 5h = Duty = 6/8 (75.0%) 6h = Duty = 7/8 (87.5%) 7h = Reserved
7-5	CHPFREQ	R/W	0h	Chopping Clock Frequency. 0h = Divide by 1 (no prescale, = 12.5 MHz at 100 MHz VCLK4) 1h = Divide by 2 (6.25 MHz at 100 MHz VCLK4) 2h = Divide by 3 (4.16 MHz at 100 MHz VCLK4) 3h = Divide by 4 (3.12 MHz at 100 MHz VCLK4) 4h = Divide by 5 (2.50 MHz at 100 MHz VCLK4) 5h = Divide by 6 (2.08 MHz at 100 MHz VCLK4) 6h = Divide by 7 (1.78 MHz at 100 MHz VCLK4) 7h = Divide by 8 (1.56 MHz at 100 MHz VCLK4)
4-1	OSHTWTH	R/W	0h	One-Shot Pulse Width. 0h = 1 x VCLK4 / 8 wide (= 80 nS at 100 MHz VCLK4) 1h = 2 x VCLK4 / 8 wide (= 160 nS at 100 MHz VCLK4) 2h = 3 x VCLK4 / 8 wide (= 240 nS at 100 MHz VCLK4) 3h = 4 x VCLK4 / 8 wide (= 320 nS at 100 MHz VCLK4) 4h = 5 x VCLK4 / 8 wide (= 400 nS at 100 MHz VCLK4) 5h = 6 x VCLK4 / 8 wide (= 480 nS at 100 MHz VCLK4) 6h = 7 x VCLK4 / 8 wide (= 560 nS at 100 MHz VCLK4) 7h = 8 x VCLK4 / 8 wide (= 640 nS at 100 MHz VCLK4) 8h = 9 x VCLK4 / 8 wide (= 720 nS at 100 MHz VCLK4) 9h = 10 x VCLK4 / 8 wide (= 800 nS at 100 MHz VCLK4) Ah = 11 x VCLK4 / 8 wide (= 880 nS at 100 MHz VCLK4) Bh = 12 x VCLK4 / 8 wide (= 960 nS at 100 MHz VCLK4) Ch = 13 x VCLK4 / 8 wide (= 1040 nS at 100 MHz VCLK4) Dh = 14 x VCLK4 / 8 wide (= 1120 nS at 100 MHz VCLK4) Eh = 15 x VCLK4 / 8 wide (= 1200 nS at 100 MHz VCLK4) Fh = 16 x VCLK4 / 8 wide (= 1280 nS at 100 MHz VCLK4)
0	CHPEN	R/W	0h	PWM-chopping Enable. 0h = Disable (bypass) PWM chopping function. 1h = Enable chopping function.

19.4.29 DCTRIPSEL Register (Offset = 60h) [reset = 0h]

DCTRIPSEL is shown in [Figure 19-91](#) and described in [Table 19-52](#).

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Figure 19-91. DCTRIPSEL Register

15	14	13	12	11	10	9	8
DCBLCOMPSEL				DCBHCOMPSEL			
R/W-0h				R/W-0h			
7	6	5	4	3	2	1	0
DCALCOMPSEL				DCAHCOMPSEL			
R/W-0h				R/W-0h			

Table 19-52. DCTRIPSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	DCBLCOMPSEL	R/W	0h	Digital Compare B Low Input Select. Defines the source for the DCBL input. The TZ signals, when used as trip signals, are treated as normal inputs and are defined as active high or active low. Values not shown are reserved. If a device does not have a particular comparator, then that option is reserved. 0h = /TZ1 input 1h = /TZ2 input 2h = /TZ3 input
11-8	DCBHCOMPSEL	R/W	0h	Digital Compare B High Input Select. Defines the source for the DCBH input. The TZ signals, when used as trip signals, are treated as normal inputs and are defined as active high or active low. Values not shown are reserved. If a device does not have a particular comparator, then that option is reserved. 0h = /TZ1 input 1h = /TZ2 input 2h = /TZ3 input
7-4	DCALCOMPSEL	R/W	0h	Digital Compare A Low Input Select. Defines the source for the DCAL input. The TZ signals, when used as trip signals, are treated as normal inputs and are defined as active high or active low. Values not shown are reserved. If a device does not have a particular comparator, then that option is reserved. 0h = /TZ1 input 1h = /TZ2 input 2h = /TZ3 input
3-0	DCAHCOMPSEL	R/W	0h	Digital Compare A High Input Select. Defines the source for the DCAH input. The TZ signals, when used as trip signals, are treated as normal inputs and are defined as active high or active low. Values not shown are reserved. If a device does not have a particular comparator, then that option is reserved. 0h = /TZ1 input 1h = /TZ2 input 2h = /TZ3 input

19.4.30 DCACTL Register (Offset = 62h) [reset = 0h]

DCACTL is shown in [Figure 19-92](#) and described in [Table 19-53](#).

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Figure 19-92. DCACTL Register

15	14	13	12	11	10	9	8
RESERVED						EVT2FRC_SY NCSEL	EVT2SRCSEL
R-0h						R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
RESERVED				EVT1SYNCE	EVT1SOCE	EVT1FRC_SY NCSEL	EVT1SRCSEL
R-0h				R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 19-53. DCACTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9	EVT2FRC_SYNCSEL	R/W	0h	DCAEVT2 Force Synchronization Signal Select. 0h = Source Is Synchronous Signal. 1h = Source Is Asynchronous Signal.
8	EVT2SRCSEL	R/W	0h	DCAEVT2 Source Signal Select. 0h = Source Is DCAEVT2 Signal. 1h = Source Is DCEVTFILT Signal.
7-4	RESERVED	R	0h	Reserved
3	EVT1SYNCE	R/W	0h	DCAEVT1 SYNC Enable. 0h = SYNC generation is disabled. 1h = SYNC generation is enabled.
2	EVT1SOCE	R/W	0h	DCAEVT1 SOC Enable. 0h = SOC generation is disabled. 1h = SOC generation is enabled.
1	EVT1FRC_SYNCSEL	R/W	0h	DCAEVT1 Force Synchronization Signal Select. 0h = Source Is Synchronous Signal. 1h = Source Is Asynchronous Signal.
0	EVT1SRCSEL	R/W	0h	DCAEVT1 Source Signal Select. 0h = Source Is DCAEVT1 Signal. 1h = Source Is DCEVTFILT Signal.

19.4.31 DCBCTL Register (Offset = 64h) [reset = 0h]

DCBCTL is shown in [Figure 19-93](#) and described in [Table 19-54](#).

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Figure 19-93. DCBCTL Register

15	14	13	12	11	10	9	8
RESERVED						EVT2FRC_SY NCSEL	EVT2SRCSEL
R-0h						R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
RESERVED				EVT1SYNCE	EVT1SOCE	EVT1FRC_SY NCSEL	EVT1SRCSEL
R-0h				R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 19-54. DCBCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-10	RESERVED	R	0h	Reserved
9	EVT2FRC_SYNCSEL	R/W	0h	DCBEVT2 Force Synchronization Signal Select. 0h = Source Is Synchronous Signal. 1h = Source Is Asynchronous Signal.
8	EVT2SRCSEL	R/W	0h	DCBEVT2 Source Signal Select. 0h = Source Is DCBEVT2 Signal. 1h = Source Is DCEVTFILT Signal.
7-4	RESERVED	R	0h	Reserved
3	EVT1SYNCE	R/W	0h	DCBEVT1 SYNC Enable. 0h = SYNC generation is disabled. 1h = SYNC generation is enabled.
2	EVT1SOCE	R/W	0h	DCBEVT1 SOC Enable. 0h = SOC generation is disabled. 1h = SOC generation is enabled.
1	EVT1FRC_SYNCSEL	R/W	0h	DCBEVT1 Force Synchronization Signal Select. 0h = Source Is Synchronous Signal. 1h = Source Is Asynchronous Signal.
0	EVT1SRCSEL	R/W	0h	DCBEVT1 Source Signal Select. 0h = Source Is DCBEVT1 Signal. 1h = Source Is DCEVTFILT Signal.

19.4.32 DCFCTL Register (Offset = 66h) [reset = 0h]

DCFCTL is shown in [Figure 19-94](#) and described in [Table 19-55](#).

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Figure 19-94. DCFCTL Register

15	14	13	12	11	10	9	8
RESERVED				RESERVED			
R-0h				R-0h			
7	6	5	4	3	2	1	0
RESERVED	RESERVED	PULSESEL		BLANKINV	BLANKE	SRCSEL	
R-0h	R-0h	R/W-0h		R/W-0h	R/W-0h	R/W-0h	

Table 19-55. DCFCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-13	RESERVED	R	0h	Reserved
12-8	RESERVED	R	0h	Reserved for TI Test
7	RESERVED	R	0h	Reserved
6	RESERVED	R	0h	Reserved for TI Test
5-4	PULSESEL	R/W	0h	Pulse Select For Blanking and Capture Alignment. 0h = Time-base counter equal to period (TBCTR = TBPRD) 1h = Time-base counter equal to zero (TBCTR = 0x0000) 2h = Reserved 3h = Reserved
3	BLANKINV	R/W	0h	Blanking Window Inversion. 0h = Blanking window is not inverted. 1h = Blanking window is inverted.
2	BLANKE	R/W	0h	Blanking Window Enable. 0h = Blanking window is disabled. 1h = Blanking window is enabled.
1-0	SRCSEL	R/W	0h	Filter Block Signal Source Select. 0h = Source Is DCAEVT1 Signal. 1h = Source Is DCAEVT2 Signal. 2h = Source Is DCBEVT1 Signal. 3h = Source Is DCBEVT2 Signal.

19.4.33 DCCAPCTL Register (Offset = 68h) [reset = 0h]

DCCAPCTL is shown in [Figure 19-95](#) and described in [Table 19-56](#).

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Figure 19-95. DCCAPCTL Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED						SHDWMODE	CAPE
R-0h						R/W-0h	R/W-0h

Table 19-56. DCCAPCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-2	RESERVED	R	0h	Reserved
1	SHDWMODE	R/W	0h	TBCTR Counter Capture Shadow Select Mode. 0h = Enable shadow mode. The DCCAP active register is copied to shadow register on a TBCTR = TBPRD or TBCTR = zero event as defined by the DCFCTL[PULSESEL] bit. CPU reads of the DCCAP register return the shadow register contents. 1h = Active Mode. In this mode the shadow register is disabled. CPU reads from the DCCAP register always returns the active register contents.
0	CAPE	R/W	0h	TBCTR Counter Capture Enable. 0h = Disable the time-base counter capture. 1h = Enable the time-base counter capture.

19.4.34 DCFOFFSET Register (Offset = 6Ah) [reset = 0h]

DCFOFFSET is shown in [Figure 19-96](#) and described in [Table 19-57](#).

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Figure 19-96. DCFOFFSET Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OFFSET															
R-0h															

Table 19-57. DCFOFFSET Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	OFFSET	R	0h	Blanking Window Offset. Valid values: 0-FFFFh These 16-bits specify the number of TBCLK cycles from the blanking window reference to the point when the blanking window is applied. The blanking window reference is either period or zero as defined by the DCFCTL[PULSESEL] bit. This offset register is shadowed and the active register is loaded at the reference point defined by DCFCTL[PULSESEL]. The offset counter is also initialized and begins to count down when the active register is loaded. When the counter expires, the blanking window is applied. If the blanking window is currently active, then the blanking window counter is restarted.

19.4.35 DCFOFFSETCNT Register (Offset = 6Ch) [reset = 0h]

DCFOFFSETCNT is shown in [Figure 19-97](#) and described in [Table 19-58](#).

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Figure 19-97. DCFOFFSETCNT Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OFFSETCNT															
R-0h															

Table 19-58. DCFOFFSETCNT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	OFFSETCNT	R	0h	Blanking Offset Counter. Valid values: 0-FFFFh These 16-bits are read-only and indicate the current value of the offset counter. The counter counts down to zero and then stops until it is re-loaded on the next period or zero event as defined by the DCFCTL[PULSESEL] bit. The offset counter is not affected by the FREE and SOFT emulation bits. That is, it always continues to count down if the device is halted by an emulation stop.

19.4.36 DCFWINDOW Register (Offset = 6Eh) [reset = 0h]

DCFWINDOW is shown in [Figure 19-98](#) and described in [Table 19-59](#).

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Figure 19-98. DCFWINDOW Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
WINDOW							
R/W-0h							

Table 19-59. DCFWINDOW Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Reserved
7-0	WINDOW	R/W	0h	Blanking Window Width. Valid values: 0-FFh These 16-bits specify the width of the blanking window in TBCLK cycles. The blanking window begins when the offset counter expires. When this occurs, the window counter is loaded and begins to count down. If the blanking window is currently active and the offset counter expires, the blanking window counter is restarted. The blanking window can cross a PWM period boundary. 0h = No blanking window is generated.

19.4.37 DCFWINDOWCNT Register (Offset = 70h) [reset = 0h]

DCFWINDOWCNT is shown in [Figure 19-99](#) and described in [Table 19-60](#).

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Figure 19-99. DCFWINDOWCNT Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
WINDOWCNT							
R-0h							

Table 19-60. DCFWINDOWCNT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Any writes to these bit(s) must always have a value of 0.
7-0	WINDOWCNT	R	0h	Blanking Window Counter. Valid value: 0-FFh These 8 bits are read only and indicate the current value of the window counter. The counter counts down to zero and then stops until it is re-loaded when the offset counter reaches zero again.

19.4.38 DCCAP Register (Offset = 72h) [reset = 0h]

DCCAP is shown in [Figure 19-100](#) and described in [Table 19-61](#).

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Figure 19-100. DCCAP Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCCAP															
R-0h															

Table 19-61. DCCAP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	DCCAP	R	0h	Digital Compare Time-Base Counter Capture. Valid values: 0-FFFFh To enable time-base counter capture, set the DCCAPCLT[CAPE] bit to 1. If enabled, reflects the value of the time-base counter (TBCTR) on the low-to-high edge transition of a filtered (DCEVTFLT) event. Further capture events are ignored until the next period or zero as selected by the DCFCTL[PULSESEL] bit. Shadowing of DCCAP is enabled and disabled by the DCCAPCTL[SHDWMODE] bit. By default this register is shadowed. If DCCAPCTL[SHDWMODE] = 0, then the shadow is enabled. In this mode, the active register is copied to the shadow register on the TBCTR = TBPRD or TBCTR = zero as defined by the DCFCTL[PULSESEL] bit. CPU reads of this register return the shadow register value. If DCCAPCTL[SHDWMODE] = 1, then the shadow register is disabled. In this mode, CPU reads return the active register value. The active and shadow registers share the same memory map address.